THE HORIZONTAL GEOMETRY OF PRODUCTION NETWORKS

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Abstract

Modern economies are organized as dense production networks, yet the transmission of shocks is conventionally understood through observed Input-Output linkages and vertical cascade effects. This paper argues that this view is incomplete. Propagation also operates through horizontal complementarities: the extent to which sectors share upstream suppliers or downstream buyers. These interdependencies shape a latent network of connections, linking also otherwise unconnected sectors, and generate simultaneous responses to idiosyncratic shocks through structural demand and supply similarities embedded in the Input-Output system. Aggregate volatility then results from widespread correlations that prevent shocks from averaging out. This diffusion channel acts as a complement to vertical propagation, offering a unified view of comovement rooted in the horizontal geometry of common interactions rather than network trade intensities. Using Input-Output data, I show that sectoral responses vary systematically with horizontal relations, forging novel implications for the transmission of sectoral and aggregate shocks.

JEL: C67, D57, E32, J21

KEYWORDS: production networks, Input-Output economy, horizontal transmission, network distance, sectoral comovement, aggregate fluctuations

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1. INTRODUCTION

A hallmark of contemporary production systems is the comovement of economic activity across sectors. Recent advances in the production network literature have demonstrated how independent, idiosyncratic shocks propagate through observed Input-Output linkages, generating simultaneous adjustments in productions and contributing to aggregate fluctuations. Here, the presence of vertical "cascade effects" diffuses the shock along parallel supply chains via the tangle of sectoral ties.[1] Yet, a shock often travels through hidden network corridors: co-movement may nonetheless emerge through common downstream buyers (*e.g.*, February 2, 2026: the HBM4 memory breakthrough reshuffles demand across competing chip providers) or common upstream suppliers (*e.g.*, the 2025 AI-driven surge in cloud demand for DRAM and HBM chips strained capacity for other downstream users).

This paper argues that production networks connect sectors not only through existing trade linkages, but also by demand and supply similarities embedded in their common structure of inter-sectoral trade. I conceptualize these interdependencies as *network economic distances* (*e.g.*, Conley and Dupor 2003). Specifically, sectors exhibit "factor input demand" distance when they source intermediate inputs from the same upstream suppliers, and "factor input supply" distance when they sell to the same downstream buyers. Viewed through this lens, an Input-Output structure not only reflects *vertical* interdependencies, where transmitting shocks across sectors reflects their intensity of being inter-connected (*e.g.*, Shea 2002), but also *horizontal* complementarities, as network participants are additionally linked through structural demand and supply connections from a resembling Input-Output geometry.

Network-based comovement is therefore endogenous. Since sectors mutually adjust in response due to shared economic relationships, horizontal geometries become a source of systematic comovement across economically proximate sectors. The propagation of shocks depends not only on the sequentiality proper of observed linkages, but also on the exposure to the same Input-Output neighbourhoods: two sectors may display correlated responses even in the absence of strong vertical connections simply because they are economically connected through common buyers or suppliers. Hence, horizontal distances generate heterogeneous exposures to both idiosyncratic and aggregate disturbances. The question is therefore not only whether shocks propagate through production networks, but also whether the horizontal geometry of these networks matters for determining who is exposed to whom, and by how much.

Under this perspective, aggregate volatility reflects not only how idiosyncratic shocks are amplified, but also the way the network structure induces correlated be-

[1] Central insight is that a positive shock to a sector enhances its productive performance, triggering: *(i)* a *downstream effect*, as buyers have an incentive to expand their demand from that sector; *(ii)* an *upstream effect*, as higher production in the shocked sector raises its demand of intermediate inputs, benefiting its suppliers; and *(iii)* vertical *network spillovers*, where the initial effects ripple beyond immediate trading partners, benefiting indirectly connected sectors. Still through observed chains, the transmission of shocks proves less straightforward when non unitary elasticities of substitution limit the ability to adjust input combinations (*e.g.*, Baqaee and Farhi 2019). On the verticality of Input-Output economies refer to Pasinetti (1973).

haviours. When a shock hits, it propagates vertically by cascading from suppliers to customers through input requirements, generating recursive feedback loops and sequential amplification along parallel supply chains (*e.g.*, Acemoglu, Carvalho, et al. 2012). At the same time, it propagates via horizontal complementarities, as sectors sharing buyers or suppliers adjust simultaneously to similar cost or supply conditions. These responses do not generate cascades but induce systematic comovement across economic activities, preventing shocks from averaging out and allowing aggregate fluctuations to emerge from the network's horizontal geometry itself.

Indeed, building a coherent theoretical framework that emphasizes the role of demand- and supply-driven network economic distances on sectoral comovement and aggregate dynamics within an Input-Output economy, and validating its main predictions using sector-level data, constitute the central novelties of this paper.

Section 2 lays down the conceptual foundation for horizontal complementarities. On the input side, sectors buying from the same upstream suppliers reveal simultaneous adjustments as they rebalance labour and intermediate inputs. On the output side, sectors selling to the same downstream buyers face relative price adjustments, and thus revenue shifts when downstream demand changes. Theorem 1 establishes a "dual representation" of an Input-Output economy: the matrix of technical coefficients is a primitive object that embeds not only observed (direct and indirect) production linkages, but also horizontal relations from shared demand and supply structures. These horizontal interdependencies also bound otherwise disconnected nodes: as follows from Theorem 2, even if don't directly trade, the existence of a common neighbour makes network participants economically connected.

Section 3 interprets the horizontal transmission of asymmetric shocks. Under demand linkages, opposite comovement arises only when intermediate inputs cannot be fully substituted, as a positive shock to one downstream user bids up the price of a shared supplier and scales other buyers' demand. By contrast, in the supply-based case, opposite comovement is immediate and mechanical, as a sector capturing demand from a shared downstream buyer reduces activity among competing suppliers after a positive shock that effectively lowers its relative price. In this sense, demand linkages make comovement ambiguous and dependent on substitution patterns in downstream demand, whereas supply linkages make it transparent and directly tied to upstream competition for downstream markets. Crucially, the elasticity of substitution across intermediate inputs does not alter the direction of comovement.

In developing this perspective, Theorem 3 emerges: sectoral interdependencies in terms of common demand or supply relationships retain an independent force, distinct from the intensity of inter-sectoral trade flows. This separation strengthen the importance of horizontal complementarities, since these are not themselves weighted by the vertical Input-Output structure. In addition, Theorem 4 establishes how, for very large networks, horizontal effects from shared buyers and suppliers proliferate, making horizontal spillovers to dominate shock transmission over vertical chains.

Since horizontal complementarities operate through adjustments in factors of production, the analysis turns to employment comovement and its role on aggregate cycles. Dynamic patterns hinge on network distances: nearby sectors move in opposite directions due to shared Input-Output relations, causing diverging employment responses that dampen aggregate fluctuations, whereas distant sectors tend to comove as the vertical propagation prevails, generating broadly aligned employment dynamics. Theorem 5 then formalizes that horizontal interdependencies alone generate persistent aggregate fluctuations, thereby breaking the standard diversification argument by Lucas (1977): with many sectors sharing upstream or downstream connections, shocks propagate by the means of correlated behaviours, inducing mutual responses that accumulate at the aggregate level.

As a validation of theoretical insights, Section 4 uses U.S. sectoral and Input-Output data to simulate GDP dynamics. Sustained aggregate volatility stems from common upstream/downstream connections, where the structural properties of the horizontal propagation induce aggregate fluctuations to mean revert at a slower rate of decay than the one predicted by the standard diversification. Delving into the mechanics of horizontal patterns reveals how they rely on diffuse interdependencies rather than a few vertical propagation hubs, and are strengthened as network sparsity increases, lending empirical support to Theorems 4 and 5 and their corollaries. A Bartik-style estimation further investigates whether horizontal geometries shape the transmission of aggregate disturbances when controlling for vertical Input-Output exposures. The relevant propagation channel depends on the nature of the disturbance: aggregate productivity shocks are primarily transmitted through supply-based horizontal relationships, whereas aggregate government spending shocks are amplified mainly through demand-based horizontal connections. Horizontal exposures therefore contain economically meaningful information about shock transmission not captured by conventional Leontief inverse linkages.

Section 5 implements the two-stage approach in Barattieri and Cacciatore (2023) to study sectoral comovement. The empirical results are threefold. Firstly, an increase in the set of circulating intermediate inputs of a sector leads to a comparatively smaller rise in its employment with minimal but alternate effects from similar changes in distanced sectors. Secondly, sectors closer under common supply distance exhibit opposite employment comovement: an increase in employment in nearby sectors tends to reduce employment in the sector; for closely demand-linked sectors, the responses are often ambiguous (increases for some, reductions for others), but mostly pointing to a dampening effect on the vertical shock's transmission. Thirdly, positive comovement emerges among more distantly demand- and supply-related sectors, since sectoral employment rises in response to increases in farther ones.

To conclude, Section 6 examines further implications of the network's horizontal geometry and how incorporating it can improve the design of policy interventions.

Overall, the discussion provides empirical support for the theoretical predictions: horizontal complementarities in terms of similar demand and supply relationships complement the canonical transmission of shocks, revealing the multiple channels through which vertical and horizontal structures of sectoral interdependencies shape the distribution and the propagation of economic activity across networked economies.

Literature.– The objective of this paper is to advance both theoretically and empirically the understanding of how Input-Output economies work. On the theoretical side, it builds on the modern and rapidly expanding literature on production networks. In the wake of Gabaix (2011)'s "granular hypothesis", the seminal observation of Long and Plosser (1983) – that comovements across sectors are not dictated by a shared disturbance but rather by sectoral interdependencies –, has been modernized through a series of papers of the early 2010s (*e.g.*, Acemoglu, Carvalho, et al. 2012; Carvalho and Gabaix 2013; Carvalho 2014; Barrot and Sauvagnat 2016). This renewed perspective gave rise to studies on production networks related to efficient economies (*e.g.*, Baqaee and Farhi 2019; vom Lehn and Winberry 2022; Liu and Tsyvinski 2024), monetary policy and nominal rigidities (*e.g.*, La'O and Tahbaz-Salehi 2022; Rubbo 2023; Ghassibe and Nakov 2025), inefficiencies (*e.g.*, Jones 2011; Grassi 2017; Baqaee 2018; Baqaee and Farhi 2020), policy-oriented issues (*e.g.*, Liu 2019; Grassi and Sauvagnat 2019; Lane 2025), endogenous network formation (*e.g.*, Oberfield 2018; Acemoglu and Azar 2020; Ghassibe 2024; Taschereau-Dumouchel 2026), economic growth (*e.g.*, Hausmann and Hidalgo 2011; Gualdi and Mandel 2019; McNerney et al. 2022), and international contexts (*e.g.*, Caliendo et al. 2022; Qiu et al. 2026; Huo et al. 2025). A common mechanism explored in these works emphasizes the vertical propagation, where shocks originating in upstream sectors transmit downstream, amplifying their effects and contributing to aggregate outcomes.[2] My contribution broadens this perspective by introducing horizontal linkages from network "economic" distances, measured by common upstream or downstream Input-Output relationships. These novel interdependencies forge horizontal complementarities that bind even disconnected sectors, allowing asymmetric shocks to propagate across the network in ways beyond traditional and vertical supply chains.

On the empirical side, relatively few studies examine the role of sectoral production networks on macroeconomic outcomes (*e.g.*, Acemoglu, Akcigit, et al. 2015; Ghassibe 2021; Barattieri and Cacciatore 2023; Barattieri, Cacciatore, and Traum 2023; Monti and Van Keirsbilck 2025), with some illustrative spillovers for horizontal transmission (*e.g.*, Boehm and Oberfield 2018; Dragomirescu-Gaina and Elia 2021; Huremović et al. 2025). From this standpoint, the paper contributes novel empirical evidence on the role of Input-Output linkages in shaping interlinked dynamics, particularly in explaining the comovement of employment across sectors. Synchronized employment patterns are an important characteristic of business cycles, as most sectors tend to move together over time (*e.g.*, Christiano and Fitzgerald 1998). Several studies explore the nature and the sources of this sectoral comovement. Cooper and Haltiwanger (1990) find that sectoral employment levels are positively correlated,

[2] As noted by Shea (2002), a positive sectoral supply shock has a price (decrease in its price level, and reduction in its nominal expenditure for intermediate inputs) and a quantity (its supply to other sectors increase, sector's production rises, thereby increasing its input demand) effects. These opposing forces affect upstream suppliers but, under Cobb-Douglas production function, they exactly offset each other and leaving intermediate inputs demand unchanged, thereby only affecting downstream suppliers. In general, only demand shocks can propagate upstream (*e.g.*, Acemoglu, Akcigit, et al. 2015; Barrot and Sauvagnat 2016; Ferrari 2024).

in ways not fully explained by aggregate shocks. Cassou and Vázquez (2014) attribute high employment comovement to similar shock transmission channels. In Yedid-Levi (2016) comovement is stronger along the extensive (number of workers) than the intensive (hours per worker) margin. Room for structural changes is in Kim (2020), where positive technology shocks in manufacturing raise employment in both manufacturing and services. While these studies provide valuable insights, they do not explicitly address how production networks shape employment comovement. By contrast, my paper focuses on short-run employment variations within a networked framework. Empirically, the paper contributes by *(i)* advancing the comprehension of shocks' transmission via network "economic" distances, and *(ii)* providing unprecedented evidence on employment comovement from a production network perspective.

2. FOUNDATIONAL THEORY

Preliminaries.– The economy is populated by a finite set of sectors, $\{s, s', s'', \dots, m\} \in \Phi$, with Φ_s denoting the set of all sectors not including sector-s. Each produces a single commodity using labour and a set of circulating intermediate inputs from other sectors. The whole economic system is thus represented by an $m \times 1$ vector of sectoral production outputs, $\boldsymbol{y} = \left[y(s) > 0\right]$, an $m \times 1$ vector of sectoral employment levels, $\boldsymbol{n} = \left[n(s) > 0\right]$, each associated to its output elasticity of labour $\alpha(s) \in (0,1)$, and by an $m \times m$ square matrix, $\boldsymbol{X} = \left[x(s,s') \geq 0\right]$, indicative of the inter-sectoral trade of intermediate input quantities, with $x(s,s') = 0$ stating that sector-s does not make use of the commodity produced by sector-s'. Finally, an $m \times m$ square matrix, $\boldsymbol{\mathcal{A}} = \left[\alpha(s,s') \geq 0\right]$, defines the intensity of the commodities produced by sectors as intermediate inputs in other sectors; denote its Leontief inverse transformation as $\boldsymbol{\mathcal{L}} = \left(\boldsymbol{I} - \boldsymbol{\mathcal{A}}\right)^{-1} = \left[\ell(s,s') \geq 0\right]$, with $\boldsymbol{I}$ being an $m \times m$ identity matrix.

Under these specifications, a perfectly competitive final good producer combines the production of different sectors, $Y = \prod_{s\in\Phi} y(s)^{\omega(s)} \propto \boldsymbol{\omega}'\boldsymbol{y}$, where $\boldsymbol{\omega}$ is a $1 \times m$ vector of $\omega(\cdot) \in (0,1)$, weighting the relative importance of each sectoral commodity in the definition of aggregate output Y. Total consumption is an aggregator over final demands: let $\boldsymbol{c} = \left[c(s)\right]$ be an $m \times 1$ vector of sectoral consumption levels by households, with generic element $c(\cdot) > 0$, it holds that $C = \prod_{s\in\Phi} c(s)^{\omega(s)} \propto \boldsymbol{\omega}'\boldsymbol{c}$.

Within a unit mass, household-i is getting utility from consumption of all sectoral commodities and leisure dis-utility from working. Utility function is

$$\mathcal{U}_i := f^u\Big(\left\{c_i(s)\right\}_{\forall s\in\Phi}, n_i(s)\Big)$$

with individual sectoral consumption and labour supplied characterized by a price, $p(s)$ and $w(s)$, respectively, the latter chosen as the numeraire.

A representative firm characterizes each sector-s, whose production function is

$$y(s) = z(s) f^y\Big(n(s)\ ,\ \left\{x(s,s')\right\}_{\forall s'\in\Phi}\Big)$$

where $z(s)$ is an exogenous Hicks-neutral sectoral productivity.

Assumption 1 (Production technology requirements) *As main characteristics of a production function: (i) it has constant returns to scale in either* $x(s,s' \in \Phi)$ *and* $n(s)$*, so that production inputs shares sum to one,* $\alpha(s) + \sum_{s'} \alpha(s,s') = 1$*; (ii) it is differentiable, continuous, strictly quasi-concave, homogeneous of degree one, and increasing in* $z(s)$*,* $x(s,s' \in \Phi)$*, and* $n(s)$*; (iii) the case* $f^y(0\ ,\ \cdot)$ *is ruled out as labour input is essential to production; (iv) at least two elements of* $x(s,s' \in \Phi)$ *have to be positive, thus* $f^y(\cdot\ ,\ 0)$ *cannot exist; consequentially, (v) part of the sectoral output is directly produced by the sector,* $x(s,s) > 0$*,* $\forall s \in \Phi$*, that is a production plan with a bundle of intermediate inputs as* $f^y(\cdot\ ,\ x(s,s' \in \Phi_s))$ *is precluded.*

The first two sub-assumptions are common in production technology. By way of *A1.iii*, workers cannot be fully substituted by any combination of intermediate inputs (so that sectoral production is finite), and changes in the purchased quantity of intermediate inputs would determine subsequent variations in the labour force size. The assumptions characterizing the bundle of intermediates, *A1.iv* and *A1.v*, ensure the sector not only to participate in the network, but also that part of the production process is roundabout: for sector-s delivering its commodity $y(s)$ requires to be at least vertically connected (buying from one upstream sector), while imposing that a fraction of intermediate inputs must be directly produced within the sector.

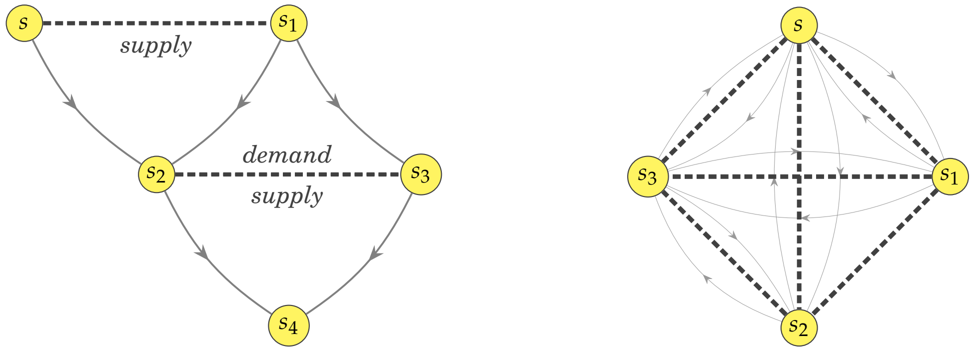
Each representative firm is seeking to maximize profits in a perfectly competitive environment, and optimality conditions for both demanded labour and intermediate inputs are then continuous functions as $n(s) = f^y_{n(s)}(y(s), p(s), w(s))$ and $x(s,s') = f^y_{x(s,s')}(y(s), p(s), p(s'))$, so that their combination delivers

$$x(s,s') = f\Big(\alpha(s,s'), p(s')\ ;\ \alpha(s), n(s), w(s)\Big) \tag{1}$$

From the outlined economy, the sectoral Input-Output equilibrium is defined by final consumption and network-based allocations: $y(s) = c(s) + \sum_{s' \in \Phi} x(s',s)$.

Distances.– In a networked economy, comovement of production inputs is endogenously induced. Each sector both relies on and supplies to others, so that changes in one sector affect not only its own optimal input composition but also the demand and supply conditions faced by other sectors. Sectors that trade with similar partners therefore face shared network exposure, providing a natural basis for a rigorous comparison. Paraphrasing Conley and Dupor (2003), two types of network "economic" distances between sectors, sketched in Panel 1a of Figure 1, can be defined: *(a) factor input demand*, when sectors are buying their intermediate inputs from similar upstream sectors; and *(b) factor input supply*, when sectors are selling their commodities as intermediate inputs to similar downstream sectors.

Network connections extend beyond their flows of intermediate inputs. These additional interdependencies highlight the complexity of inter-sectoral dynamics and complement the *Leontief inverse*, a reminiscent result from Leontief (1936): it captures how sectors not directly trading are indirectly related – *e.g.*, a sector buying from an upstream one introjects such sector's purchases as well the other of more upstream sectors –, emphasizing the depth of vertical linkages through layers of



(A) DISTANCE DEFINITION

(B) DISTANCE NETWORK

FIGURE 1: HORIZONTAL INPUT-OUTPUT GEOMETRIES

Note: the figure presents a stylized production network and its corresponding network of sectoral distances. Production network linkages are depicted with oriented thin lines, while dashed thick lines indicate the distance relationships.

Input-Output relationships. Network-based economic distances, by contrast, consider how participants are mapped horizontally as well: sectors are additionally related because they are relying on the same suppliers or serving the same customers. These broader notions of linkages provide a richer understanding about systemic interdependencies and explain *horizontal complementarities* in demand and supply that classical Input-Output analysis overlook. They reveal how seemingly unrelated sectors can experience synchronized fluctuations, and how the structure of common upstream or downstream connections shapes the propagation of independent and idiosyncratic shocks. In other words, the notion of network interdependencies is not confined to "who trades with whom", but also emerges from "who depends on whom".

Panel 1b sketches a stylized economy associated to its distance-based network: distances effectively double Input-Output linkages and bound disconnected sectors.

EXAMPLE 1 (Distance in the network) *Consider an economy populated by four sectors, $\{s, s_1, s_2, s_3\} \in \Phi$, where some trade with all others, while some do not. In Panel 1b, focus on the pair $\{s_1, s_2\}$: they do not mutually trade, so $\alpha(s_1, s_2) = 0 = \alpha(s_2, s_1)$. Nevertheless, their linkages with all the remaining sectors generate a common structure of inter-sectoral trade. Yet, while the underlying trade and distance structures are unique, the horizontal metrics vary under alternative specifications of network distance, as Examples 2-3 clarify.*

Finally, I embed these notions of horizontal geometry into an Input-Output equilibrium. The same primitive structure governing the trade of commodities not only determines how connections spread through production chains, but also how sectors are positioned relative to each other in the space of interdependencies. In this sense, the network matrix induces both a dynamic vertical structure and a geometric delineation of sectoral similarities. The following result formalises this dual role.

THEOREM 1 (On the dual representation) *Consider the equilibrium of an Input-Output economy, $\boldsymbol{y} = \boldsymbol{\mathcal{A}} \boldsymbol{y} + \boldsymbol{c}$. The matrix $\boldsymbol{\mathcal{A}}$ admits two distinct representations:*

– *vertical system, an equilibrium mapping governed by* $\mathcal{L} = \sum_{k=0}^{\infty} \mathcal{A}^k$ *encoding multiple production layers for circulating commodities;*
– *horizontal embedding, a collection of Input-Output profiles* $\{\boldsymbol{\alpha}(s)\}_{s\in\Phi}$ *encoding a geometric representation of sectoral similarities;*

Both representations are fully determined by the primitive Input-Output matrix.

Proof in Appendix A.1.

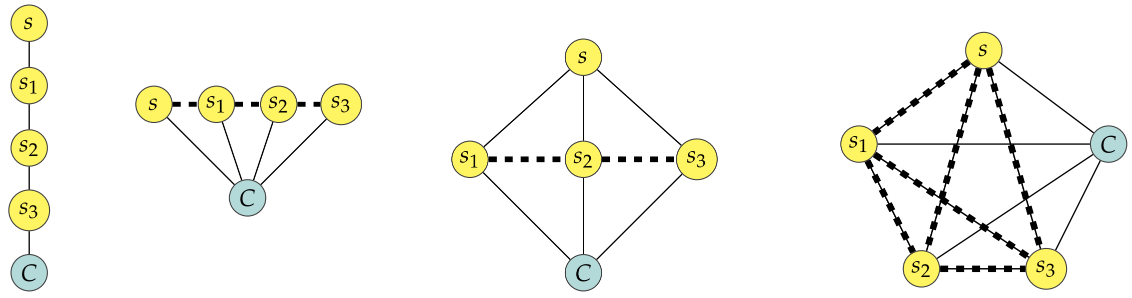
The theorem states that a single Input-Output matrix generates two distinct but tightly connected objects. On one hand, its Leontief inverse captures vertical dynamics: how commodities are produced by the means of other commodities (Sraffa 1960) through successive layers of production from direct to increasingly indirect effects along length-k production chains. On the other hand, the same matrix defines a horizontal geometry: sectors are related insofar as they share similar demand and supply relationships, interacting with common segments of the network. This induces a set of sectoral profiles $\boldsymbol{\alpha}(s) = [\alpha(s,s), \ldots, \alpha(s,m)]$, for all $s \in \Phi$, which embeds sectors according to their commonalities in the Input-Output space. Key is that these objects are not separated but different features of the same primitive structure: one capturing recursive chains, the other describing the overlap of network linkages.

2.1. Stylized economies and horizontal geometry

Before turning to technicalities, I illustrate the economic relevance of the network's horizontal geometry through highly stylized production structures, sketched in Figure 2. I consider three benchmark configurations: *(i) sequential economies*, where connections occurs purely along a vertical chain (sector-by-sector towards the final consumer) or, alternatively, purely horizontally (distinct sectors supplying only the final consumer); *(ii) rhomboid economies*, where sectors depend on a common upstream supplier while simultaneously selling to the final consumer; and *(iii) star economies*, with all sectors mutually connected while serving the final consumer.

Sequential economies.– Consider the "snake" in the left graph of Figure 2a. Connections are purely vertical, the upstream commodities flow sequentially along the supply chain, with no scope for horizontal interdependencies. By contrast, in the "half-spider" on the right graph, sectors sell directly to the final consumer without trading among themselves. This structure is a characteristic of standard multisector models with no inter-sectoral trade, and supply-based horizontal complementarities are central: even without explicit trade linkages, paired sectors are tied together by their shared exposure to final demand. Adjustments in one sector reshapes consumers' expenditure across all others, moving horizontally through final demand reallocation and generating simultaneous responses absent any network structure.

Rhomboid economies.– A simple network in Figure 2b combines both sequential economies into a unique structure. This configuration blends vertical and horizontal propagation, as commodities travel vertically (via sector-specific supply chain) and horizontally across sectors simultaneously exposed to common supplier and final demand. Rhomboid economies thus epitomize both types of horizontal complementari-



(A) SEQUENTIAL (B) RHOMBOID (C) STAR

FIGURE 2: BENCHMARK ECONOMIES

Note: the figure illustrates some benchmark economic systems and their associated horizontal geometry (dashed) highlighting common Input-Output structures (solid). Sectors are in yellow, while the teal circles with C represent the final consumer.

ties, as even sectors not directly trading with each other become tightly synchronized through their shared dependence on upstream and downstream markets.

Star economies.– Figure 2c classifies three archetypes. In fully symmetric and balanced star networks, where all sectors buy and sell with identical intensity, network distances collapse to zero: propagation is only vertical, but the network structure becomes irrelevant as sectoral shocks average out (*e.g.*, Lucas 1977; Acemoglu, Carvalho, et al. 2012). In partially symmetric star networks, trade intensities in buying and selling are equal but sector-specific; vertical and horizontal dimensions coexist, though the two distance measures coincide. Finally, asymmetric star networks (*i.e.*, different trade intensities across sectors) generate parallel vertical and horizontal channels, with comovement emerging endogenously from network distances: horizontal complementarities in production and shared exposure to final demand weave a dense web of horizontal interdependencies, and sectors with differing upstream and downstream connections experience unrelated effects.

These illustrative examples represent extreme cases. In more realistic, highly interconnected systems, the boundaries between these benchmark economies blur: vertical supply chains coexist with horizontal interdependencies through shared buyers and suppliers, and sectors participate simultaneously in multiple propagation layers.

2.2. RECOVERING DEMAND AND SUPPLY DISTANCES

Consider the case in which two sectors, say $\{s, s'\}$, are buying intermediate inputs from the same sector, say s^*. In this scenario, by substituting out the combined optimality conditions in eq. (1) for such sectors purchasing from the same upstream seller then one obtains, after total *log*-differentiation, that

$$d \log \mathcal{B}_{s^* \rightarrow}(s') = \frac{\alpha(s, s^*)}{\alpha(s', s^*)}\, d \log \mathcal{B}_{s^* \rightarrow}(s) \tag{2}$$

Given $\gamma^{\mathcal{B}}_{(\cdot)} = \frac{1}{\log \overline{x}(\cdot, s^*)}$ and $\delta^{\mathcal{B}}_{(\cdot)} = \frac{\log \overline{n}(\cdot)}{\log \overline{x}^2(\cdot, s^*)}$ being steady-state basis for production inputs, the differential elements $d \log \mathcal{B}_{s^* \rightarrow}(s) = \gamma^{\mathcal{B}}_{s}\, d \log n(s) - \delta^{\mathcal{B}}_{s}\, d \log x(s, s^*)$ and $d_{s^* \rightarrow} \log \mathcal{B}(s') = \gamma^{\mathcal{B}}_{s'}\, d \log n(s') - \delta^{\mathcal{B}}_{s'}\, d \log x(s', s^*)$ identify the *log*-difference of employment levels and intermediate inputs in the two sectors buying from the same sector. Subscript "$s^* \rightarrow$" reads as "purchasing from sector-s^*" by sectors in brackets.

Iteratively, the above condition can be rewritten accounting for all sectors to which the pair is buying from:

$$\boldsymbol{\mathcal{F}}\left[s,s'\right] = \left(\frac{\alpha\left(s,s\right)}{\alpha\left(s',s\right)}, \frac{\alpha\left(s,s'\right)}{\alpha\left(s',s'\right)}, \ldots, \frac{\alpha\left(s,m\right)}{\alpha\left(s',m\right)}\right) = \boldsymbol{d}^{fd}\left[s,s'\right]$$

where, given $\mathcal{F}\left[s,s'\right] = d\log\mathcal{B}_{s^*\rightarrow}\left(s'\right) / d\log\mathcal{B}_{s^*\rightarrow}\left(s\right)$, the indicator $\boldsymbol{\mathcal{F}}\left[s,s'\right]$ is a $1 \times m$ vector of changes in the ratio of any difference in production inputs within the pair of sectors, and $\boldsymbol{d}^{fd}\left[s,s'\right]$ is the associated $1 \times m$ vector of their relative intermediate usage intensity when sectors $\{s,s'\}$ are buying from others, yielding a unique value, $d^{fd}\left[s,s'\right]$. Stacking such condition across all sectors, then $\boldsymbol{\mathcal{F}} = \boldsymbol{\mathcal{D}}^{fd}$. In other words, the interdependence of inputs of production of the pair is determined by their distance in the production network. In fact, each cell of the $m \times m$ matrix $\boldsymbol{\mathcal{F}}$ displays the differential variation across factors of production between the row-sector and the column-sector, while matrix $\boldsymbol{\mathcal{D}}^{fd}$ is constructed as

$$\underset{m\times m}{\boldsymbol{\mathcal{D}}^{fd}} = \begin{bmatrix} \left(\frac{\alpha(s,s)}{\alpha(s,s)},\ldots,\frac{\alpha(s,m)}{\alpha(s,m)}\right) & \left(\frac{\alpha(s,s)}{\alpha(s',s)},\ldots,\frac{\alpha(s,m)}{\alpha(s',m)}\right) & \cdots & \left(\frac{\alpha(s,s)}{\alpha(m,s)},\ldots,\frac{\alpha(s,m)}{\alpha(m,m)}\right) \\ \left(\frac{\alpha(s',s)}{\alpha(s,s)},\ldots,\frac{\alpha(s',m)}{\alpha(s,m)}\right) & \left(\frac{\alpha(s',s)}{\alpha(s',s)},\ldots,\frac{\alpha(s',m)}{\alpha(s',m)}\right) & \cdots & \left(\frac{\alpha(s',s)}{\alpha(m,s)},\ldots,\frac{\alpha(s',m)}{\alpha(m,m)}\right) \\ \vdots & \vdots & \ddots & \vdots \\ \left(\frac{\alpha(m,s)}{\alpha(s,s)},\ldots,\frac{\alpha(m,m)}{\alpha(s,m)}\right) & \left(\frac{\alpha(m,s)}{\alpha(s',s)},\ldots,\frac{\alpha(m,m)}{\alpha(s',m)}\right) & \cdots & \left(\frac{\alpha(m,s)}{\alpha(m,s)},\ldots,\frac{\alpha(m,m)}{\alpha(m,m)}\right) \end{bmatrix}$$

The greater the value of each cell, the closer the sectors are within the production network.[3] This concept of "network proximity" is foundational on the theory side while, in the empirics, the interpretation is inverted: entries of $\boldsymbol{\mathcal{D}}^{fd}$ are the sum over a sequence of *Euclidean distances* across the elements of the intensity ratios,

$$d^{fd}\left[s,s'\right] = \sqrt{\left[\alpha\left(s,s\right)-\alpha\left(s',s\right)\right]^2 + \left[\alpha\left(s,s'\right)-\alpha\left(s',s'\right)\right]^2 + \cdots + \left[\alpha\left(s,m\right)-\alpha\left(s',m\right)\right]^2}$$

thereby representing a certain network "economic" distance between two sectors, with larger values denoting how dissimilar they are in terms of shared Input-Output structure when buying from upstream sellers. Henceforth, matrix $\boldsymbol{\mathcal{D}}^{fd}$ is reflecting the proper distance in common demand mapping sectors, whose characterizing elements, $d^{fd}\left[s,s'\right] > 0, \forall s,s' \in \Phi$, are symmetric outside the main diagonal (the distance from s to s' is the same as from s' to s), and $diag\left(\boldsymbol{\mathcal{D}}^{fd}\right) = 0$.

Example 2 (Factor input demand metrics) *Consider the pair of sectors $\{s_1,s_2\}$ in an Input-Output matrix mirroring Example 1, where the common sectors from which both are purchasing are $\{s,s_3\}$, and assume fictional trade intensities for sector-*

[3] A lower trade intensity ratio in eq. (2) implies that a shock to s transmits mildly to s'. For $\alpha\left(s,s^*\right) < \alpha\left(s',s^*\right)$, the resulting change in its relative quantities, $d\log\mathcal{B}_{s^*\rightarrow}\left(s\right)$, affects that of s', $d\log\mathcal{B}_{s^*\rightarrow}\left(s'\right)$, less than proportionally, signaling low horizontal complementarity. Conversely, if $\alpha\left(s,s^*\right) > \alpha\left(s',s^*\right)$, then $d_{s^*\rightarrow}\left[s,s'\right] > 1$, resulting in a more-than-proportional horizontal complementarity. The same logic applies to eq. (3).

s_1 to be $\alpha(s_1, s) = 0.2$ and $\alpha(s_1, s_3) = 0.1$, while that of sector-s_2 are $\alpha(s_2, s) = 0.1$ and $\alpha(s_2, s_3) = 0.2$. Accordingly, the metrics expressing their factor input demand distance relation is composed of $\boldsymbol{d}^{fd}[s_1, s_2] = (.2, .1), (.1, .2)$ when buying from sector-s and sector-s_3, respectively. Demand-based distance value is positive even in the absence of direct trade linkage between considered sectors in the pair.

Consider now the case in which two sectors, say $\{s, s'\}$, are selling their own commodity as an intermediate input to the same sector, say s^*. In this scenario, by substituting out the combined optimality conditions in eq. (1) for such sectors trading to the same downstream buyer, total *log*-differentiation leads to

$$d\log \mathcal{Q}_{\to s^*}[s, s'] = \frac{\alpha(s^*, s)}{\alpha(s^*, s')}\, d\log \mathcal{P}_{\to s^*}[s', s] \tag{3}$$

Given $\gamma^{\mathcal{Q}}_{(\cdot)} = \frac{1}{\log \overline{x}(s^*,\cdot)}$, $\delta^{\mathcal{Q}}_{(\cdot)} = \frac{\log \overline{x}(s^*,\neq\cdot)}{\log \overline{x}^2(s^*,\cdot)}$, $\gamma^{\mathcal{P}}_{(\cdot)} = \frac{1}{\log \overline{p}(\cdot)}$ and $\delta^{\mathcal{P}}_{(\cdot)} = \frac{\log \overline{p}(\neq\cdot)}{\log \overline{p}^2(\cdot)}$ being steady-state components for intermediate inputs and their associated price levels, then differential elements $d\log \mathcal{Q}_{\to s^*}[s, s'] = \gamma^{\mathcal{Q}}_{s'}\, d\log x(s^*, s) - \delta^{\mathcal{Q}}_{s'}\, d\log x(s^*, s')$ and $d\log \mathcal{P}_{\to s^*}[s', s] = \gamma^{\mathcal{P}}_{s}\, d\log p(s') - \delta^{\mathcal{P}}_{s}\, d\log p(s)$ identify the *log*-difference of intermediate inputs and price levels between both sectors selling to the same downstream sector. Subscript "$\to s^*$" reads as "selling to sector-s^*" by sectors in brackets.

Following the previous logic scheme, then $\boldsymbol{\mathcal{R}} = \boldsymbol{\mathcal{D}}^{fs}$. In other words, the interdependence of marginal revenues for these sectors is determined by their distance in the network. When selling to the same downstream sector, the quantity sold of sector-s, given that of any sector-s', is determined by changes in their relative price and scaled by their relative trade intensity.

EXAMPLE 3 (Factor input supply metrics) *Consider the pair of sectors $\{s_1, s_2\}$ in an Input-Output matrix mirroring Example 1, where the common sectors to which both are selling are $\{s, s_3\}$, and assume fictional trade intensities for sector-s_1 to be $\alpha(s, s_1) = 0.4$ and $\alpha(s_3, s_1) = 0.3$, while that of sector-s_2 are $\alpha(s, s_2) = 0.6$ and $\alpha(s_3, s_2) = 0.6$. Accordingly, the metrics expressing their factor input demand distance relations is composed of $\boldsymbol{d}^{fs}[s_1, s_2] = (.4, .3), (.6, .6)$ when selling to sector-s and sector-s_3, respectively. Supply-based distance value is positive even in the absence of direct trade linkage between considered sectors in the pair.*

An important aspect to consider when introducing economic distances in the production network is that, under the same configuration of the inter-sectoral trade structure, sectors can be in a different "distance relation" depending on whether they are buying from or selling to other sectors. Moreover, all sectors are simultaneously connected by network distances due to common demand and supply.

THEOREM 2 (On the configuration of network "economic" distances) *Let the set of sectors being Φ, and $\alpha^j(\cdot)$, for $j = \{fd, fs\}$, the normalized Input-Output fixed coefficients. Then, demand- and supply-based distance matrices satisfy: (i) for any $s, s' \in \Phi$ there exists a common neighbour-s^*, so that both $\boldsymbol{\mathcal{D}}^{fd}$ and $\boldsymbol{\mathcal{D}}^{fs}$ are non-negative and full; (ii) a given Input-Output structure uniquely determines $\boldsymbol{\mathcal{D}}^{fd}$ and*

$\mathcal{D}^{fs}$, which differ entrywise unless the corresponding bilateral trade with the common neighbour coincides; and (iii) if $\alpha^j(\cdot)$'s are non-symmetric, then all entries of $\mathcal{D}^{fd}$ and $\mathcal{D}^{fs}$ differ; conversely, if $m \geq 4$, then both share at least one identical entry whenever at least $m-2$ sectors trade symmetrically with a common sector and at least one additional symmetric trade relation elsewhere in the network exists.

Proof in Appendix A.1.

The first two clearly emerge in Examples 2-3 and Figure 1. The final property clarifies when the distance between two sectors is uniquely determined: if two sectors buy and sell the same relative quantity of intermediate inputs with a common third sector, then their factor input demand and factor input supply distances coincide. Their actual quantities may differ; what matters is that, for each sector individually, its purchases or sales with the common sector match in relative terms.[4]

COROLLARY 1 (Coinciding distances) *For $m \geq 4$, if a common sector trades symmetrically with $m' = m-2$ others, then $\frac{m'(m'-1)}{2}$ sectors exhibit symmetric distances.*

3. AN INSPECTION INTO THE HORIZONTAL TRANSMISSION

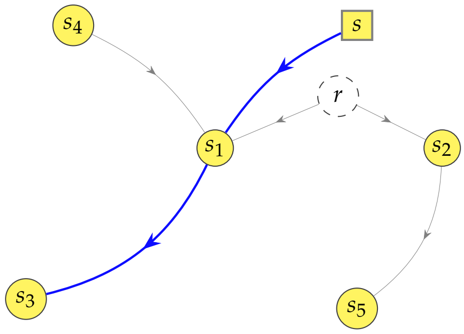
Outlined horizontal geometry introduces novel implications for the propagation of asymmetric, independent and idiosyncratic shocks across the network. Crucially, the transmission depends not only on the size of existing connections among sectors, but also on their network "economic" distance. As shown in Figure 3, a shock originating in a given sector propagates vertically through downstream buyers along the supply chain, while concurrently transmitting horizontally. Notably, this horizontal transmission materializes even between sectoral pairs that are not directly connected, reflecting their shared demand and/or supply relationships.[5]

EXAMPLE 4 (Propagating behaviour under horizontal distances) *Consider an economy populated by $\{s, s_1, s_2, s_3, s_4, s_5, r\} \in \Phi$ sectors, with r representing the rest of the Input-Output network. Some trade with others, while some others do not: s_1 is buying from $\{s, s_4, r\}$ while selling to s_3; s_2 trades from r and to s_5. Suppose a supply shock to sector-s originates. In Panel 3a, the shock propagates downstream: it directly affects s_1 and indirectly reaches s_3, yielding a purely vertical transmission with others unaffected.*[6] *Unknown remain the potential effects of induced adjustments in s_1, even*

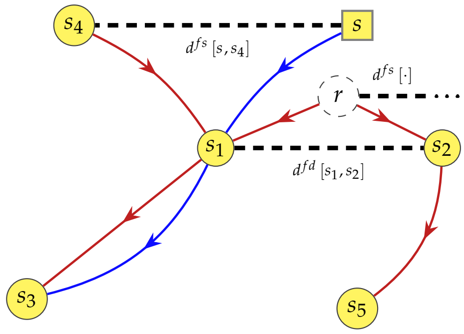
[4] A "normalized" network refers to an Input-Output matrix once rows or columns are scaled by their sum. Entries thus express relative and not absolute values, required to build $\mathcal{D}^{fd}$ and $\mathcal{D}^{fs}$.

[5] The example is presented as the more intuitive case of a sectoral supply shock propagating downstream, affecting sectors that depend on the initially shocked sector. Yet, the same underlying logic applies in the case of a demand shock, where the direction of a transmission is reversed, with the sectoral shock propagating upstream as sectors increase demand for intermediate inputs from their suppliers. In the presence of an already-existing Input-Output linkage between distanced sectors, the transmission is progressively amplified.

[6] Upstream responses vanish because price and quantity effects offset each other (*e.g.*, Acemoglu, Carvalho, et al. 2012). Yet, the horizontal mechanism uncovers these latent transmission channels via common network structures instead of tracking non-linearities in production (*e.g.*, Baqaee and Farhi 2019).



(A) VERTICALITY



(B) VERTICALITY AND DISTANCES

FIGURE 3: HORIZONTALITY OF A TRANSMISSION

Note: stylized example of how a sectoral shock propagates across the production network. Coloured and oriented bold-curves trace the propagation mechanism: vertical transmission is in blue, while additional transmission paths from the horizontal geometry are in red. Oriented thin-grey curves indicate further existing Input-Output (I-O) connections, and network economic distances through common nodes are represented by dashed lines. The underlying network structure is identical in both panels. Panel 3a shows a purely vertical propagation of a shock to sector-s, following standard sectoral supply-chain logics. Panel 3b depicts a richer mechanism, where the shock also spreads horizontally through sectors sharing common upstream suppliers or downstream buyers.

though it shares linkages with others. Panel 3b illustrates both vertical and horizontal propagation: a shock to s not only affects its immediate trading connections but also alters its interaction with s_4: changes in s_1 triggered by the initial shock generate an horizontal effect that links $\{s, s_4\}$ through their common supply-based connection with s_1. By analogy, this reasoning falls under supply distances of $\{s, s_4\}$ with the rest of other sectors-r selling to s_1, $d^{fs}[\cdot]$. With due demand-based scheme, analogous is the logic underlying $d^{fd}[s_1, s_2]$, simultaneously buying from r. Subsequent adjustments in $\{s_4, r\}$ feed back into s_1, compounding the indirect propagation towards s_3, and into s_2: reverberating the initial shock on s, via $\{s_1, r\}$ to s_2, an additional vertical effect further downstream, from s_2 to s_5, is generated.

Example 4 illustrates how incorporating factor input demand and factor input supply network economic distances, and therefore horizontal complementarities, fundamentally reshapes the vertical transmission of micro-level shocks. The new propagation stems from similar demand/supply trade relationships between sectors sharing a comparable Input-Output structure (*i.e.*, common upstream suppliers or downstream buyers), and this paper is mostly on the conceptualisation of the $d[\cdot]$-types of linkages. Stated differently, distance-based complementarities that emerge along the network's horizontal architecture alter the vertical transmission by either compounding or mitigating the propagating behaviours of a micro-originated shock.

3.1. REDUCED FORMS FOR DISTANCE-LEAD COMOVEMENT

A shadowed implication of the framework developed so far is that intermediate inputs generally function as substitutes, allowing sectors to freely adjust their input mix in response to shocks. However, the horizontal geometry of a network can embed two aspects: *(i)* vertical propagation through the supply chain may remain the dominant channel, and what may appear as a horizontal transmission can in fact be the by-product of vertical propagation; and *(ii)* intermediate inputs complementarity in production may alter the transmission of shocks. This tension between horizontal complementarities at the network level and substitutability at the input level sets the stage for a more precise characterization of how sectoral comovement occurs.

In the context of factor input supply, alleviating the tension is doable since competition for common downstream buyers yields a more direct and unambiguous form of horizontal interdependence. Conversely, the structure of upstream connections renders demand-based linkages ambiguous to interpret, since the coexistence of a dual transmission makes the direction and the strength of induced comovement subtle.

Consider a single-input (purchasing from only one upstream supplier, with no substitution possibilities), and a multiple-input (sourcing from multiple suppliers) cases, and a Constant Elasticity of Substitution (CES) among intermediate inputs.

PROPOSITION 1 (Demand-driven comovement) *In the single-input case, comovement in demand implies $\delta_{s^*\rightarrow}[s,s'] \propto -\sigma\,\psi^{fd}$. In the multi-input CES case, then $\delta_{s^*\rightarrow}[s,s'] \propto -\,\sigma\,[1-e\,(s,s^*)\,]\,\psi^{fd}$, with relative intermediate inputs changes from common supplier, $\delta_{s^*\rightarrow}[s,s'] = \frac{\partial \log x(s,s^*)}{\partial \log x(s',s^*)}$, jointly determined by substitution elasticity σ, downstream sectoral expenditure share $e\,(\cdot)$, and upstream pass-through ψ^{fd}. Opposite comovement occurs if $\psi^{fd} > 0$.*

Proof in Appendix A.2.

Opposite comovement between downstream sectors arises through upstream adjustments in supply. Intuitively, when s' experiences a positive shock it might bids up the price of inputs from the common supplier-s^*, $\psi^{fd} = \frac{\partial \log p(s^*)}{\partial \log x(s',s^*)} > 0$, through increased demand. If this higher input cost makes production more expensive for s, its demand from s^* falls, generating opposite comovement. Whether this happens hinges on the complementarity between s^*'s input and the others purchased by s: when $\sigma > 0$, any price increase in s^* alters s's input mix; in the limit $\sigma \rightarrow \infty$, inputs become perfect substitutes and s fully shifts away from s^*, generating no comovement with s'. In the general multi-input CES case, the effect is further scaled by sector-s's expenditure share on s^*, $e\,(s,s^*)$, so that sectors more reliant on s^* are strongly affected. Thus, demand-driven comovement reflects not automatic crowding-out, but the interplay of upstream price pass-through (from vertical propagation), and downstream demand adjustments (indirectly generated by horizontal complementarities).

Considering instead supply-based distances, the reallocation of downstream demand across competing upstream suppliers is a clear channel for comovement.

PROPOSITION 2 (Supply-driven comovement) *In the single-input case no condition exists for comovement in supply due to the absence of competition for downstream markets. In the multi-input CES case it holds $\delta_{\rightarrow s^*}[s,s'] \propto \sigma\, e\,(s^*,s')\;\psi^{fs}$, with relative intermediate inputs changes to common buyer, $\delta_{\rightarrow s^*}[s,s'] = \frac{\partial \log x(s^*,s)}{\partial \log x(s^*,s')}$, jointly determined by substitution elasticity σ, sectoral revenues for upstream supplier $e\,(\cdot)$, and downstream pass-through ψ^{fs}. Opposite comovement occurs if $\psi^{fs} < 0$.*

Proof in Appendix A.2.

Opposite comovement between upstream suppliers mechanically arises from competition for downstream buyers. Intuitively, when s' receives a positive shock it lowers its relative price, capturing a larger share of common buyer-s^*'s demand from adjustments in upstream price, $\psi^{fs} = \frac{\partial \log p(s')}{\partial \log x(s^*, s')} < 0$. This revenue reallocation necessarily comes at the expense of the competing supplier-s, generating opposite comovement even when $\sigma \to \infty$ (shifting away of s^*'s demand from s, increased towards s' due to relative price effect). The magnitude depends on two forces: the elasticity of substitution σ across sector-s^*'s inputs controls how easily s^* reallocates spending, while the expenditure share $e(s^*, s')$ scales the importance of s' for sector-s^*.

Discussion.– A networked economy inherently embeds complementarities in intermediate inputs – scaling the magnitude but not the direction of comovement –, as sectors depend on overlapping sets of buyers and suppliers. In the demand-distance case, comovement is ambiguous: it reflects a combination of vertical propagation along the supply chain and horizontal complementarities from shared suppliers. In the supply-distance case, comovement is transparent: the horizontal transmission directly arises from the reallocation of inputs to adjustments in relative prices, and the interaction between the two dimensions is mechanically determined.[7]

Importantly, the discrepancy between the ambiguous effects in demand and the clear responses in supply is in accordance with non-linear network theories (*e.g.*, Atalay 2017; Baqaee and Farhi 2019), where downstream complementarities – and hence complementarities between purchased intermediate inputs – matter the most. This dimension is embodied into the supply-based distances through the downstream pass-through effect, which create horizontal interdependencies among sectors selling to the same buyers: when one supplier experiences a positive shock, downstream complementarities of common buyers cause other suppliers to adjust their production in response, and the propagation operates directly through the interdependence of upstream inputs used together in downstream production. Such amplification is largely absent under demand-based distances, since the upstream pass-through reflects only substitution patterns across common upstream suppliers.

3.2. Vertical and horizontal non-dependence

A key step in the analysis is to establish a dimensional separation.

Theorem 3 (On the weighting of network "economic" distances) *In an Input-Output economy, the horizontal dimension of a network (defined by demand- and supply-based distance matrices) is not weighted by its vertical dimension (defined by the directed or Leontief inverse matrix).*

Proof in Appendix A.2.

The theorem postulates that, within an Input-Output economy, the horizontal geometry of the network – captured by demand- and supply-driven distances – remains

[7] Baqaee and Farhi (2019) develop a framework that emphasizes non-linear amplification; mine summarizes similar complementarity forces that drive non-linear dynamics by the means of horizontal distances.

unweighted and independent of its trade intensities, encoded in the Input-Output matrix. This distinction is crucial for understanding the economics of production networks: while the vertical structure reflects the strength of sectoral relationships, the horizontal structure identifies demand/supply complementarities across sectors related through common upstream suppliers or downstream buyers. In other words, by separating the horizontal from the vertical geometry, the theorem clarifies that whatever distance matrix is not merely a transformation of the Input-Output matrix, but a complementary object that reveals otherwise hidden propagation channels.

Nevertheless, as linkages densify and the network expands, horizontal complementarities become increasingly important (with more sectors sharing common suppliers and buyers, interdependencies proliferate).

THEOREM 4 (On the importance of network "economic" distances) *Consider a network of size-m with Input-Output matrix $\boldsymbol{\mathcal{A}}_m := \boldsymbol{\mathcal{V}}_m \cup \boldsymbol{\mathcal{D}}_m$, where $\boldsymbol{\mathcal{V}}_m$ encodes vertical buyer-supplier ties and $\boldsymbol{\mathcal{D}}_m$ encoding horizontal complementarities. Let $\boldsymbol{\varepsilon}_m$ be a vector of independent shocks, and let $\boldsymbol{\mathcal{P}}_m^{(\mathcal{V})}$ and $\boldsymbol{\mathcal{P}}_m^{(\mathcal{D})}$ denote the associated matrices for propagation channels. Then, as the network size grows, the vertical propagation vanishes in spectral norm: $\left\| \boldsymbol{\mathcal{P}}_m^{(\mathcal{V})} \boldsymbol{\varepsilon}_m \right\|_2 \Big/ \left\| \boldsymbol{\mathcal{P}}_m^{(\mathcal{D})} \boldsymbol{\varepsilon}_m \right\|_2 \xrightarrow[m\to\infty]{} 0.$*

Proof in Appendix A.2.

In other words, for larger production networks (*e.g.*, firm-to-firm network), vertical buyer-supplier chains become too sparse and elongated to transmit shocks effectively, while horizontal complementarities grow denser and increasingly bind sectors together. As more network participants buy and sell, shared buyers/suppliers become more likely, and the more important horizontal complementarities become. In the limit where network size-$m \to \infty$, horizontal effects dominate: bigger networks transmit shocks not as cascading supply chains, but as clusters of horizontally-related sectors. This dominance persists even when vertical supply chains themselves expand endogenously as the network grows in size.

COROLLARY 2 (Endogenous expansion) *In growing networks, the horizontal geometry asymptotically dominates shock propagation even as vertical chains expand.*

3.3. SECTORAL COMOVEMENT AND AGGREGATE BEHAVIOUR

Equilibrium adjustments in production networks are central to understand the origins of aggregate business cycles. Within this perspective, an important question emerges: to what extent does employment comovement shape the dynamics of aggregate fluctuations in output? From the model's equilibrium conditions, the impact of a certain sector-specific shock on real Gross Domestic Product (GDP) can be decomposed into its propagation on aggregate Total Factor Productivity (TFP) and its effect on aggregate employment – exploiting the Divisia Index (*e.g.*, vom Lehn and Winberry 2022). Fluctuations in aggregate real GDP growth are the result of

$$d\log Y = \sum_{s\in\Phi} \Big(\lambda\,(s)\; d\log z\,(s) + \nu\,(s)\; d\log n\,(s) \Big) \tag{4}$$

where $\lambda(s) = \frac{p(s)\,y(s)}{PY}$ is the *Domar weight* of sector-s, and $\nu(s) = \frac{p^Y(s)\,y^Y(s)}{PY}$ its *value-added Domar weight*, with $p^Y(s)\,y^Y(s)$ representing the nominal value-added. These terms measure, respectively, the sectoral ratios of the gross output and value-added to GDP. As such, both capture each sector's relative importance in the economy and its weight in aggregate fluctuations and business cycle dynamics.

Employment comovement thus constitutes a central phenomenon to investigate under inter-sectoral trade network: the cyclical behaviour of the economy origins from both changes in sectoral TFP, $d\log z(s)$, and in sectoral employment, $d\log n(s)$. Focusing on employment comovement offers distinct advantages. It provides a directly observable measure of sectoral activity, avoiding the interpretability (whether there is shock transmission or shocks correlation; Huo et al. 2025) and the identification challenges associated with productivity.As such, through the lens of eq. (4), different values of network distances govern the sign of the shock's transmission:

$$d\log Y = \sum_{s\in\Phi} \lambda(s)\; d\log z(s) + \sum_{s\in\Phi} \begin{cases} \nu(s)\;\Delta\,\text{comovement} & \textit{under major distance} \\ \nu(s)\;\Delta\,\text{reallocation} & \textit{under minor distance} \end{cases}$$

In fact, the model outlined in Section 2 delivers the following insight into employment comovement.

PROPOSITION 3 (Propagation of sectoral variations) *The overall production network effects on sectoral employment are summarized by:*

$$d\,log\,\boldsymbol{n} = \Theta\Big\{d\,log\,C + d\,log\,\boldsymbol{\mathcal{S}} + d\,log\,\boldsymbol{\mathcal{V}} + d\,log\,\boldsymbol{\mathcal{D}}(j)\Big\}, \quad for \quad j = \{fd, fs\}$$

where $\boldsymbol{\mathcal{V}}$ and $\boldsymbol{\mathcal{D}}(j)$ represent the vertical and horizontal effects, respectively, with $\boldsymbol{\mathcal{D}}(fd) = \boldsymbol{\mathcal{D}}^{fd}\Big[d\,log\,\boldsymbol{n} - (\boldsymbol{\mathcal{I}}'_\Phi\, d\,log\,\boldsymbol{n})\,\mathbf{1}\Big]$ and $\boldsymbol{\mathcal{D}}(fs) = \boldsymbol{\mathcal{D}}^{fs}\Big[d\,log\,\boldsymbol{p} - (\boldsymbol{\mathcal{I}}'_\Phi\, d\,log\,\boldsymbol{p})\,\mathbf{1}\Big]$.

Proof in Appendix A.5.

Mitigated by a bundle of structural parameters (Θ), changes in sectoral employment are a function of aggregate consumption (C), both sector-specific productivity and consumption of its good ($\boldsymbol{\mathcal{S}}$), other sectors' productivities, employment levels, and intermediate inputs usage ($\boldsymbol{\mathcal{V}}$) whose magnitudes depend on the Input-Output matrix, and network "economic" distance effects ($\boldsymbol{\mathcal{D}}$) for both demand (through employment differentials) or supply (through relative price differentials) horizontal linkages. Square parentheses denote the difference between the change in all other sectors and the sector's own change. For example, considering sector-s employment, $d\log\boldsymbol{n} - (\boldsymbol{\mathcal{I}}'_s\, d\log n(s))\mathbf{1} = \big[d\log n(s) - d\log n(s), \ldots, d\log n(m) - d\log n(s)\big]$ where $\boldsymbol{\mathcal{I}}_s$ is an $m\times 1$ indexing vector that selects sector-s and $\mathbf{1}$ an $m\times 1$ vector of ones, both replicating the reference sector's change across all sectors, ensuring that the horizontal propagation term captures relative deviations. For all sectors:

$$\boldsymbol{\mathcal{D}}^{fd}\Big[d\log\boldsymbol{n} - (\boldsymbol{\mathcal{I}}'_\Phi\, d\log\boldsymbol{n})\,\mathbf{1}\Big] = \begin{bmatrix} \sum_{s'\in\Phi} d^{fd}[s,s']\,\big[d\log n(s') - d\log n(s)\big] \\ \vdots \\ \sum_{s\in\Phi} d^{fd}[m,s]\,\big[d\log n(s) - d\log n(m)\big] \end{bmatrix}$$

The interpretation of Proposition 3 follows Subsection 3.1: sectors sharing upstream suppliers exhibit ambiguous comovement, as employment changes ($d \log \boldsymbol{n}$) enter in both vertical and horizontal components; sectors sharing downstream buyers undergo relative-price reallocation of supply, producing opposite comovement. Moreover, vertical propagation prevails under low proximity (high distance, low theoretical $\mathcal{D}^j$), leading to positive sectoral comovement. For low distances, horizontal forces dominate, thereby inducing opposite comovement.

Finally, it is worth assessing the extent to which the mere horizontal geometry of networked economies contributes to aggregate fluctuations.

THEOREM 5 (On aggregate cycles) *Consider an Input-Output economy made of $\{s, \ldots, m\} \in \Phi$ sectors. Let $\lambda(s)$ denote sector-s's Domar weight, and assume no sector is dominant, in the sense that $\lambda(s) \asymp \frac{1}{m}$, $\forall s \in \Phi$. Beside vertical propagation, if sectors share upstream suppliers or downstream buyers, then idiosyncratic shocks generate non-vanishing aggregate GDP fluctuations:* $\liminf_{m\to\infty} \; var(Y_m) > 0$.

Proof in Appendix A.2.

Aggregate cycles need not originate from economically dominant sectors nor from vertical production chains: as many horizontally-related sectors move in response, shocks do not average out, and aggregate volatility survives even as the economy grows. Even when sectors are small, idiosyncratic shocks generate persistent aggregate volatility under horizontal linkages: fluctuations from sectoral responses are induced by shared Input-Output structures rather than through usual economic centrality or vertical propagation (more in Subsection 4.1). Horizontal interdependences thus prevent shocks from averaging out in aggregate.

COROLLARY 3 (Granular sectoral weights) *Economically large sectors amplify aggregate fluctuations induced by horizontal network propagation.*

Allowing for heterogeneous Domar weights – so that some sectors represent a larger fraction of aggregate GDP – does not alter the horizontal transmission mechanism identified in Theorem 5. Instead, they strengthen its aggregate impact: when a highly horizontally connected sector is also economically large, its shock propagates across the horizontal cluster and carries greater weight in aggregate output.

Horizontality also shapes the effectiveness of the aggregate diversification rate.

COROLLARY 4 (Rate of decay) *Let Ψ be the set of horizontally connected sectors whose cardinality is $|\Psi|$. Horizontal geometry alone breaks the standard diversification argument: $var(Y_m) = \Omega\left(|\Psi|/m\right)$. Consequently, aggregate volatility decays at slower rate than $\mathcal{O}(1/m)$, the standard diversification rate.*

The declining speed of aggregate GDP volatility reflects the strength of horizontal complementarities. When small, sectors behave almost independently and horizontal propagation diversifies, with negligible aggregate outcomes (*e.g.*, Lucas 1977). As horizontal cluster expands, shocks affect more sectors simultaneously: diversification weakens as aggregate volatility is bounded below $\Omega(\cdot)$, but exceeding the standard diversification rate $\mathcal{O}(\cdot)$. When the network size scales horizontal similarities, sectoral responses are broadly synchronized and aggregate volatility persists.

Vertical and horizontal transmissions thus reflect distinct but complementary mechanisms. Vertical propagation operates through the Input-Output matrix: shocks cascade along supply chains via observed linkages, generating amplifications that scale up mechanically with sectoral importance. Granularity therefore magnifies aggregate responses by strengthening vertical cascades. Horizontal propagation, instead, acts on comovement: sectors sharing many suppliers or buyers respond jointly, without inducing cascades, but generating systematic correlation that survives aggregation. Granularity amplifies these spillovers only when large sectors belong to dense horizontal clusters. As in the vertical framework (*e.g.*, Acemoglu, Carvalho, et al. 2012), aggregate fluctuations decay more slowly under network interconnections: the law of large numbers implied by the standard diversification argument is weakened due to novel horizontal effects, and aggregate volatility is more persistent.

4. Data and Horizontal Anatomy

From the United States' Bureau of Economic Analysis (BEA), available data display information, for 3-digit U.S. 2017 North American Industrial Classification System (NAICS) sectors, on value-added, employment, net exports, and inter-sectoral Input-Output (I-O) matrices. The (balanced) panel data built is made of 65 private sectors over the period 1998-2022. For additional details refer to Appendix B.

Distance measures are derived from the promptly-downloadable *directed* network and not from its *Leontief inverse*. While sparsity is preserved under both representations, the directed matrix ensures distances to stem from immediate (direct and indirect) Input-Output intensities, including only pure correlating behaviours and preserving the strictly observed interdependencies, without introducing any Leontief inverse amplification. Each cell $d^j[s,s'] \in [0,1]$ of matrix $\boldsymbol{\mathcal{D}}^j$, for $j = \{fd, fs\}$ and $\forall s,s' \in \Phi$, identifies the Euclidean distance (square root of the sum for squared differences of network intensities; see Section 2) between dyads of sectors in demand or supply relative to common sectors.

The empirical estimation of Section 5 will consider network distances at their *extensive margin* for classification purposes. Such matrices are labelled as $\boldsymbol{\mathcal{D}}^j_{ext}$, whose generic entry is $d^j_{ext}[s,s'] = \{1,2,\ldots,d_{max}\}$. Differently from the theory, values closer to one correspond to shorter horizontal distances. As for their characterization, network distance-d among any two sectors $\{s,s'\}$ is computed as a *shortest path problem* from *intensive margin* distance matrices, $\boldsymbol{\mathcal{D}}^j$, implementing an algebraic version (*i.e.*, using matrix multiplication) of the traditional Dijkstra (1959)'s or Roy (1959)-Floyd (1962)-Warshall (1962)'s types of algorithms, since it is *(i)* suitable for discrete-value adjacency matrices – as the ones for intensive margin distances –, and *(ii)* fairly efficient for small to medium-sized graphs – as a sector-level production network.

Appendix B reports the technical version of the implemented algorithm which, in plain words, reads as follows.

(Shortest path algorithm) *Central idea is to find, for each single node, all the other nodes that can be reached in one step, then in two steps, and so on. Each time a node is reached for the first time, it records the number of steps it took to get there – this is the shortest path approach. In other words, find the shortest number of steps it takes to get from every node to every other node in a network. In particular:*

(a) identify the direct (one-step) connections between any pair of nodes;

(b) for not-connected pairs, increase the path length by 1 in each round (two steps, then three steps, etc.). Multiply the graph by itself to discover which new pairs of nodes are now connected through longer paths;

(c) if two nodes becomes connected, record the length as the shortest distance;

(d) repeat this process until any new reachable pairs of nodes is found. For any pair of nodes that are still uncategorised, set their distance to infinity.

4.1. Aggregate dynamics

To illustrate the implications of horizontal geometry for business cycles, I simulate aggregate GDP responses to sectoral shocks exploiting the network's vertical (Input-Output chains) and horizontal (shared buyers/suppliers) structures, relying on observed Input-Output and intensive margin distance matrices from the 2007 BEA tables. Both network dimensions remain unweighted, as dictated by Theorem 3.

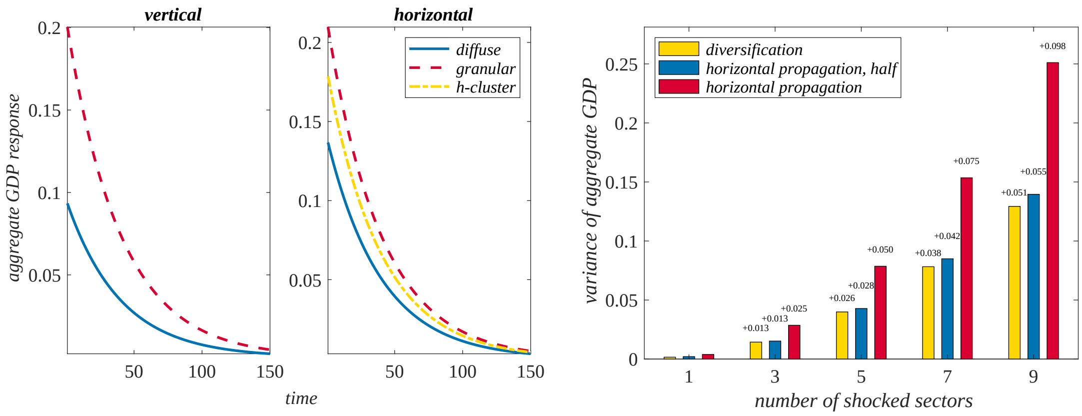
Sectoral outputs evolve as $\boldsymbol{y}_t = \boldsymbol{\mathcal{P}}^{(\mathcal{V},\mathcal{D})}\,\boldsymbol{\varepsilon}_t$, where matrix $\boldsymbol{\mathcal{P}}^{(\cdot)}$ encodes how sector-specific shocks in $\boldsymbol{\varepsilon}$ propagate, following a mean-reverting AR(1) process with persistence ρ. Aggregate Impulse Response Functions (IRFs) trace the path $\{Y_t\}_{t=0}^{T}$ following an initial shock in $t = 0$, and evolve over time according to

$$Y_t = \boldsymbol{\lambda}'\,\boldsymbol{y}_t \quad \longleftrightarrow \quad Y_t = \boldsymbol{\lambda}'\,\boldsymbol{\mathcal{P}}^{(\mathcal{V},\mathcal{D})}\left(\rho_t\,\boldsymbol{\varepsilon}_0\right)$$

where $\boldsymbol{\lambda}$ bundles Domar weights (*i.e.*, relative sectoral contribution to GDP). The resulting series for Y_t provide dynamic responses that summarize either cascading or distance-lead effects across the observed network structure, shown in Figure 4.

Panel 4a directly mirrors the logic of Theorem 5 and its Corollary 3. Vertical propagation, governed by the Leontief inverse, amplifies shocks primarily through parallel supplier-customer chains.[8] Horizontal propagation, instead, does not rely on Input-Output flows but depends on the geometry of production similarities: idiosyncratic shocks diffuse strongly across sectors with common network linkages, even in the absence of direct trade, and aggregate effects depend on the distribution of economic proximity. In both cases, granularity mechanically increases aggregate volatility because economically large sectors transmit shocks proportionally more

[8] Sectoral shocks average out in aggregate for "balanced" – *i.e.*, almost symmetric – Input-Output networks (*e.g.*, Acemoglu, Carvalho, et al. 2012), and not necessarily for equal Domar weights.



(A) RESPONSE

(B) VOLATILITY

FIGURE 4: AGGREGATE FLUCTUATIONS

Note: under "diffuse", sectoral Domar weights are evenly distributed; if concentrated, the economy is vertically "granular"; a not vertically dominant yet strong horizontally connected sector lies under "h-cluster". Numbers above bars indicate the incremental change in aggregate output variance relative to the previous one. *Source*: BEA and own calculations.

strongly. Horizontal spillovers are mitigated when large Domar weights are associated with sectors embedded in large horizontal clusters: a sector that is not central in the standard Input-Output network but is strongly connected to many others through shared linkages (the "h-cluster") generates milder aggregate effects than a scenario in which equally large Domar weights are concentrated in vertically central sectors (the "granular" setup). Indeed, this transmission mechanism is not driven by upstream/downstream similarity alone, but by the alignment between sectoral importance and the network's horizontal geometry.

Additionally, from the y-axis, horizontal propagation generates slightly larger aggregate responses than vertical connections. This outcome is not a numerical artifact, but a direct consequence of the economic mechanisms embedded in: reminiscent from Figure 3, vertical transmission amplifies shocks through recursive cascades but produces "localized" effects along specific supply chains, whereas horizontal propagation increases systemic covariance, creating widespread comovement throughout the production network without confined feedback loops or cascading amplification.

Panel 4b illustrates how horizontal propagation modifies the conventional diversification argument. When shocks are transmitted through horizontal spillovers, aggregate volatility increases more sharply than in the diversification scenario. Even a partial (50%) horizontal spread amplifies fluctuations beyond such benchmark with no network effects, while full horizontal propagation generates the largest variance, resulting in the slowest rate of decay. Corollary 4 finds support: horizontal complementarities prevent idiosyncratic shocks from averaging out, so aggregate fluctuations persist and intensify with the strength of network interconnectedness.

An alternative test is to decompose the contribution of the network distance structure to aggregate GDP variance. I compute the spectral norm of the horizontal propagation matrix, which measures the maximal amplification of idiosyncratic shocks through the network; larger values imply a slower rate of decay. The results indicate that horizontal complementarities generate aggregate dynamics beyond the

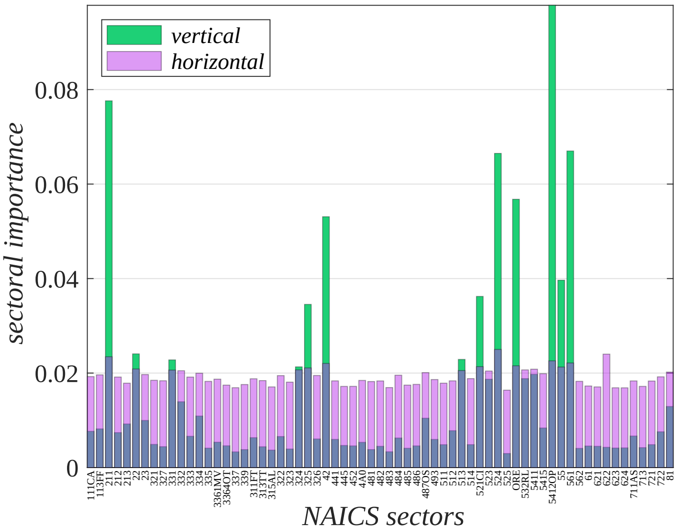


FIGURE 5: SECTORAL INFLUENCE IN SHOCK TRANSMISSION

Note: importance of each sector in reverberating idiosyncratic shocks through their vertical and horizontal network structures as dictated by $\mathcal{P}^{(\mathcal{V},\mathcal{D})}$ U.S. Input-Output propagation matrices in year 2007. *Source*: BEA and own calculations.

benchmark predicted by the standard diversification argument. Specifically, the fact that $\left\|\mathcal{P}^{(\mathcal{D})}\right\|_2 = 0.156 > 0.124 = \left\|\mathcal{P}^{(Lucas,1977)}\right\|_2$ illustrates how structural interdependence among sectors sharing suppliers or buyers prevents idiosyncratic shocks from averaging out, in contrast to the classical $1\big/\sqrt{m}$ convergence predicted by Lucas (1977). This comparatively slow decay implies that aggregate volatility is less likely to be smoothed out in economies with strong horizontal complementarities, highlighting systemic vulnerabilities, and supporting the claim of Corollary 4.

4.2. PROPAGATING PATTERNS

I turn to analyse the mechanics of the horizontal transmission. Exploiting the observed propagation matrices, Figure 5 visualizes the sectoral propagation patterns using centrality measures from spectral norms, to be read in terms of how important each sector is for the transmission of shocks considering the dual network structures. The vertical transmission highlights few key sectors that act as propagation hubs:[9] asymmetric shocks originating in these sectors are recursively amplified along parallel supply chains, generating strong cascading effects, while most other sectors transmit shocks only weakly. In contrast, the horizontal propagation system spreads shocks uniformly across sectors: each sector contributes to co-movement with others, but without recursive amplification, producing widespread responses. Overall, the image demonstrates the distinction between cascade-driven amplification along key nodes and diffuse propagation across economically similar sectors: vertical chains concentrate and amplify shocks along central nodes, whereas horizontal complementarities induce diffuse comovement without feedback loops.

[9] Systemic role in providing inputs – *e.g.*, energy (211), risk intermediation (524), knowledge-intensive services (5412OP), and organizational support (561) – that underpin a wide range of production activities.

TABLE 1: DIMENSIONAL DOMINANCE

	network sparsity				
	15% $\tau = 0.5\%$	30% $\tau = 1\%$	40% $\tau = 1.5\%$	50% $\tau = 2\%$	95% $\tau = 5\%$
vertical propagation	.2133	.1995	.1803	.1497	.0762
horizontal propagation	.1371	.1559	.1718	.1731	.1746

Spectral norms for a 3-digit U.S. Input-Output network in year 2007. Due to tresholding-τ, transactions are removed and sparsity is approximated. Source: BEA and own calculations.

Table 1 then investigates how vertical and horizontal geometries respond to varying levels of network sparsity. By progressively removing weaker transactions (increasing the threshold-τ for the volume of total sectoral shares), the network becomes sparser and vertical propagation declines, reflecting the diminishing role of cascade effects. In contrast, horizontal propagation remains remarkably stable across thresholds, and it gradually becomes the predominant channel of shock transmission as τ increases, so that horizontal complementarities dominate in sparser networks. These results illustrate that, even when vertical linkages (*i.e.*, observed trade intensities) are trimmed, horizontal interdependencies sustain substantial propagation, empirically clarifying Theorem 4: as Input-Output economies become sparser, shocks increasingly travel along clusters of horizontally related sectors rather than along elongated vertical chains, confirming the principle of horizontal dominance.

4.3. NETWORK EXPOSURE AND AGGREGATE SHOCKS

The discussion so far has shown that horizontal geometries capture dimensions of economic exposures that are unconditionally distinct from vertical Input-Output linkages. If these capture economically meaningful dimensions of interdependence, they should also explain differences in sectoral responses to aggregate shocks "above and beyond" the information already embedded in Leontief inverse measures, retaining a predictive power that conditionally complements them.

A Bartik (1991) shift-share design comes to the rescue. Consider an aggregate shock that affects all sectors simultaneously. Since the shock is widespread, identification cannot come from time-series variation alone. Instead, the relevant source of variation is cross-sectional: sectors may be differently exposed to the same aggregate disturbance because of their Input-Output exposures. Formally, I first construct a measure of exposure by combining the weight-ω of each network structure with a measure of predetermined sectoral sensitivity $q(s)$ to the aggregate disturbance

$$exposure^{(\mathcal{V},\mathcal{D}^j)}(s) = \sum_{s' \in \Phi_s} \omega_{s,s'}^{(\mathcal{V},\mathcal{D}^j)} \, q(s')$$

for $j = \{fd, fs\}$. This formulation encompasses all network structures and, depending on the application, the weight may correspond to the vertical production network, or the horizontal demand- and supply-based network. Intuitively, a sector is highly exposed when it is strongly connected to sectors that are themselves highly sensitive to the aggregate disturbance. Exposure therefore measures how embedded a sector is to the other sectors' sensitivities relative to the aggregate disturbance.

A natural concern is that horizontal and vertical exposures may be highly collinear and partly reflect similar features, since both belong to the same primitive Input-Output matrix (Theorem 1). Hence, the component of horizontal exposure that is orthogonal to vertical one is residualised from the auxiliary projection

$$exposure^{\mathcal{D}j}(s) = \beta^{cons}_{exposure} + \beta_{\mathcal{V}}\, exposure^{\mathcal{V}}(s) + \gamma_s' \mathcal{Z}(s) + r^{\mathcal{D}j}(s)$$

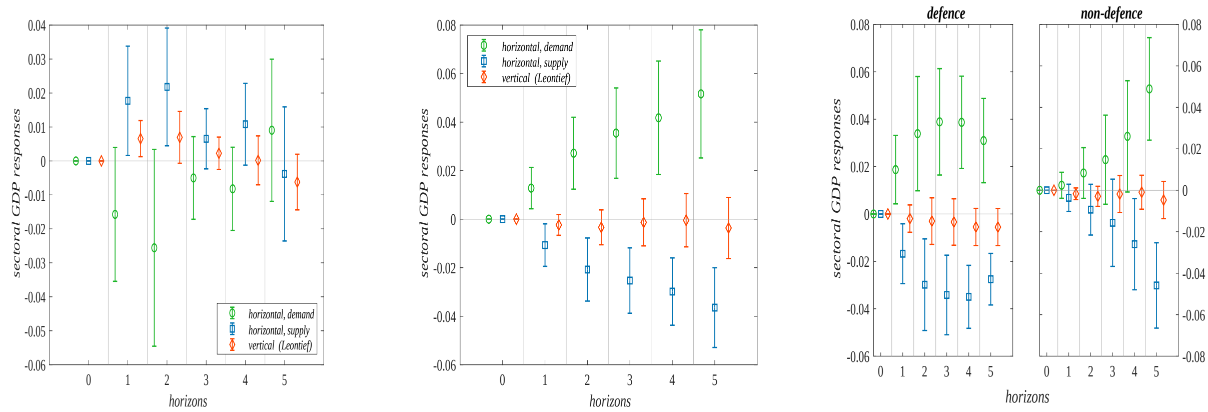
where $\mathcal{Z}(s)$ are sectoral controls (*e.g.*, size, Domar weights, network importance). Let G_t denote an aggregate shock. The Bartik exercise is the Local Projection (LP)

$$\begin{aligned}\Delta y_{t+h}(s) &= \beta_h^{\mathcal{D}fd}\left(G_t \times \widehat{r}^{\mathcal{D}fd}(s)\right) + \beta_h^{\mathcal{D}fs}\left(G_t \times \widehat{r}^{\mathcal{D}fs}(s)\right) \\ &\quad + \beta_h^{\mathcal{V}}\left(G_t \times exposure^{\mathcal{V}}(s)\right) + \phi(t+h) + \phi_h(s) + v^y_{t+h}(s)\end{aligned}$$

given $h = \{1, \ldots, \hbar\}$ time-horizons, where $\phi_h(s)$ and $\phi(t+h)$ denote sector and time fixed effects, respectively. The interaction term captures the identified intensity with which the aggregate disturbance affects each sector through the network. Importantly, the coefficients β do not measure the dynamic response of the economy to the aggregate shock. Instead, each measures whether sectors with greater vertical/horizontal network exposure systematically react more or less strongly to the same shock than sectors with lower exposure. Consequently, the identifying variation is cross-sectional: a positive coefficient implies that highly exposed sectors react more strongly to the aggregate disturbance, whereas a negative coefficient indicates that network exposure dampens the transmission of the shock; the sequence of such coefficients across horizons traces the evolution of these cross-sectional differences over time. Therefore, the key empirical content of Figure 6 is not the dynamic unconditional effect of horizontal distance, but rather the per-period conditional effect of horizontal distance after controlling for vertical Input-Output exposure. Note that the aggregate shock itself is absorbed by time fixed effects, implying that the identification comes entirely from differences in exposure across sectors.

Aggregate TFP shock.– I first consider an aggregate TFP disturbance, obtained as the unexpected component of aggregate total factor productivity after removing persistence and low-frequency trends, where sectoral sensitivity measures capture how close each sector is to sectors that are historically sensitive to aggregate TFP innovations.[10] The estimated cross-sectional LP coefficients in Panel 6a reveal a sharp contrast across transmission channels. While the Leontief-based exposure displays some predictive power at short horizons, its magnitude is smaller and its effects dissipate rapidly. By contrast, the supply-based measures generate larger and more

[10] Using the annual series from Fernald (2014), I extract the TFP innovation as the fitted residual from the AR(1) regression with time trend: $\log TFP_t = \beta^{cons}_{tfp} + \beta_{tfp} \log TFP_{t-1} + \beta_{time} \times t + \varepsilon^{tfp}_t$. This series is also exploited to construct sectoral exposures. First, I estimate sector-specific historical sensitivities to aggregate TFP innovations, capturing how strongly the sector tends to react when aggregate productivity unexpectedly changes. These sensitivities are then propagated through the alternative network structures considered.



(A) TFP

(B) GOVERNMENT SPENDING

FIGURE 6: "NETWORKING UP" AGGREGATE SHOCKS

Note: cross-sectional coefficients measure how differently sectors respond positively (amplification) or negatively (attenuation), due to vertical and horizontal network exposures, to the same aggregate shock, as in one standard deviation. Bars correspond to intervals at 90% of bootstrapped confidence level, clustered by sectors. *Source*: various and own calculations.

persistent differences in sectoral responses: sectors that are more closely connected, through common downstream buyers, to TFP-sensitive sectors react more strongly to aggregate productivity shocks over several horizons. Demand-based horizontal exposure does not generate statistically significant effects. A positive aggregate TFP shock hits asymmetrically due to pre-determined exposures, increasing productivity more in some sectors than others. Through the horizontal supply channel, more productive sectors expand production and capture a larger share of downstream demand, leading to a reallocation of expenditure across competing suppliers. This competitive displacement implies that aggregate demand is satisfied by increasingly efficient producers, so that total economic activity rises despite unchanged downstream demand, as production is concentrated in higher-productivity sectors. By contrast, the horizontal demand channel does not play a significant role, since TFP shocks operate through productivity differences rather than own demand expansion: they do not systematically increase intermediate input demand, making upstream propagation through shared suppliers weak, non-systematic, and of second-order importance. These findings suggest that horizontal network structures constitute an independent and economically relevant dimension of production networks.

Government spending shock.– Panel 6b, instead, plots the coefficients for government spending disturbance, constructed similarly to the TFP case.[11] The results reveal a markedly different pattern across propagation channels. Government spending shocks diffuse not only through directly affected industries but also through common Input-Output relationships, and appear to diffuse across sectors sharing common suppliers, generating positive horizontal spillovers in consistency with the intrinsic demand-based nature of these shocks, while sectors sharing com-

[11] Pre-determined exposures are constructed from the 1997 U.S. Input-Output table as the government purchases in the sectoral supply of intermediates, capturing the extent to which a sector is connected to government demand and therefore likely to benefit from expansions in aggregate public expenditure.

mon buyers exhibit weaker responses, consistent with competitive reallocations of downstream demand. Reassuringly, the results are also robust to disentangling aggregate government expenditure into defence and non-defence components, whose pre-determined spending categories are concentrated in targeted industries: despite the different cross-sectional allocation of network exposures, the pattern of sectoral responses is preserved. These type of shocks operate as positive demand shocks that propagate through the Input-Output network via two distinct horizontal channels. On the demand side, higher input demand from directly affected sectors increases demand faced by shared upstream suppliers. These suppliers can adjust either through quantities or prices. If they respond by expanding production at given prices, the increase in input supply can benefit other sectors connected through the same supplier, or leave their production unaffected. If instead they raise prices, downstream sectors may either continue purchasing at higher cost or substitute toward alternative suppliers, reallocating production across the network. In both cases, the adjustment of common upstream suppliers to higher downstream demand generates positive spillovers as it propagates through Input-Output linkages or substitution effects, and leads to an overall expansion in economic activity. On the supply side, targeted spending induces sectors to expand their production to meet higher government demand. This allows these sectors to meet their supply more effectively, affecting how downstream expenditure is allocated across upstream suppliers. Given the unaltered demand of common buyers, higher absorption by directly exposed sectors crowds out others serving the same customers: horizontally exposed sectors experience negative spillovers through competitive pressures for downstream markets. In this case, there is no compensating expansion from shared downstream demand, so that effects are driven by within-buyer reallocation, resulting in overall dampened responses in horizontally exposed sectors.

Taken together, these results reinforce the broader message of the paper. Aggregate productivity shocks primarily propagate through the supply-based horizontal structure, while aggregate fiscal shocks operate through a distinct network dimension, namely the demand-based horizontal geometry. The fact that the dominant propagation channel varies across shocks suggests that horizontal geometries do not merely replicate classical Input-Output effects, but instead capture economically meaningful dimensions of sectoral interdependence that become relevant under different types of aggregate disturbances, and contain additional information on shock propagation that is not fully captured by standard vertical production linkages.

5. Empirics

In this empirical section sector-level data for the United States economy are exploited to test the predictions about sectoral comovement in Section 3. Estimation relies on the two-stage procedure outlined by Barattieri and Cacciatore (2023): first, I perform a series of sectoral panel Fixed-Effect (FE) regressions to isolate the exogenous dynamics in a given input of production not due to other (sector-specific, other sectors-specific and aggregate) factors; then, each estimated residual is exploited as an identified structural shock into a Local Projection (LP) analysis, in order to asses the network-based short-run cumulative variations in sectoral employment.

5.1. Intermediate inputs and sectoral employment

Sections 2 and 3 rest on the premise that adjustments in the stock of circulating intermediate inputs lead to synchronized sectoral employment responses. Considering variations within a sector and across its economic neighbours, the first step isolates changes in input sourcing plausibly orthogonal to employment dynamics. The following panel fixed-effect regression over $s \in \Phi$ sectors is estimated:

$$\begin{aligned} x_t(s) = \beta_x^{cons} + \beta_n\, \dot{n}_t(s) \;+\; \sum_d \beta_{n(d)}\, \dot{n}_t\big(\Phi_s, d\big) \;+\; \sum_k \beta_{n(k)}\, \dot{n}_t\big(\Phi_s, k\big) \\ +\; \gamma_s' \mathcal{Z}_t(s) \;+\; \Gamma_s' \dot{\mathcal{Z}}_t(s) \;+\; \gamma' \mathcal{Z}_t \;+\; \Gamma' \dot{\mathcal{Z}}_t \;+\; \phi(s) \;+\; u_t^x(s) \end{aligned} \tag{5}$$

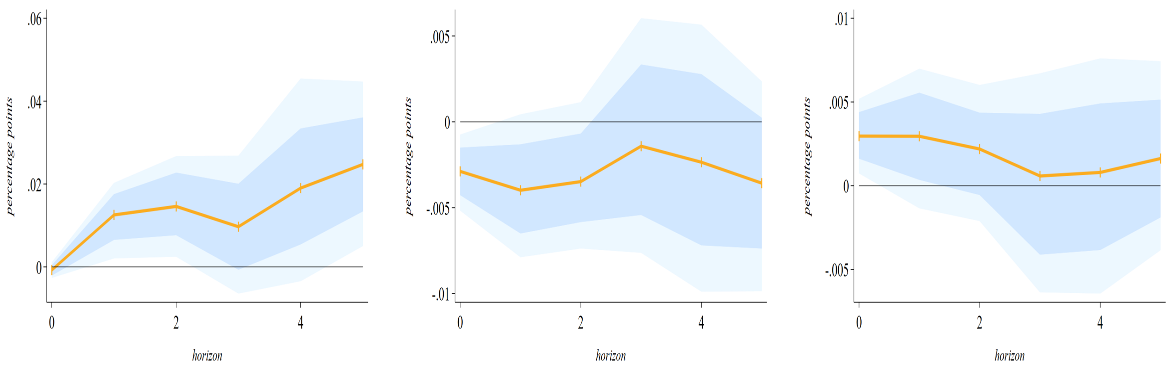
where $x_t(s)$ identifies the set of intermediate inputs bought by sector-s (as a share of its total production), while $\dot{n}_t(s)$, $\dot{n}_t\big(\Phi_s, d\big)$, and $\dot{n}_t\big(\Phi_s, k\big)$ represent its employment growth, that of its economically closer and further sectors, and that of its upstream and downstream sectors, respectively. The latter two measures are defined as

$$\begin{aligned} \dot{n}_t\big(\Phi_s, d\big) &= \sum_{s' \neq s \in \Phi_s} \Delta\, n\big(s', d\big) \qquad\qquad , \quad \forall d \in \boldsymbol{\mathcal{D}}_{ext}^j \quad and \quad j = \{fd, fs\} \\ \dot{n}_t\big(\Phi_s, k\big) &= \sum_{s' \neq s \in \Phi_s} \ell\,(s, s')\; \Delta\, n\big(s', k\big) \quad , \quad \forall k = \{up, dw\} \end{aligned}$$

where $\Delta n(\cdot) = n_t(\cdot) - n_{t-1}(\cdot)$. As for vertical propagation, each change in employment is weighted by the sales-based Leontief inverse relation, $\ell\,(s, s') = \big[1 - \alpha\,(s, s')\big]^{-1}$, between sector-$s$ and each of the other sectors whenever these are located upstream or downstream to sector-s, thereby capturing direct and indirect network exposure. Differently, changes in employment under extensive margin network demand or supply distances seize all the horizontal relationships across sectors which, in line with Theorem 3, are not leveraged by the Leontief inverse weights.

A set of control variables characterizes the second line of eq. (5). $\mathcal{Z}_t(s)$ and $\dot{\mathcal{Z}}_t(s)$ comprise the labour force size of the sector, both its level and growth rate of value-added, and its net exports that control for changes in demand/supply not due to internal factors. The set of aggregate controls, $\mathcal{Z}_t$ and $\dot{\mathcal{Z}}_t$, is made of aggregate value-added and aggregate employment in size as well as in growth terms. Finally, while β's are coefficients to be estimated, $\phi(s)$ imposes fixed-effects to control for unobserved heterogeneity across sectors. Standard errors are clustered at sector-level. For each sector, the estimated residual $\widehat{u}_t^x(s)$ identifies the exogenous variation in the share of production of sector-s that is bought from Input-Output connections.

Impulse Response Functions (IRFs) of sectoral employment to changes in the sector-specific intermediate inputs rely on the estimation of Local Projections. Given $h = \{1, \dots, \hbar\}$ time-horizons, and for all sectors $\{s, s', \dots, m\} \in \Phi$, a battery of panel FE predictive regressions, consisting in estimating the identified sector-specific struc-



(A) SECTOR-SPECIFIC (B) $d = 1$ (C) $d = 2$

FIGURE 7: SECTORAL EMPLOYMENT RESPONSE TO INTERMEDIATE INPUTS

Note: given the *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment to a 1% increase in the sector-specific set of intermediate inputs as a share of its value-added for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. In particular, Panel 7a represents the sector-level employment response to changes in its intermediate inputs, while Panel 7b-7c to changes in the set of intermediates in closer (distance equal to 1) and further (distance equal to 2) sectors. The solid-orange line corresponds to the average response of employment across sectors, while shadow-blue and shadow-light blue areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eqs. (6)-(7). *Source*: BEA and own calculations.

tural shock, $\widehat{u}_t^x(s)$, on the cumulative difference in sectoral levels, is perfomed:[12]

$$\dot{n}_{t+h}^{cum}(s) = \beta_{\widehat{u}^x,h}\ \widehat{u}_t^x(s)\ +\ \phi(t+h)\ +\ \phi_h(s)\ +\ v_{t+h}^x(s) \tag{6}$$

with $\dot{n}_{t+h}^{cum}(s) = n_{t+h}(s) - n_{t-1}(s)$ denoting the cumulative change in sector-s employment at each horizon. Estimated coefficient $\beta_{\widehat{u}^x,h}$ is the response of the variation in the cumulative difference of sectoral employment at time $t+h$ given the identified shock hitting in period-t. Prediction error term $v_{t+h}^x(s)$ is horizon-specific, while standard errors and bootstrapped Confidence Interval (CI) of $\beta_{\widehat{u}^x,h}$ are clustered by sectors. To avoid measurement errors due to unobserved heterogeneity in response, sectoral fixed-effects $\phi_h(s)$, and horizon fixed-effects $\phi(t+h)$ – that remove common trends in the comovement of sectoral employment –, are imposed.[13]

Outcomes from eq. (6) estimate the cumulative responses of sectoral employment to changes in their set of intermediate inputs. Analogous estimation is performed to variations in nearby sectors, considering the d^{th}-distance between sectors:

$$\dot{n}_{t+h}^{cum}(s) = \beta_{\widehat{u}^x(d),h} \sum_{s'\neq s} \widehat{u}_t^x(s',d)\ +\ \phi(t+h)\ +\ \phi_h(s)\ +\ v_{t+h}^x(s,d) \tag{7}$$

Performed estimation of eqs. (6) and (7), under *Leontief inverse* network structure and *factor input supply* distances, delivers Figure 7. The cumulative (short-term) re-

[12] Cumulative variables are likely to be highly correlated with the error term because they embed past values – bequeathing past errors – of the variable of interest, leading to biased and inconsistent estimates. Differencing cumulative measures soothes this endogeneity concern.

[13] Differently from the first-stage, where I control common trends over time using aggregate variables, $\phi_h(t+h)$ removes the possibility of common trend in the *response* of sector-specific employment to idiosyncratic shocks; the possibility that employment responses are influenced by a common trend is ruled out.

sponse of sector-specific employment levels to a 1% change in its output share of intermediate input is in the first panel: averaging across sectors, the increase induces a positive statistically-significant change in sectoral employment, the central rebalancing effects from circulating inputs in Sections 2-3. Each additional unit of intermediate inputs requires less than one worker so that factors of production are at least substitute, as the Cobb-Douglas specification instructs.

Changes in intermediate inputs from other sectors do not yield consistent outcomes. Results are almost always not statistically significant at 90% except when the shock hits, and the magnitude of the variation is negligible. Interesting is to note the different response between Panels 7b and 7c: for intermediate inputs changes in closer sectors there is an average negative effect, while changes in further sectors are positively related with variations in sectoral employment.[14] Such observations might suggest that certain types of intermediate inputs are more essential than others in production (*e.g.*, Carvalho and Voigtländer 2015), but it is actually the type of economic distance that determines their importance both in the production of sectoral output and in their network centrality of being larger buyers or suppliers.

5.2. Sectoral comovement of employment

I now analyse how sectoral employment adjusts in response to employment changes in other sectors, depending on their network economic distance. The previous identification and estimation scheme is developed, but now purging the dynamics of sector-specific employment with changes in its and other sectors' value-added, since employment is not exogenous to growth rates of sectoral output (*e.g.*, Hornstein and Praschnik 1997; Acemoglu, Akcigit, et al. 2015; Sandqvist 2017).

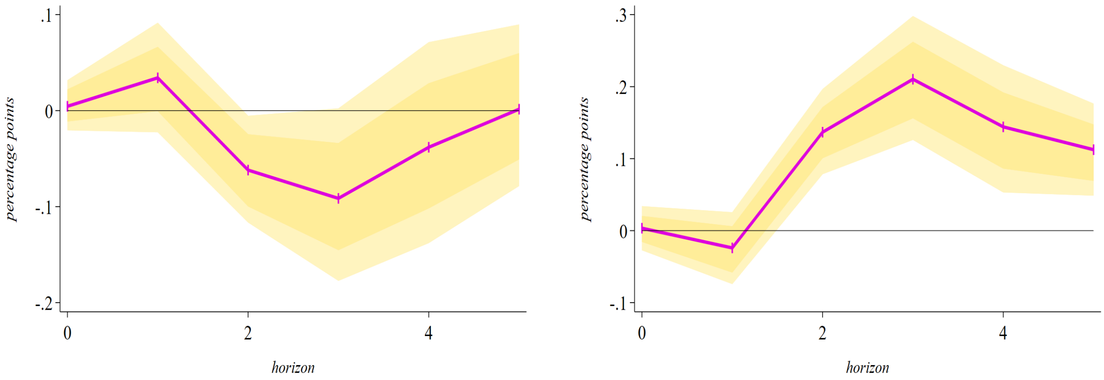
Identification.– To identify exogenous variations in sectoral employment, $n_t(s)$, a panel fixed-effects regression over each sector $s \in \Phi$ is estimated:

$$\begin{aligned} n_t(s) = \beta_n^{cons} + \beta_y\, \dot{y}_t(s) \,+\, \sum_d \beta_{y(d)}\, \dot{y}_t(\Phi_s, d) \,+\, \sum_k \beta_{y(k)}\, \dot{y}_t(\Phi_s, k) \\ +\, \gamma_s' \mathcal{Z}_t(s) \,+\, \Gamma_s' \dot{\mathcal{Z}}_t(s) \,+\, \gamma' \mathcal{Z}_t \,+\, \Gamma' \dot{\mathcal{Z}}_t \,+\, \phi(s) \,+\, u_t^n(s) \end{aligned} \tag{8}$$

where $\dot{y}_t(s)$ is the value-added growth for sector-s, $\dot{y}_t(\Phi_s, d)$ identifies the value-added growth of its related sectors according to network distances, and $\dot{y}_t(\Phi_s, k)$ represents the value-added growth of its upstream and downstream sectors. Each measure is defined as that of eq. (5) but using $y(s)$ instead of employment. Moreover, for this specification, sector-level controls, $\mathcal{Z}_t(s)$ and $\dot{\mathcal{Z}}_t(s)$, comprise the labour force size of the sector, its value-added level and net exports, while aggregate controls, $\mathcal{Z}_t$ and $\dot{\mathcal{Z}}_t$, are analogous to eq. (5). Estimated coefficients are the β's, given sectoral fixed-effects, $\phi(s)$, and sectoral standard errors.[15] Sector-specific estimated residual

[14] In Appendix C, under *Leontief inverse* and *factor input demand* network distances, I show how that the same impact of Panel 7a still holds (Panel C.1a of Figure C.1), while results for distances' effects (Panels C.1b and C.1c) identify stronger but opposite results: increasinf for closer sectors, declining for further ones.

[15] In both identifications I exclude time fixed-effect while controlling for variations in aggregate variables.



(A) $d = 1$

(B) $d = 2$

FIGURE 8: EMPLOYMENT COMOVEMENT UNDER DEMAND LINKAGES

Note: given the *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Panel 8a represents the sector-level employment response to employment changes in closer (distance equal to 1) sectors, while Panel 8b plots the response to employment changes in further (distance equal to 2) ones. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (9). *Source*: BEA and own calculations.

$\widehat{u}_t^n(s)$ from eq. (8) identifies the exogenous variation in sectoral employment.

Local Projection analysis.– Sectoral comovement is defined as the response of employment in a given sector to changes in employment of other sectors:

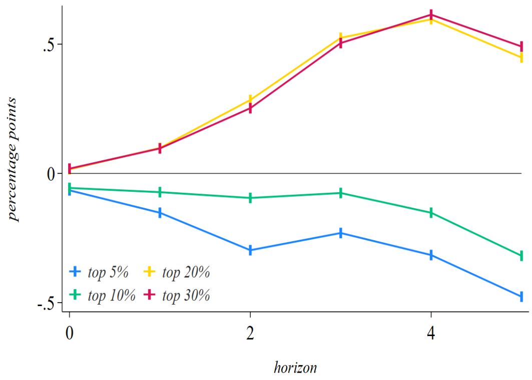
$$\dot{n}_{t+h}^{cum}(s) = \beta_{\widehat{n}(d),h} \sum_{s' \neq s} \widehat{u}_t^n(s', d) \; + \; \phi(t+h) \; + \; \phi_h(s) \; + \; \nu_{t+h}^n(s, d) \tag{9}$$

with $\dot{n}_{t+h}^{cum}(s) = n_{t+h}(s) - n_{t-1}(s)$ denoting the cumulative change in sector-s employment at each horizon given the identified employment shock in closer or further sectors at the d^{th}-distance. Coefficient $\beta_{\widehat{n}(d),h}$ is the h-step-ahead response of sector-specific employment in cumulative difference due to the identified shock $\widehat{u}_t^n(s', d)$. Still, sector and horizon fixed effects, $\phi_h(s)$ and $\phi(t+h)$ respectively, are imposed to remove unobserved heterogeneity and common trends across sectors in the comovement of sectoral employment. Prediction error term $\nu_{t+h}^n(s, d)$ is horizon- and distance-specific in each predictive panel regression, with standard errors and bootstrapped Confidence Interval (CI) of $\beta_{\widehat{n}(d),h}$ clustered by sectors.

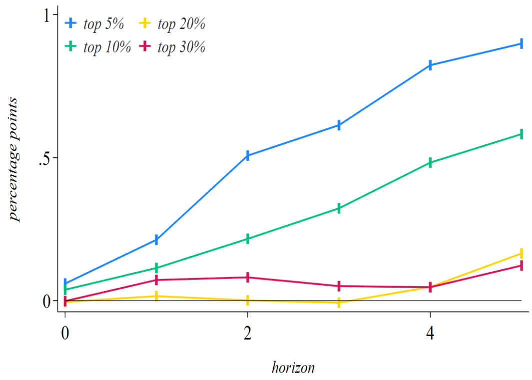
5.2.1. *Results for factor input demand distances*

Considering sectors whose distance is defined in terms of buying from the same set of upstream suppliers, estimating eq. (9) yields Figure 8. As already emphasized in discussing Proposition 1, demand-based complementarities are inherently ambiguous to interpret since they depend on the joint action of vertical propagation and horizontal spillovers: when buyers can easily substitute inputs, a positive shock in

Comovement between aggregate and sectoral variables is well established (*e.g.*, Stock and Watson 1999; Huffman and Wynne 1999; Rebelo 2005; DiCecio 2009).



(A) $d = 1$



(B) $d = 2$

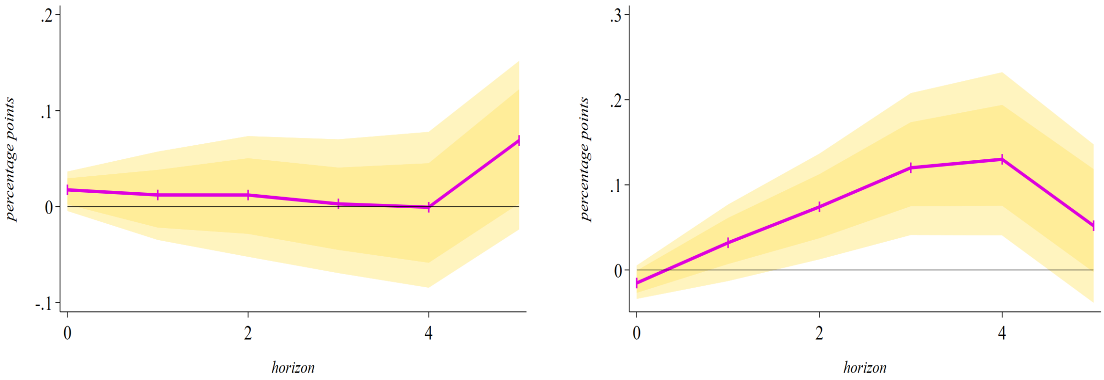
FIGURE 9: DEMAND LINKAGES AND COMOVEMENT IN HIGHLY INTERLINKED SECTORS

Note: given the *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Panel 9a represents the sector-level employment response to employment changes in closer (distance equal to 1) sectors, while Panel 9b plots the response to employment changes in further (distance equal to 2) ones. Each line corresponds to the average response of employment across sectors with different numbers of linkages, robust to 68% and 90% significance levels of bootstrapped Confidence Interval (CI) computed from eq. (9). If a plot is not appearing, it means there is no response for the specified distance value. *Source*: BEA and own calculations.

one sector strengthens the demand for the shared supplier and raises the upstream price, thereby crowding out the other sector; by contrast, when inputs are complements, the increase in the supplier's price erode the demand competitiveness of the other sector, dampening comovement. Where these forces conflict or remain weak, observed correlations are muted or ambiguous.

These mechanisms echo in observed responses. Panels 8a and 8b show that a 1% increase in employment in closely linked sectors ($d = 1$) produces an ambiguous response, whereas more distant sectors ($d = 2$) consistently display positive comovement. However, sectoral shocks in Panel 8a do not produce any statistically significant effects, as the estimated responses fail to reach significance at the 90% confidence levels. This lack may be driven by including all sectors indiscriminately, as sectors with fewer inter-sectoral linkages might introduce noise or bias to the estimation, masking and diluting the true magnitude of measured comovement in sectoral employment: restricting the sample to sectors with most Input-Output linkages could reveal more meaningful propagation effects, in line with Theorem 4, where network effects are more pronounced.

Nevertheless, ambiguous patterns emerge when partitioning sectors into finer subsets by the number of Input-Output linkages (Figure 9): comovement depends on the specific cluster considered. This confirm that demand-driven horizontal complementarities and vertical propagation operate jointly, often obscuring one another. Such mixed evidence ultimately reflects the inherent ambiguity of demand-based distances discussed in Subsection 3.1: because they capture substitution patterns across upstream intermediate goods rather than genuine horizontal complementarities, horizontal propagation remains weak and easily masked by vertical effects.



(A) $d = 1$ (B) $d = 2$

FIGURE 10: EMPLOYMENT COMOVEMENT UNDER SUPPLY LINKAGES

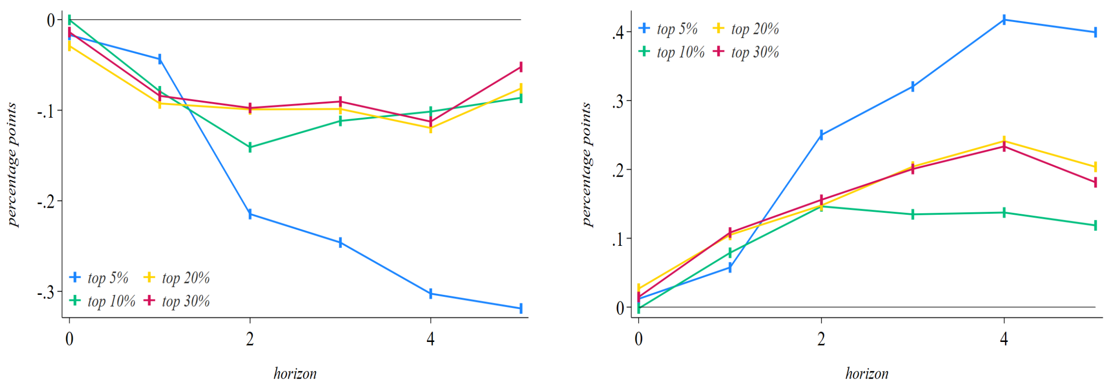
Note: given the *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Panel 10a represents the sector-level employment response to employment changes in closer (distance equal to 1) sectors, while Panel 10b plots the response to employment changes in further (distance equal to 2) ones. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (9). *Source*: BEA and own calculations.

Observed demand-driven comovement broadly aligns with the theory developed in Sections 2 and 3. Positive comovement is most likely to emerge when sectors are "economically" distant, since vertical effects dominate and the conflicting role of horizontal spillovers is attenuated. For mostly connected sectors, however, the balance between vertical and horizontal channels is more delicate, producing the mixed responses observed in the data. This suggests that observed employment comovement mirrors an interplay of vertical and horizontal propagation, thereby highlighting the inherent complexity in interpreting demand-based complementarities.

5.2.2. *Results for factor input supply distances*

Considering now sectors whose distance is defined in terms of selling to the same set of downstream buyers, Figure 10 plots the responses from the estimation of eq. (9). Interpreting the effect of further ($d = 2$) sectors' variations in employment yields the same results of the factor input demand case: from Panel 10b, a 1% increase in their employment increases sectoral employment as well, thus determining positive comovement and a positive transmission of the sector-idiosyncratic shock. Different is the case of closer ($d = 1$) sectors, where it appears a positive and then a negative cumulative effect on sector-specific employment with a small magnitude.

The lack of significance in Panel 10a might have two complementary interpretations: *(i)* changes in employment in nearby sectors have no measurable impact on others when the sectors are linked through a common set of downstream buyers; and *(ii)* the factor input supply network distance seems to dampen the vertical propagation of shocks across the production network, thereby weakening the expected comovement in production inputs across sectors. Specifically, when sectors share



(A) $d = 1$ (B) $d = 2$

FIGURE 11: SUPPLY LINKAGES AND COMOVEMENT IN HIGHLY INTERLINKED SECTORS

Note: given the *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Panel 11a represents the sector-level employment response to employment changes in closer (distance equal to 1) sectors, while Panel 11b plots the response to employment changes in further (distance equal to 2) ones. Each line corresponds to the average response of employment across sectors with different numbers of linkages, robust to 68% and 90% significance levels of bootstrapped Confidence Interval (CI) computed from eq. (9). *Source*: BEA and own calculations.

the same downstream buyers, a change in their relative price faced by the common buyer leads to opposite changes in their relative supply of intermediate inputs. This generates an inverse comovement in sectoral production levels and, consequently, an opposite comovement in sectoral employment levels. In the context of Panel 10a, these opposing forces roughly cancel each other out when averaged across all sectors, resulting in the non-significant response. Accordingly, clearer effects might be correctly identified when considering highly interlinked sectors.

Impulse responses for the clustered analysis are presented in Figure 11. The results for major network distances, shown in Panel 11b, largely mirror those found using the full sample, but with a notably higher magnitude in the point estimates responding to the structural employment shock. More strikingly, minor supply distances in Panel 11a fully confirm Proposition 3: any increase in employment levels within closely linked sectors induces a negative transmission to other nearby sectors, resulting in an opposite comovement in sectoral employment. This inverse relationship grows stronger as the analysis narrows to an even smaller group of sectors with increasingly dense linkages: the closer the sectors are in the production network when supplying downstream, the more pronounced these opposing employment responses become. In sum, restricting the analysis to highly interlinked sectors (Theorem 4) reveals that the Input-Output structure and its horizontal geometry fundamentally condition either the direction and the strength of sectoral comovement.

All these theoretical and empirical results are in line with the non-linear theories in Atalay (2017) and Baqaee and Farhi (2019): downstream complementarities in intermediate inputs are crucial to generate opposite responses from an idiosyncratic shock – *i.e.*, the downstream demand pass-through effect from the supply-driven network economic distance is accordingly generating opposite comovement (Proposition 2). Yet, such pass-through is found to be weaker when sectors have dissimilar Input-Output structure in supply.

5.3. Robustness

Results on comovement hold along different "economically-based" robustness checks, sequentially presented in Appendix C. Initially, I shift the focus towards peripheral sectors within the production network. Much of the existing literature holds on the role of central sectors, typically measured using Bonacich-Katz centrality (*e.g.*, Carvalho 2014). The underlying premise is that a positive shock in a central sector transmits outward, potentially inducing broader economic expansions – *i.e.*, microeconomic shocks can scale up to macroeconomic consequences through network centrality. From a comovement perspective, periods of economic expansion (contraction) are expected only when a positive (negative) shock in a central sector effectively propagates to more peripheral sectors: fluctuating aggregate activity is not merely a direct realization of shocks to central nodes, but rather it stems from the manner in which such asymmetric shocks ripple across the network to other, more peripheral sectors.[16] If network distances were irrelevant, comovement would be expected universally. However, should demand- and supply-side linkages prove significant, the comovement patterns ought to remain consistent with those documented thus far.

I further account for changes in sectoral inventories, since they introduce a short-run margin that may buffer the network-based transmission: when inventories are available, sectors smooth shocks through previously accumulated stocks, temporarily decoupling production adjustments from contemporaneous changes in the purchased intermediate input bundle. As a result, the transmission of shocks through shared Input-Output structures should be attenuated, breaking the simultaneity of horizontal complementarities and weakening the role of both demand- and supply-based network economic distances in shaping short-run comovement. By contrast, in the absence of inventories, changes in economically nearby sectors are expected to translate directly into the usual horizontal propagation effects.

Finally, I test the robustness of the presented findings by employing alternative base years for the U.S. production network. My main analysis is anchored at the 2007 Input-Output structure, which conveniently divides the sample period into two equal sub-periods. Theoretically, the results should remain valid provided that the production network structure exhibits limited evolution over time.[17] Empirically, I replicate the full set of analyses using network data from other benchmark years.

Centrality scores.– Figures C.6 and C.7 present the impulse responses of em-

[16] Stated differently, it is not the initial shock itself that generates business cycle fluctuations, but the subsequent consequences it produces as it transmits across the network to less central nodes.

[17] A condition broadly supported in the empirical literature. For instance, Carvalho (2014) and Acemoglu, Ozdaglar, et al. (2016) document relative stability of U.S. sectoral production network and its role on shock propagation. Classical Input-Output analysis (*e.g.*, Leontief 1986; Miller and Blair 2022) also emphasises the persistence of production linkages in the short to medium run.

ployment to factor input demand and supply distances, respectively, considering only the peripheral sectors of the U.S. production network. Results are broadly consistent with those discussed in earlier sections: opposite comovement under supply distances, and positive comovement for shocks originated in further sectors. The only noteworthy exception concerns the response of employment to changes in closer sectors under demand-based distances: rather than exhibiting the ambiguity observed for all sectors (overlapping of vertical and horizontal dimensions), inspecting horizontal complementarities in demand for more peripheral sectors generate effects that resemble those observed under supply-based distances, with opposite comovement occurring across sectors sharing the same network structure in terms of buying from common suppliers. This refines the observed pattern under direct demand-based measures of Figure 8, suggesting that opposite employment comovement among sectors purchasing from similar upstream suppliers largely occurs in sectors not playing a key role in the propagation of shocks.[18] These outcomes create a bridge between eq. (4) and the insights of Theorem 5: positive shocks in central sectors transmit unevenly to peripheral ones according to different network economic distance values, thereby inducing different types of aggregate behaviours.

Inventories.– The outlined hypothesis is strongly supported by the supply-based patterns in Figure C.8. Horizontal complementarities vanish for sectors holding inventories, and the transmission of a positive shock reverts to its vertical logic.[19] By contrast, in contexts without inventories, the baseline results are preserved, with positive shocks generating opposite comovement across sectors with nearly similar Input-Output structures. This sharp contrast indicates that horizontal effects are operative when production adjustments are directly tied to contemporaneous changes in circulating intermediate inputs, rather than when accumulated in storage. Inventories act as a short-run buffer that temporarily deactivates the horizontal propagation channel without altering the underlying network structure. These findings provide direct evidence that the documented horizontal complementarities reflect structural economic mechanisms, arising from contemporaneous adjustments through shared linkages, and not mechanical features of the production network.

Different base years.– All the documented results, derived using the 2007 U.S. tables, remain robust if replicated for alternative benchmark years (2002 and 2012), as depicted in Figures C.9, C.10, and C.11, supporting the inherent ambiguity of demand-based distances and the clearer, more systematic comovement patterns under supply-based distances. This temporal consistency suggests that the structural features of the U.S. production network have remained remarkably stable over time, documenting the enduring nature of Input-Output linkages and the slow-moving evolution of production networks, thus lending further credibility to the analysis.

[18] In line with the discussion in Baqaee and Farhi (2019): within a sector, complementarities ("horizontal" in my framework) across intermediate inputs attenuate the aggregate effects of a positive idiosyncratic shock.

[19] Over the sample period, inventories are observed in NAICS agriculture (11), extraction (21), and manufacturing (31-33) sectors, as well as in transportation (481-484), publishing (511), and recording (512) industries.

6. Future Policy Perspectives

The Input-Output structure of the economy shapes how shocks propagate and contributes to business cycle dynamics. This paper has explored how the structure of production networks allows to push further the traditional focus on vertical transmission to emphasize the role of horizontal economic relationships. By analysing shared network connections, the analysis reveals that it is not merely the existence of Input-Output linkages, but rather the demand and supply geometry of these linkages that fundamentally governs the transmission of micro-originated shocks and the resulting macroeconomic comovement patterns. Several are the contributions that horizontal complementarities can provide to the ongoing literature exploiting a production network perspective to form and deliver policy prescriptions.[20]

Fiscal policy.– Horizontal complementarities play a critical role when considering the specific composition of public expenditure since certain sectors may be disproportionately exposed to changes in government demand. The evidence presented in Subsection 4.3 suggests that fiscal shocks propagate differently across horizontal geometries, implying that two spending programs of identical magnitude may therefore generate substantially different outcomes depending on the targeted industries. Incorporating demand- and supply-driven network distances into fiscal policy analysis could thus provide a more accurate characterization of how sector-specific government interventions diffuse throughout the economy than traditional Input-Output measures alone (*e.g.*, Barattieri, Cacciatore, and Traum 2023; Bouakez et al. 2025). Tariff policies operate through similar network mechanisms, not by raising demand directly, but rather altering relative prices (*e.g.*, Barattieri and Cacciatore 2023; Antonova et al. 2025; Clausing and Obstfeld 2025), effectively acting as revenue reallocation shocks (*i.e.*, my supply-based distances), lending further perspectives from network's horizontal geometries.

Industrial policy.– Two contemporaneous papers analyse how production networks amplify the impact of industrial policies supporting strategic sectors to promote sustained economic growth:[21] central are the notions of "distortionary effects" (market imperfections accumulate through backward demand linkages; Liu 2019) and "downstream spillovers" (positive effects on buyers of targeted sectors; Lane 2025), and both perspectives exploit the verticality of sectoral connections. Yet, how do industrial policies affect untargeted sectors exhibiting similar Input-Output

[20] On the theory side, horizontal complementarities can provide valuable insights into: *(i)* the functioning of global production networks (*e.g.*, Caliendo et al. 2022; Huo et al. 2025), extending beyond merely vertical supply chains; *(ii)* the complexity of firm-to-firm production networks (*e.g.*, di Giovanni et al. 2018; Boehm, Flaaen, et al. 2019), as demand- and supply-based distances help to reveal relationships between nodes that would otherwise appear unconnected; and *(iii)* the endogenous formation of production networks, since "economically" closer sectors tend to share similar Input-Output structures and might rely on common intermediate inputs that are more essential than others (*e.g.*, Carvalho and Voigtländer 2015).

[21] Explanations related to cross-country income differences, Input-Output economies and industrialization are presented in Ciccone (2002) and Fadinger et al. (2022). Conley and Ligon (2002) study how cross-country dependence and GDP growth rates are shaped by "economic distance".

structures to the targeted ones? This is where the horizontal geometry matters.

Keynesian transmission mechanism.– Complementarities (either in consumption or production) might turn an asymmetric supply shock, affecting a subset of sectors and reducing their demand for other sectors, into a demand-like shock at the aggregate level (the "Keynesian supply shock"; Guerrieri et al. 2022). Production network is a natural field to study this mechanism (*e.g.*, Cesa-Bianchi and Ferrero 2021) and, reminiscent from Subsection 2.1 (Figure 2b), horizontal complementarities among sectors can help to clearly, easily and linearly identify demand and supply forces within the network that guide the Keynesian transmission. This extended mechanism is important for interventions aimed at stabilizing business cycles, acting as a suitable extension of the production network framework for monetary policy (*e.g.*, Pasten et al. 2020; Kalemli-Özcan et al. 2025).

Finally, by embedding similarities in common structures of Input-Output linkages, the horizontal dimension transcends the confines of production networks, rendering it applicable to a broader class of economic interdependencies including studies analysing investment (*e.g.*, vom Lehn and Winberry 2022), credit (*e.g.*, Delli Gatti et al. 2010), or financial (*e.g.*, Elliott et al. 2014; Huremovic et al. 2025) networks.

7. Concluding Remarks

Initial inquiries into production networks chiefly centred on how disaggregation induced by inter-sectoral trade generates comovement of macroeconomic variables. Reviving this foundational perspective, this paper moves beyond traditional vertical linkages by theoretically introducing and empirically validating the horizontal geometry of Input-Output economies, measured through network "economic" distances reflecting similarities between sectors in terms of common upstream suppliers or downstream buyers. Constitutive premise is that sectors are densely interconnected not only by their observed trade flows but also through shared inter-sectoral relationships, providing a richer understanding of the complex architecture underlying any capitalist economy. Demand-based distances promote an overlap of horizontal and vertical propagation, as shocks transmitted through shared upstream suppliers interact with parallel supply chain effects. By contrast, supply-based distances directly threaten the design of vertical transmission, as revenue reallocation among sectors selling to the same downstream buyers counteracts the standard propagation of shocks. These mechanisms operate independently of trade intensities, offering a powerful refinement to prevailing network analysis. As for aggregate fluctuations, horizontal propagation induces correlated adjustments across sectors sharing suppliers or buyers, generating persistent aggregate comovement absent any cascading effect. By conceptualising these horizontal complementarities, the paper enriches the understanding of how independent and idiosyncratic shocks permeate the production system. Ultimately, it reveals that the demand/supply architecture of an Input-Output structure and its horizontal geometry critically shape the transmission of shocks, offering novel insights into the networked nature of economic activity.

Bibliography

Acemoglu, D., U. Akcigit, and W. Kerr (2015), "Networks and the macroeconomy: an empirical exploration", *NBER Macroeconomics Annual*, 30:273–335

Acemoglu, D. and P. D. Azar (2020), "Endogenous production networks", *Econometrica*, 88(1):33–82

Acemoglu, D., V. M. Carvalho, A. Ozdaglar, and A. Tahbaz-Salehi (2012), "The network origins of aggregate fluctuations", *Econometrica*, 80(5):1977–2016

Acemoglu, D., A. Ozdaglar, and A. Tahbaz-Salehi (2016), "Networks, shocks, and systemic risk". in: *The Oxford Handbook of the Economics of Networks*, pp. 569-608, edited by Yann Bramoullé, Andrea Galeotti and Brian W. Rogers. New York: Oxford University Press

Antonova, A., L. Huxel, M. Matvieiev, and G. Muller (2025), "The propagation of tariff shocks via production networks", *CEPR Discussion Paper n. 20305*

Atalay, E. (2017), "How important are sectoral shocks?", *American Economic Journal: Macroeconomics*, 9(4):254–280

Baqaee, D. R. (2018), "Cascading failures in production networks", *Econometrica*, 86(5):1819–1838

Baqaee, D. R. and E. Farhi (2019), "The macroeconomic impact of microeconomic shocks: beyond Hulten's Theorem", *Econometrica*, 87(4):1155–1203

— (2020), "Productivity and misallocation in general equilibrium", *Quarterly Journal of Economics*, 135(1):105–163

Barattieri, A. and M. Cacciatore (2023), "Self-harming trade policy? Protectionism and production networks", *American Economic Journal: Macroeconomics*, 15(2):97–128

Barattieri, A., M. Cacciatore, and N. Traum (2023), "Estimating the effects of government spending through the production network", *NBER Working Paper n. 31680*

Barrot, J.-N. and J. Sauvagnat (2016), "Input specificity and the propagation of idiosyncratic shocks in production networks", *Quarterly Journal of Economics*, 131(3):1543–1592

Bartik, T. J. (1991), *Who Benefits from State and Local Economic Development Policies?* Kalamazoo, MI: W.E. Upjohn Institute for Employment Research

Boehm, C. E., A. Flaaen, and N. Pandalai-Nayar (2019), "Input linkages and the transmission of shocks: firm-level evidence from the 2011 Tōhoku earthquake", *Review of Economics and Statistics*, 101(1):60–75

Boehm, J. and E. Oberfield (2018), "Growth and the fragmentation of production", *Working Paper n. 885 Meeting Papers from Society for Economic Dynamics*

Bouakez, H., O. Rachedi, and E. Santoro (2025), "The sectoral origins of heterogeneous spending multipliers", *Journal of Public Economics*, 248:105404

Caliendo, L., F. Parro, and A. Tsyvinski (2022), “Distortions and the structure of the world economy”, *American Economic Journal: Macroeconomics*, 14(4):274–308

Carvalho, V. M. (2014), “From micro to macro via production networks”, *Journal of Economic Perspectives*, 28(4):23–48

Carvalho, V. M. and X. Gabaix (2013), “The Great Diversification and its undoing”, *American Economic Review*, 103(5):1697–1727

Carvalho, V. M. and N. Voigtländer (2015), “Input diffusion and the evolution of production networks”, *NBER Working Paper n. 20025*

Cassou, S. P. and J. Vázquez (2014), “Employment comovements at the sectoral level over the business cycle”, *Empirical Economics*, 46:1301–1323

Cesa-Bianchi, A. and A. Ferrero (2021), “The transmission of Keynesian supply shocks”, *CEPR Discussion Paper n. 16430*

Christiano, L. J. and T. J. Fitzgerald (1998), “The business cycle: it’s still a puzzle”, *Economic Perspectives*, 22(4):56–83

Ciccone, A. (2002), “Input chains and industrialization”, *Review of Economic Studies*, 69(3):565–587

Clausing, K. and M. Obstfeld (2025), “Tariffs as fiscal policy”, *CEPR Discussion Paper n. 20677*

Conley, T. G. and B. Dupor (2003), “A spatial analysis of sectoral complementarity”, *Journal of Political Economy*, 11(2):311–352

Conley, T. G. and E. Ligon (2002), “Economic distance and cross-country spillovers”, *Journal of Economic Growth*, 7:157–187

Cooper, R. and J. Haltiwanger (1990), “Inventories and the propagation of sectoral shocks”, *American Economic Review*, 80(1):170–190

Delli Gatti, D., M. Gallegati, B. Greenwald, A. Russo, and J. E. Stglitz (2010), “The financial accelerator in an evolving credit network”, *Journal of Economic Dynamics and Control*, 34(9):1627–1650

di Giovanni, J., A. A. Levchenko, and I. Mejean (2018), “The micro origins of international business-cycle comovement”, *American Economic Review*, 108(1):82–108

DiCecio, R. (2009), “Sticky wages and sectoral labor comovement”, *Journal of Economic Dynamics and Control*, 33:538–553

Dijkstra, E. W. (1959), “A note on two problems in connexion with graphs”, *Numerische Mathematik*, 1:269–271

Dragomirescu-Gaina, C. and L. Elia (2021), “Technology shocks and sectoral labour market spill-overs”, *Economics Letters*, 201

Elliott, M., B. Golub, and M. O. Jackson (2014), “Financial networks and contagion”, *American Economic Review*, 104(10):3115–53

Fadinger, H., C. Ghiglino, and M. Teteryatnikova (2022), "Income differences, productivity, and Input-Output networks", *American Economic Journal: Macroeconomics*, 14(2):367–415

Fernald, J. G. (2014), "A quarterly, utilization-adjusted series on Total Factor Productivity", *Federal Reserve Bank of San Francisco Working Paper 2012-19*

Ferrari, A. (2024), "Inventories, demand shocks propagation and amplification in supply chains", *CEPR Discussion Paper n. 19716*

Floyd, R. W. (1962), "Algorithm 97: shortest path", *Communications of the ACM*, 5(6):345

Gabaix, X. (2011), "The granular origins of aggregate fluctuations", *Econometrica*, 79(3):733–772

Ghassibe, M. (2021), "Monetary policy and production networks: an empirical investigation", *Journal of Monetary Economics*, 119:21–39

— (2024), "Endogenous production networks and non-linear monetary transmission", *BSE Working Paper n. 1449*

Ghassibe, M. and A. Nakov (2025), "Business cycles with pricing cascades", *CEPR Discussion Paper n. 20605*

Grassi, B. (2017), "IO in I-O: size, industrial organization, and the Input-Output network make a firm structurally important", *IGIER Working Paper n. 619*

Grassi, B. and J. Sauvagnat (2019), "Production networks and economic policy", *Oxford Review of Economic Policy*, 35(4):638–677

Gualdi, S. and A. Mandel (2019), "Endogenous growth in production networks", *Journal of Evolutionary Economics*, 29:91–117

Guerrieri, V., G. Lorenzoni, L. Straub, and I. Werning (2022), "Macroeconomic implications of COVID-19: can negative supply shocks cause demand shortages?", *American Economic Review*, 112(5):1437–1474

Hausmann, R. and C. A. Hidalgo (2011), "The network structure of economic output", *Journal of Economic Growth*, 16:309–342

Hornstein, A. and J. Praschnik (1997), "Intermediate inputs and sectoral comovement in the business cycle", *Journal of Monetary Economics*, 40(3):573–595

Huffman, G. W. and M. A. Wynne (1999), "The role of intratemporal adjustment costs in a multisector economy", *Journal of Monetary Economics*, 43(2):317–350

Huo, Z., A. A. Levchenko, and N. Pandalai-Nayar (2025), "International comovement in the global production network", *Review of Economic Studies*, 92(1):365–403

Huremovic, K., G. Jimenez, E. Moral-Benito, J.-L. Peydro, and F. Vega-Redondo (2025), "Production and financial networks in interplay", *Working Paper*

Huremović, K., F. Nutarelli, F. Serti, and F. Vega-Redondo (2025), "Learning to import through production networks", *Working Paper*

Jones, C. I. (2011), "Intermediate goods and weak links in the theory of economic development", *American Economic Journal: Macroeconomics*, 3(2):1–28

Kalemli-Özcan, S., C. Soylu, and M. A. Yildirim (2025), "Global networks, monetary policy and trade", *NBER Working Paper n. 33686*

Kim, B.-G. (2020), "Sectoral shifts and comovements in employment", *Economics Letters*, 192:109208

La'O, J. and A. Tahbaz-Salehi (2022), "Optimal monetary policy in production networks", *Econometrica*, 90(3):1295–1336

Lane, N. (2025), "Manufacturing revolutions: industrial policy and industrialization in South Korea", *Quarterly Journal of Economics*, 140(3):1683–1741

Leontief, W. W. (1936), "Quantitative input and output relations in the economic systems of the United States", *Review of Economics and Statistics*, 18(3):105–125

— (1986), *Input-Output Economics*. New York: Oxford University Press

Liu, E. (2019), "Industrial policies in production networks", *Quarterly Journal of Economics*, 134(4):1883–1948

Liu, E. and A. Tsyvinski (2024), "A dynamic model of input-output networks", *Review of Economic Studies*, 91(6):3608–3644

Long, J. B. and C. I. Plosser (1983), "Real business cycles", *Journal of Political Economy*, 91(1):39–69

Lucas, R. E. (1977), "Understanding business cycles", *Carnegie-Rochester Conference Series on Public Policy*, 5:7–29

McNerney, J., C. Savoie, F. Caravelli, V. M. Carvalho, and D. J. Farmer (2022), "How production networks amplify economic growth", *Proceedings of the National Academy of Sciences (PNAS)*, 119(1):e2106031118

Miller, R. E. and P. D. Blair (2022), *Input-Output Analysis: Foundations and Extensions*. 3rd edition, Cambridge, UK: Cambridge University Press

Monti, F. and L. Van Keirsbilck (2025), "The transmission of shocks across sectors and the dynamics of sectoral prices", *CEPR Discussion Paper n. 20621*

Oberfield, E. (2018), "A theory of Input-Output architecture", *Econometrica*, 86(2):559–589

Pasinetti, L. L. (1973), "The notion of vertical integration in economic analysis", *Metroeconomica*, 25(1):1–29

Pasten, E., R. Schoenle, and M. Weber (2020), "The propagation of monetary policy shocks in a heterogeneous production economy", *Journal of Monetary Economics*, 116:1–22

Qiu, Z., Y. W. Wang, L. Xu, and F. Zanetti (2026), "Monetary policy in open economies with production networks", *Journal of Monetary Economics*, 159:103918

Rebelo, S. (2005), "Real Business Cycle models: past, present and future", *The Scandinavian Journal of Economics*, 107(2):217–238

Roy, B. (1959), "Transitivité et connexité", *Comptes rendus de l'Académie des Sciences*, 249:216–218

Rubbo, E. (2023), "Networks, Phillips curves, and monetary policy", *Econometrica*, 91(4):1417–1455

Sandqvist, A. P. (2017), "Dynamics of sectoral business cycle comovement", *Applied Economics*, 49(47):4742–4759

Shea, J. (2002), "Complementarities and comovements", *Journal of Money, Credit and Banking*, 34(2):412–433

Sraffa, P. (1960), *Production of Commodities by Means of Commodities: Prelude to a Critique of Economic Theory*. London: Cambridge University Press

Stock, J. H. and M. W. Watson (1999), "Business cycle fluctuations in U.S. macroeconomic time series". in: *Handbook of Macroeconomics*, vol. 1, pt. A, pp. 3-64, edited by John B. Taylor and Michael Woodford. North-Holland: Elsevier

Taschereau-Dumouchel, M. (2026), "Cascades and fluctuations in an economy with an endogenous production network", *Review of Economic Studies*, 93(2):1354–1392

vom Lehn, C. and T. Winberry (2022), "The investment network, sectoral comovement, and the changing U.S. business cycle", *Quarterly Journal of Economics*, 137(1):387–433

Warshall, S. (1962), "A theorem on Boolean matrices", *Journal of the ACM*, 9(1):11–12

Yedid-Levi, Y. (2016), "Why does employment in all major sectors move together over the business cycle?", *Review of Economic Dynamics*, 22:131–156

Appendix

(Outline) *In this appendix I report all the material complementary to the main text; it is made of additional tables and figures, and of discussions on further analytical results. Section A deepens further all the theoretical findings; in particular, Subsection A.1 derives all the salient elements of the theoretical foundation for production network distances in Section 2; Subsection A.2 proves the theorems in Section 3, while Subsections A.3-A.5 derive and prove all the characterizing features of the outlined Input-Output general equilibrium model. Section B consolidates and further discusses the data presented in Section 4. Finally, Section C complements and enriches the exposition of the empirical findings in Section 5. Proofs are presented to be intuitive and informative, but will not be long-winded write-ups if published.*

A. Theoretical results and model derivation

A.1. Proofs for theoretical results on network economic distances

(Proof of Theorem 1) *The following proves that a single primitive Input-Output structure induces both a dynamic propagation operator along production stages and a structural representation of sectoral proximities in the production network space.*

Step 1: Input-Output equilibrium and the Leontief inverse.– *Let $G = (m, \Lambda)$ be a finite directed weighted graph, with indicator-m representing the number of nodes in the network, and $\Lambda = \{i, j \in m \times m : \alpha_{ij} \geq 0\}$ the set of directed edges representing non-negative linkages between nodes. I start from the standard Input-Output equilibrium: $y_i = \sum_{j \in \Phi} x_{ij} + c_i$. When technical coefficients are $\alpha_{ij} = \frac{x_{ij}}{y_j}$, then $y_i = \sum_{j \in \Phi} \alpha_{ij} y_j + c_i$, so that $\boldsymbol{y} = \boldsymbol{\mathcal{A}} \boldsymbol{y} + \boldsymbol{c}$, where $\boldsymbol{\mathcal{A}} = [\alpha_{ij}] \in \mathbb{R}^{m \times m}$ is the $m \times m$ matrix of technical coefficients, $\boldsymbol{y}$ is the $m \times 1$ vector of sectoral commodities, and $\boldsymbol{c}$ is that for their final demand. Rearranging terms, net output equals final demand after accounting for intermediate input requirements: $(\boldsymbol{I} - \boldsymbol{\mathcal{A}})\boldsymbol{y} = \boldsymbol{c}$. Assuming a spectral radius $rds\left(\boldsymbol{\mathcal{A}}\right) < 1$, I obtain $\boldsymbol{y} = \left(\boldsymbol{I} - \boldsymbol{\mathcal{A}}\right)^{-1}\boldsymbol{c}$, so that the Leontief inverse – capturing all direct and indirect connections through layers of production chains – is simply $\boldsymbol{\mathcal{L}} = \left(\boldsymbol{I} - \boldsymbol{\mathcal{A}}\right)^{-1}$, delivering $\boldsymbol{y} = \boldsymbol{\mathcal{L}}\,\boldsymbol{c}$. A well-known property is that the Leontief inverse admits the expansion*

$$\boldsymbol{I} + \boldsymbol{\mathcal{A}} + \boldsymbol{\mathcal{A}}^2 + \cdots + \boldsymbol{\mathcal{A}}^k \quad \longleftrightarrow \quad \boldsymbol{\mathcal{L}} = \sum_{k=0}^{\infty} \boldsymbol{\mathcal{A}}^k.$$

with each power-k of $\boldsymbol{\mathcal{A}}$ corresponding to an additional step in the production network (direct inputs, inputs of inputs, and so on). At the sector level it becomes $y_i = \sum_s \ell_{is} c_s$, where $\ell_{is} = \sum_{k=0}^{\infty} \left(\mathcal{A}^k\right)_{is}$ captures all direct and indirect propagation from sector-s to sector-i, so that the output in sector-i depends on adjustments along the production chains starting from sector-s. Define $\boldsymbol{\ell}_i = [\ell_{i1}, \ldots, \ell_{ij}, \ell_{is}, \ldots, \ell_{im}]$. Each term in the summation represents production chains of length-k, so $\boldsymbol{\mathcal{L}} = [\boldsymbol{\ell}_i]$ encodes the full vertical propagation through the production network.

Step 2: Structural network space.*– Each sector-i is associated with its Input-Output profile vector $\boldsymbol{a}_i = [\alpha_{i1}, \ldots, \alpha_{ij}, \alpha_{is}, \ldots, \alpha_{im}]$ which induces a geometric representation of the economy: $\mathcal{E} := \{\boldsymbol{a}_1, \ldots, \boldsymbol{a}_m\}$. The set $\mathcal{E}$ represents the primitive structural "economic" space induced by the rows/columns of matrix $\boldsymbol{\mathcal{A}}$, where each sector is characterized by its full vector of Input-Output interactions. In other words, the Input-Output matrix $\boldsymbol{\mathcal{A}}$ defines a latent geometric space where sectors are embedded according to their observed Input-Output structure. This representation is purely structural: the collection $\mathcal{E} = \{\boldsymbol{a}_i\}_{i=1}^m$ fully characterizes the matrix $\boldsymbol{\mathcal{A}}$.*

Step 3: Input-Output decomposition.*– Starting from $y_i = \sum_j \alpha_{ij} y_j + c_i$, and substituting the Leontief equilibrium $y_j = \sum_s \ell_{js} c_s$, I obtain $y_i = \sum_s \left(\sum_j \alpha_{ij}\, \ell_{js}\right) c_s + c_i$. Define the propagation operator $\mathcal{P}_{is} \equiv \sum_j \alpha_{ij}\, \ell_{js} = \left(\boldsymbol{\mathcal{A}\mathcal{L}}\right)_{is}$. Using $\boldsymbol{\mathcal{L}} = (\boldsymbol{I} - \boldsymbol{\mathcal{A}})^{-1}$, it follows that $\boldsymbol{\mathcal{A}\mathcal{L}} = \boldsymbol{\mathcal{L}} - \boldsymbol{I}$, so that $\mathcal{P}_{is} = \ell_{is} - I_{is}$. Hence, the equilibrium $y_i = \sum_s \mathcal{P}_{is} c_s + c_i$ can be written as $y_i = \sum_s \ell_{is} c_s$, where ℓ_{is} captures the total effect of adjustments in sector-s on sector-i, including all direct and indirect production linkages.*[I] *To understand the structure of this propagation, recall that each coefficient admits the path-wise expansion $\ell_{js} = \sum_{k=0}^{\infty} \left(A^k\right)_{js}$, where each term corresponds to production paths of length-k. Expanding the Neumann series, $\mathcal{P}_{is}$ can be decomposed as*

$$\mathcal{P}_{is} = \sum_{k=0}^{\infty} \sum_j \alpha_{ij} \left(A^k\right)_{js} = \alpha_{is} + \sum_j \alpha_{ij} \alpha_{js} + \sum_{k \geq 2} \sum_j \alpha_{ij} \left(A^k\right)_{js}$$

The zero-order term yields $\sum_j \alpha_{ij} I_{js} = \alpha_{is}$ and captures direct linkages. Differently, the first-order term reveals a key structural feature: different source sectors are compared through their existing connection with the same set of intermediate sectors-j. This provides a geometric interpretation of the Input-Output matrix: its shared-connections decomposition implies that sectors can be represented by their profiles over intermediaries, inducing a notion of horizontal similarity based on overlap in Input-Output linkages rather than sequential propagation. Consider two propagation paths originating from nodes $\{s, m\}$ and ending at a common destination-i, $\mathcal{P}_{is} = \sum_j \alpha_{ij} \ell_{js}$ and $\mathcal{P}_{im} = \sum_j \alpha_{ij} \ell_{jm}$. Both expressions are decomposed over the same set of intermediate sectors-j, using identical local Input-Output coefficients α_{ij}. This common structure implies that $\{s, m\}$ are represented through their simultaneous connection to a shared intermediate inputs space. The Input-Output matrix $\boldsymbol{\mathcal{A}}$ therefore induces not only a propagation mechanism over k-stages, but also a geometric structure over sectors via their similar profiles over intermediaries.[II] *Such overlap is naturally captured by bilinear forms over the rows (or columns) of the Input-Output matrix.*

[I] Note that the term c_i is now embedded in the identity component of the Leontief inverse. In fact, such term is inside ℓ_{is} since it now includes I_{is}, so the component $+c_i$ is absorbed.

[II] The same decomposition can define a dual notion of horizontal similarity. In fact, fixing dual destinations, say $\{i, n\}$, yields a dual geometry. Still, the Input-Output matrix induces multiple geometric structures depending on the direction of comparison, all arising from the same decomposition over intermediaries-j.

Lemma *1 **(Gram row-representation of Input-Output profiles)** Let $\mathcal{A} \in \mathbb{R}^{m\times m}$ be an Input-Output matrix, and let $\boldsymbol{a}_i$ denote its i-th row. Then the matrix $\mathcal{A}\mathcal{A}^\top$ satisfies $\langle \boldsymbol{a}_i, \boldsymbol{a}_s\rangle$. In particular, it is symmetric positive semidefinite and defines a Gram matrix over the set of Input-Output profiles $\mathcal{E} = \{\boldsymbol{a}_i\}_{i=1}^m$. Hence, the matrix $\mathcal{A}$ canonically induces a geometric structure over sectors through its rows.*

Proof: by definition of matrix multiplication, the (i,s)-th entry of $\mathcal{A}\mathcal{A}^\top$ is given by $\left(\mathcal{A}\mathcal{A}^\top\right)_{is} = \sum_{j=1}^m \alpha_{ij}\alpha_{sj}$. Now observe that the right-hand side coincides with the standard inner product between the row vectors $\boldsymbol{a}_i$ and $\boldsymbol{a}_s$, namely $\langle \boldsymbol{a}_i, \boldsymbol{a}_s\rangle = \sum_{j=1}^m \alpha_{ij}\alpha_{sj}$. Therefore, $\left(\mathcal{A}\mathcal{A}^\top\right)_{is} = \langle \boldsymbol{a}_i, \boldsymbol{a}_s\rangle$. This shows that $\mathcal{A}\mathcal{A}^\top$ is the Gram matrix associated with the set of vectors $\mathcal{E} = \{\boldsymbol{a}_i\}_{i=1}^m$.

By symmetry, Lemma 1 holds when considering columns. The first-order term is[III]

$$\sum_j \alpha_{ij}\alpha_{sj} = \left(\mathcal{A}\mathcal{A}^\top\right)_{is} \quad \textit{or symmetrically for columns} \quad \sum_j \alpha_{ji}\alpha_{js} = \left(\mathcal{A}^\top\mathcal{A}\right)_{is}$$

These two objects arise from distinct algebraic operations on the same primitive matrix $\mathcal{A}$ whose characterization, by the discussion so far, results in: (i) a dynamic composition, capturing propagation along network paths; and (ii) a bilinear aggregation, capturing similarity across nodes in the space of Input-Output profiles. While the former generates vertical propagation effects through higher-order linkages, the latter bilinear term depends only on the latent overlap of Input-Output connections and therefore defines a horizontal notion for the network structure distinct from higher-order propagation effects. In particular, $\sum_j \alpha_{ij}\alpha_{sj}$ measures similarity in input structures (common upstream suppliers), while $\sum_j \alpha_{ji}\alpha_{js}$ captures similarity in output structures (common downstream buyers). These bilinear terms induce an implicit geometric structure over sectors which complements, rather than coincides with/dominates on, the higher-order propagation structure ($k \geq 2$), capturing sequential multi-step transmission along production chains. Overall, the Input-Output system admits a decomposition into direct linkages, first-order structural overlap, and higher-order propagation effects, all generated by the same underlying network matrix $\mathcal{A}$.

□

(Proof of demand-distance matrix) *Consider the economy in Section 3. The combination of optimal demand for both labour and intermediate inputs would yield*

$$x\left(s,s'\right) = \alpha\left(s,s'\right) w\left(s\right) n\left(s\right)\, \alpha\left(s\right)^{-1} \frac{1}{p\left(s'\right)}$$

Then, consider the case in which two sectors $\{s,s'\}$ are buying their own intermediate inputs from the same sector-s^*. Associated system of equations is

[III] The term $\sum_j \alpha_{ij}\alpha_{js}$ corresponds to the (i,s)-entry of $\mathcal{A}^2$ and captures sequential propagation through an intermediate sector-j. In contrast, $\sum_j \alpha_{ij}\alpha_{sj}$ corresponds to the (i,s)-entry of $\mathcal{A}\mathcal{A}^\top$ and measures overlap between the Input-Output profiles of sectors $\{i,s\}$. These two expressions therefore reflect distinct operations: compositional propagation versus bilinear similarity, both generated by the same primitive matrix $\mathcal{A}$.

$$\begin{cases} x(s,s^*) = \alpha(s,s^*)\, w(s)\, n(s)\ \alpha(s)^{-1} \dfrac{1}{p(s^*)} \\ x(s',s^*) = \alpha(s',s^*)\, w(s')\, n(s')\ \alpha(s')^{-1} \dfrac{1}{p(s^*)} \end{cases}$$

$$\longrightarrow \quad \begin{cases} x(s,s^*) = \alpha(s,s^*) \left[x(s',s^*) \dfrac{1}{\alpha(s',s^*)\, n(s')} \alpha\, p(s^*) \right] n(s)\ \alpha^{-1} \dfrac{1}{p(s^*)} \\ w = x(s',s^*) \dfrac{1}{\alpha(s',s^*)\, n(s')} \alpha\, p(s^*) \end{cases}$$

where I exploit the fact that wage is the numeraire of the economy, so that $w(s) = w$, $\forall s \in \Phi$, and, without any loss of generality (to simplify notation and build better intuition), $\alpha(s) = \alpha$, $\forall s \in \Phi$ as well. The above substituted equation will result in

$$\frac{x(s,s^*)}{x(s',s^*)} = \frac{\alpha(s,s^*)}{\alpha(s',s^*)} \frac{n(s)}{n(s')} \quad \equiv \quad \frac{n(s')}{x(s',s^*)} = \frac{\alpha(s,s^*)}{\alpha(s',s^*)} \frac{n(s)}{x(s,s^*)} \tag{A.1}$$

Now, note that solving the representative firm's problem with log-quantities will report exactly the same optimality conditions. Henceforth, it is not necessary to take logs in total log-differentiation. In other words, consider the following expression, in which both $n(\cdot)$ and $x(\cdot)$ are already expressed in logarithmic form:

$$\frac{\log n(s')}{\log x(s',s^*)} = \frac{\alpha(s,s^*)}{\alpha(s',s^*)} \frac{\log n(s)}{\log x(s,s^*)}$$

Let me define the following for notational convenience: $N(s) := \log n(s)$, $X(s,s^*) := \log x(s,s^*)$, and $d_{s^*\to}[s,s'] := \frac{\alpha(s,s^*)}{\alpha(s',s^*)}$. The expression becomes

$$\frac{N(s')}{X(s',s^*)} = d_{s^*\to}[s,s'] \frac{N(s)}{X(s,s^*)}$$

I proceed by performing a first-order Taylor expansion around the steady state. Let $\overline{N}(s)$ and $\overline{X}(s,s^*)$ denote the steady-state values of $N(s)$ and $X(s,s^*)$, respectively, and let $d\,N(s)$ and $d\,X(s,s^*)$ denote their log-deviations. Using the total differential of a ratio, it is possible to obtain

$$d\left(\frac{N(s)}{X(s,s^*)}\right) = \frac{1}{\overline{X}(s,s^*)}\, d\,N(s) - \frac{\overline{N}(s)}{\overline{X}^2(s,s^*)}\, d\,X(s,s^*)$$

and similarly for sector-s', $d\left(\frac{N(s')}{X(s',s^*)}\right) = \frac{1}{\overline{X}(s',s^*)}\, d\,N(s') - \frac{\overline{N}(s')}{\overline{X}^2(s',s^*)}\, d\,X(s',s^*)$. Since $d_{s^*\to}[s,s']$ is a constant parameter, it is possible to differentiate both sides of the original equation, $\frac{N(s')}{X(s',s^*)} = d_{s^*\to}[s,s'] \frac{N(s)}{X(s,s^*)}$:

$$d\left(\frac{N(s')}{X(s',s^*)}\right)=d_{s^*\to}[s,s']\,d\left(\frac{N(s)}{X(s,s^*)}\right)$$

Inserting differentials, I derive the following first-order log-linearized expression

$$\frac{1}{\overline{X}(s',s^*)}d\,N(s')-\frac{\overline{N}(s')}{\overline{X}^2(s',s^*)}d\,X(s',s^*)=d_{s^*\to}[s,s']\left(\frac{1}{\overline{X}(s,s^*)}d\,N(s)-\frac{\overline{N}(s)}{\overline{X}^2(s,s^*)}d\,X(s,s^*)\right)$$

which, given $d_{s^*\to}[s,s']=\frac{\alpha(s,s^*)}{\alpha(s',s^*)}$, will deliver the two-sector condition in Subsection 2.2. This result illustrates that the log-linearized response of the ratio between logged employment and logged intermediate inputs in sector-s' is proportional to that of sector-s, with the scaling factor given by their relative intensity of intermediate input usage, $\alpha(s,s^*)\,/\,\alpha(s',s^*)$. Steady-state values $\overline{N}(\cdot)$ and $\overline{X}(\cdot,s^*)$ cannot be avoided in this log-linearization due to the non-linear nature of the ratio between logarithms. Stacked across all sectors, the above condition yields the desired matrix.

□

(Proof of supply-distance matrix) *Consider the economy in Section 3. The combination of optimal demand for both labour and intermediate inputs would yield*

$$x(s,s')=\alpha(s,s')\,w(s)\,n(s)\;\alpha(s)^{-1}\frac{1}{p(s')}$$

Then, consider the case in which two sectors $\{s,s'\}$ are selling their own intermediate inputs to the same sector-s^*. Associated system of equations is

$$\begin{cases} x(s^*,s)=\alpha(s^*,s)\,w(s^*)\,n(s^*)\;\alpha(s^*)^{-1}\dfrac{1}{p(s)} \\ x(s^*,s')=\alpha(s^*,s')\,w(s^*)\,n(s^*)\;\alpha(s^*)^{-1}\dfrac{1}{p(s')} \end{cases}$$

$$\longrightarrow\quad \begin{cases} x(s^*,s)=\alpha(s^*,s)\left[x(s^*,s')\dfrac{1}{\alpha(s^*,s')\,n(s^*)}\alpha(s^*)\;p(s')\right]n(s^*)\;\alpha(s^*)^{-1}\dfrac{1}{p(s)} \\ w(s^*)=x(s^*,s')\dfrac{1}{\alpha(s^*,s')\,n(s^*)}\alpha(s^*)\;p(s') \end{cases}$$

The above substituted equation will result in

$$\frac{x(s^*,s)}{x(s^*,s')}=\frac{\alpha(s^*,s)}{\alpha(s^*,s')}\frac{p(s')}{p(s)} \tag{A.2}$$

which, if log-linearized around its steady state (note that solving the representative firm's problem with log-quantities will report exactly the same optimality conditions; thus, not necessary to take logs in total log-differentiation), closely following the proof for demand-based matrix, delivers the two-sector condition in Subsection 2.2 which, stacked across all sectors, delivers the supply-based distance matrix.

For completeness, consider the following expression

$$\frac{\log x\left(s^{*},s\right)}{\log x\left(s^{*},s'\right)}=\frac{\alpha\left(s^{*},s\right)}{\alpha\left(s^{*},s'\right)}\frac{\log p\left(s'\right)}{\log p\left(s\right)}$$

and define the following for notational convenience: $P\left(s\right):=\log p\left(s\right)$, $X\left(s^{*},s\right):=\log x\left(s^{*},s\right)$, *and* $d_{\rightarrow s^{*}}\left[s,s'\right]:=\frac{\alpha\left(s^{*},s\right)}{\alpha\left(s^{*},s'\right)}$. *Thus, the expression becomes*

$$\frac{X\left(s^{*},s\right)}{X\left(s^{*},s'\right)}=d_{\rightarrow s^{*}}\left[s,s'\right]\frac{P\left(s'\right)}{P\left(s\right)}$$

Proceed by performing a first-order Taylor expansion around the steady state. Let $\overline{X}\left(s^{*},s\right)$ *and* $\overline{P}\left(s\right)$ *denote the steady-state values of* $X\left(s^{*},s\right)$ *and* $P\left(s\right)$, *respectively, and let* $d\,X\left(s\right)$ *and* $d\,P\left(s\right)$ *to denote their log-deviations. Totally differentiating the ratio:*

$$d\left(\frac{X\left(s^{*},s\right)}{X\left(s^{*},s'\right)}\right)=\frac{1}{\overline{X}\left(s^{*},s'\right)}d\,X\left(s^{*},s\right)-\frac{\overline{X}\left(s^{*},s\right)}{\overline{X}\left(s^{*},s'\right)^{2}}d\,X\left(s^{*},s'\right)$$

In the same manner the right-hand side can be manipulated, $d\left(d\left(s^{*}\,|\,s,s'\right)\frac{P\left(s'\right)}{P\left(s\right)}\right)=d\left(s^{*}\,|\,s,s'\right)\left(\frac{1}{\overline{P}\left(s\right)}d\,P\left(s'\right)-\frac{\overline{P}\left(s'\right)}{\overline{P}^{2}\left(s\right)}d\,P\left(s\right)\right)$, *since* $d_{\rightarrow s^{*}}\left[s,s'\right]$ *is not changing. Differentiating both sides of the original equation,* $\frac{X\left(s^{*},s\right)}{X\left(s^{*},s'\right)}=d_{\rightarrow s^{*}}\left[s,s'\right]\frac{P\left(s'\right)}{P\left(s\right)}$, *one obtains*

$$\frac{1}{\overline{X}\left(s^{*},s'\right)}d\,X\left(s^{*},s\right)-\frac{\overline{X}\left(s^{*},s\right)}{\overline{X}\left(s^{*},s'\right)^{2}}d\,X\left(s^{*},s'\right)=d_{\rightarrow s^{*}}\left[s,s'\right]\left(\frac{1}{\overline{P}\left(s\right)}d\,P\left(s'\right)-\frac{\overline{P}\left(s'\right)}{\overline{P}\left(s\right)^{2}}d\,P\left(s\right)\right)$$

This result demonstrates how the log-linearized relationship between price and quantity ratios in sectors $\{s,s'\}$ *depends on their relative intensity* $d_{\rightarrow s^{*}}\left[s,s'\right]=\frac{\alpha\left(s^{*},s\right)}{\alpha\left(s^{*},s'\right)}$. *Similar to the previous proof, the steady-state values* $\overline{X}\left(s^{*},\cdot\right)$ *and* $\overline{P}\left(\cdot\right)$ *appear due to the non-linear nature (i.e., the ratio of logarithms) of the initial expression.*

□

(Proof of Theorem 2) *Let* Φ *denote the finite set of sectors in the economy. Each sector can buy intermediate inputs from other sectors and, simultaneously, can sell its output to other sectors. Define this Input-Output bilateral trade intensity as*

$$\alpha_{ij}:\Phi\times\Phi\rightarrow\left[0,\infty\right),\quad \textit{the amount of inputs purchased by } i \textit{ from } j$$

Express then each fixed coefficient α*'s in terms of share of sectoral total purchases or sales – i.e.,* $\alpha_{ij}^{fd}=\frac{\alpha_{ij}}{\sum_{j\in row}\alpha_{ij}}$ *and* $\alpha_{ij}^{fs}=\frac{\alpha_{ij}}{\sum_{j\in column}\alpha_{ij}}$;[IV] *these values consider not only "who trades with whom" but how much in relative terms.*

Define the upstream (buying intermediate input from) and downstream (selling output to) sets of neighbours of sector-s *as*

$$\Phi_{s}^{up}:=\left\{i\in\Phi:\alpha_{si}^{fs}>0\right\}\quad \textit{and}\quad \Phi_{s}^{dw}:=\left\{i\in\Phi:\alpha_{is}^{fd}>0\right\}$$

[IV] Row-normalized values show each sector's sales share to other sectors, highlighting how similarly sectors source from common upstream suppliers; analogously, column-normalized values show each sector's input share from other sectors, highlighting how similarly sectors sell to common downstream buyers. For further references, refer to Appendix B.

Simplify temporarily with $\Phi = \{i, j, s\}$: the upstream set points for the share of purchases of the common sector-s from sectors $\{i, j\}$, and the downstream set represents the share of sales of the common sector-s to sectors $\{i, j\}$. These are the same information in the Input-Output matrix, but now fully abstract. For any pair of sectors $\{i, j\} \in \Phi$, let the set of common neighbours be

$$\Phi_{i,j}^{com} := \left\{ s \in \Phi : s \in \Phi_i^{up} \cap \Phi_j^{up} \quad or \quad s \in \Phi_i^{dw} \cap \Phi_j^{dw} \right\}$$

for any common sector-s. These common neighbours are the sectors that can horizontally connect two already-connected or otherwise unconnected sectors vertically. Indeed, define the horizontal linkage (or network "economic" distance) metrics

$$d_{s\to}\left[i, j\right] := f\left(\alpha_{is}^{fd}, \alpha_{js}^{fd}\right) \quad and \quad d_{\to s}\left[i, j\right] := f\left(\alpha_{si}^{fs}, \alpha_{sj}^{fs}\right)$$

for $s \in \Phi_{i,j}^{com}$. Relative to sectors in brackets, subscript "$s \to$" reads as "purchasing from sector-s", while subscript "$\to s$" reads as "selling to sector-s". Define further the vectors of factor input demand and supply distances for the pair $\{i, j\}$ as

$$d^{fd}\left[i, j\right] := \left\{ d_{s\to}\left[i, j\right] : s \in \Phi_{i,j}^{com} \right\} \quad and \quad d^{fs}\left[i, j\right] := \left\{ d_{\to s}\left[i, j\right] : s \in \Phi_{i,j}^{com} \right\}$$

Part 1: horizontal linkage exists even if two sectors are not connected. *Assume $\alpha_{ij}^{fd,fs} = 0$ and $\alpha_{ji}^{fd,fs} = 0$. If there exists $s \in \Phi$ such that $s \in \Phi_i^{up} \cap \Phi_j^{up}$ or $s \in \Phi_i^{dw} \cap \Phi_j^{dw}$, then*

$$d^{fd}\left[i, j\right] > 0 \;\; and \;\; d^{fs}\left[i, j\right] > 0$$

quantify the horizontal linkages between $\{i, j\}$ in common upstream demand or downstream supply, respectively. Therefore, these sectors are horizontally connected through a common neighbour sector-s: even though $\{i, j\}$ don't trade directly within the production network, the fact that they both interact with s makes them economically connected; this is the "horizontal complementarity" concept.

Part 2: factor input demand and supply distances are generally different. *Let $\{i, j\} \in \Phi$ and let $s \in \Phi_{i,j}^{com}$ be a common neighbour. By definition, the network "economic" distance metrics are $d_{s\to}\left[i, j\right]$ and $d_{\to s}\left[i, j\right]$. Unless the trade is symmetric (next part of the proof), that is when $\alpha_{is}^{fd} = \alpha_{si}^{fs}$ $\forall i \in \Phi$, then $d_{s\to}\left[i, j\right] \neq d_{\to s}\left[i, j\right]$. Extending to all common neighbours $s \in \Phi_{i,j}^{com}$, it follows that*

$$d^{fd}\left[i, j\right] := \left\{ d_{s\to}\left[i, j\right] : s \in \Phi_{i,j}^{com} \right\} \quad \neq \quad d^{fs}\left[i, j\right] := \left\{ d_{\to s}\left[i, j\right] : s \in \Phi_{i,j}^{com} \right\}$$

The network "economic" distance between two sectors depends on whether one looks at it from the perspective of demand or supply between one sector and the common one, and they usually don't match; these two sets of numbers are different because buying and selling patterns are usually not symmetric. Hence, the same Input-Output structure always generates two distinct factor input demand and factor input supply distance metrics unless trade linkages with common sectors are perfectly symmetric.

Part 3: symmetric interactions imply coinciding distances for at least one common neighbour. *Suppose that for some common sector-s, the trade is perfectly symmetric: i buys the same relative amount from s as it sells to s, and same for j. If it exists a common $s \in \Phi_{i,j}^{com}$ such that $\alpha_{is}^{fd} = \alpha_{si}^{fs}$ and $\alpha_{js}^{fd} = \alpha_{sj}^{fs}$, with these being of different values, then*

$$d_{s\rightarrow}[i,j] := f\left(\alpha_{is}^{fd}, \alpha_{js}^{fd}\right) \quad = \quad f\left(\alpha_{si}^{fs}, \alpha_{sj}^{fs}\right) =: d_{\rightarrow s}[i,j]$$

and therefore $d^{fd}[i,j] \cap d^{fs}[i,j] \neq \emptyset$.[V] *In this case, the distance metrics are exactly equal: demand-based distance is the same of the supply-based distance for that neighbour – or distance vectors $\forall s \in \Phi$. If two sectors trade symmetrically with a common sector, then their economic distance looks the same whether it is measured by demand or supply. The building intuition is that symmetry in relative trade "aligns" the two perspectives of network "economic" distance, which would otherwise differ.*

Finally, it is left to prove the minimum number of sectors required for such coinciding distances to exist. Suppose the economy has m sectors, $\Phi = \{i, j, s, \ldots, m\}$, and consider a sector-$s$ acting as the common neighbour. To construct a pair of sectors $\{i, j\}$ whose factor input demand and supply distances coincide via s, it is required: (i) the common neighbour s itself, and (ii) at least two distinct sectors i and j that trade with s symmetrically. Now, compare two network sizes.

Three-sector network.*– Let the sectors be $\{i, j, s\} \in \Phi$ with $s \in \Phi_{i,j}^{com}$, so that the minimum number of sectors is $m = 2 + 1 = 3$. Under trade symmetry, I have already claimed that the two distance metrics coincide: $d_{s\rightarrow}[i,j] = d_{\rightarrow s}[i,j]$. But a general condition also asks for an additional symmetric entry between another pair of sectors (distinct from the pair linked by s). With only three sectors there is no distinct third pair available that is disjoint from $\{i, j\}$ and uses a different sector: the only other possible symmetric entries would involve the same sectors $\{i, j, s\}$ as well. Therefore, a case with $m = 3$ does not provide an independent additional symmetric entry: any symmetric pair must involve these three sectors and cannot be both an extra independent symmetric pair and keep the structure non-degenerate. Concretely, with $m = 3$ one may obtain a single coinciding distance metric via a common neighbour, but it is not general enough to satisfy the hypothesis requiring a separate additional symmetric entry independent of the same three nodes. Thus, a three-sector network is insufficient to guarantee a general statement of the theorem as formulated.*

Bigger network.*– To allow the existence of an additional symmetric entry between another pair of sectors, at least one more sector is required. Let $\{i, j, s, \ldots, n\} \in \Phi$ with $s \in \Phi_{i,j}^{com}$, and with at least one extra sector-n. Therefore, the absolute minimum network size to guarantee that there exists at least one coinciding distance vector is $m \geq 4$. Usual trade symmetry delivers $d_{s\rightarrow}[i,j] = d_{\rightarrow s}[i,j]$. Because there is at least one additional sector-n, it is possible also to impose an additional symmetric entry be-*

[V] In other words, if sectors $\{i, j\}$ buy and sell symmetrically with s but at different Input-Output intensities, then the demand and supply network "economic" distances between $\{i, j\}$ are the same.

tween some other pair: this additional symmetric entry can be chosen so it is not the same symmetry used at $\{i,j\}$ and s – i.e., it provides a distinct symmetric pair in the network. Thus, both ingredients required by the general theorem can be satisfied in a non-degenerate way when $m \geq 4$: a common neighbour symmetric with (at least) two sectors, and at least one independent symmetric pair elsewhere. In such a way, the network has enough nodes to realize the general hypothesis (common neighbour symmetric with $m-2$ sectors and a separate symmetric pair), thereby guaranteeing the existence of at least one distance vector that coincides in the constructions of demand and supply distances.

***General formulation**.– Let Φ be the set of sectors, with $|\Phi| = m$, and let the set of common neighbours of sectors i and j to be*

$$\Phi^{com}_{i,j} := \left\{ s \in \Phi \setminus \{i,j\} : s \text{ is a common neighbour of } i \text{ and } j \right\}$$

Define the demand- and supply-based network "economic" distance multi sets as $d^{fd}[i,j] := \left\{ f\left(\alpha^{fd}_{is}, \alpha^{fd}_{js}\right) : s \in \Phi^{com}_{i,j} \right\}$ and, analogously, $d^{fs}[i,j] := \left\{ f\left(\alpha^{fs}_{si}, \alpha^{fs}_{sj}\right) : s \in \Phi^{com}_{i,j} \right\}$. Assume it exists a subset $\Phi^{add}_{i,j} \subseteq \Phi^{com}_{i,j}$ such that all sectors in $\Phi^{add}_{i,j}$ satisfy bilateral symmetry with both $\{i,j\}$: $\alpha^{fd}_{is} = \alpha^{fs}_{si}$ and $\alpha^{fd}_{js} = \alpha^{fs}_{sj}$, $\forall s \in \Phi^{add}_{i,j}$. Then, the set of coinciding distances is

$$\mathcal{C}_{i,j} := d^{fd}[i,j] \cap d^{fs}[i,j] = \left\{ f\left(\alpha^{fd}_{is}, \alpha^{fd}_{js}\right) : s \in \Phi^{add}_{i,j} \right\}$$

Thus, $\mathcal{C}_{i,j} \neq \emptyset$ if $\Phi^{add}_{i,j} \neq \emptyset$: the demand-based and supply-based distance between two sectors $\{i,j\}$ coincide if and only if there is at least one sector $s \in \Phi^{add}_{i,j}$ that both trade with symmetrically. For the general statement of the theorem: if $\left|\Phi^{add}_{i,j}\right| \geq m-2$ and there exists at least one additional symmetric pair $\{i,n\} \in \Phi^2$, then

$$\exists\, i,j \in \Phi : \mathcal{C}_{i,j} \neq \emptyset$$

□

(Proof of Corollary 1 for Theorem 2) *The task is to determine how many coinciding distances should appear when $m \geq 4$. Let*

$$\Phi^{sym}_{|s} = \{i, j, \ldots, m'\} \subset \Phi \setminus \{s\}, \qquad m' = m-2$$

denote the set of sectors that trade symmetrically with the common sector s. Symmetry means that for every $i \in \Phi^{sym}_{|s}$, $\alpha^{fd}(i,s) = \alpha^{fs}(s,i)$, and similarly for any other directional equality required by the construction of $d^{fd}[\cdot]$ and $d^{fs}[\cdot]$. Recall that these equalities must hold after Input-Output matrix normalization (divide by rows for demand distance, by columns for supply distance). Which distances coincide and how many? For any unordered pair $\{i,j\} \subset \Phi^{sym}_{|s}$, the distances defined through the common neighbour s satisfy, as usual, $d_{s\to}[i,j] = d_{\to s}[i,j]$. Thus each unordered pair inside $\Phi^{sym}_{|s}$ generates one coinciding demand- and supply-based network "economic" distance. Now, the idea is to count all pairs of sectors within the set of symmetric neighbours $\Phi^{sym}_{|s}$ that share the common sector-s. Suppose that $\Phi^{sym}_{|s}$ contains

$m' = m - 2$ sectors that trade symmetrically with s. To find the coinciding distances, I consider unordered pairs $\{i, s\} \in \Phi^{sym}_{|s}$ because each pair corresponds to one entry in the distance matrix (which is symmetric). The number of ways to pick two sectors from among the m-sectors is given by the binomial coefficient $\binom{m'}{2} = \frac{m'(m'-1)}{2}$. It counts all distinct pairs of symmetric neighbours: each pair "shares" the same symmetric relation with the common sector, so that their demand- and supply-based network "economic" distances coincide.[VI] *Intuitively, for a common sector symmetric with $m-2$ neighbours, every unordered pair of those neighbours has matching demand- and supply-based distances, giving $\binom{m-2}{2}$ coinciding distance values; the formula comes from counting all distinct pairs among the $m-2$ symmetric sectors.*

□

A.2. Proofs for theoretical results on distance-based propagation

(Proof of Proposition 1) *Consider a buyer sector-i that demands an intermediate input supplied by an upstream sector-s. The buyer's technology follows a CES aggregator with elasticity of substitution $\sigma > 0$. From cost minimization, the log-demand elasticity of the intermediate with respect to its own price is given by $\frac{\partial \log x_{is}}{\partial \log p_s} = -\sigma$. Suppose a shock in sector-j increases demand for the input produced by s. Let the upstream price respond according to $\frac{\partial \log p_s}{\partial \log x_{js}} = \psi^{fd}$. By the chain rule, the effect of the demand shock on sector-i's input demand is then*

$$\frac{\partial \log x_{is}}{\partial \log x_{js}} = \frac{\partial \log x_{is}}{\partial \log p_s} \cdot \frac{\partial \log p_s}{\partial \log x_{js}} = -\left(\sigma \cdot \psi^{fd}\right)$$

Thus, the sign of the comovement depends on the product $\sigma \cdot \psi^{fd}$. If $\sigma \cdot \psi^{fd} > 0$, comovement is reversed. The mechanism operates through two key channels: (i) the price channel, where a shock to sector-j increases demand for inputs from s and, if upstream supply is inelastic, the price p_s rises, so that $\psi^{fd} > 0$; and (ii) the substitution channel, where the buyer sector-i responds by substituting away from the now more expensive input, and the magnitude of this effect depends on the elasticity of substitution σ. When both effects are active (positive ψ^{fd} and sizeable σ), sector-i reduces demand, creating opposite comovement with the shocked sector.

Consider instead a downstream sector-i that buys a finite set of intermediate inputs, bundled into a composite input from a CES aggregator, $x_i = \left(\sum_{s\in\Phi} \alpha_{is}^{\frac{1}{\sigma}} x_{is}^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}$, where x_{is} is the quantity of the intermediate supplied by upstream sector-s to downstream buyer-i, $\alpha_{is} \in (0,1)$ is the network intensity, and $\sigma > 0$ the (constant) elasticity of substitution across intermediate inputs. Define its associated (unit) price index for its composite intermediate by the usual CES formula, $P_i = \left(\sum_{s\in\Phi} \alpha_{is} p_s^{1-\sigma}\right)^{\frac{1}{1-\sigma}}$, where p_s is the price of the intermediate supplied by upstream sector-s.

[VI] If one sector-s is symmetric with exactly $m' = m-2$ other sectors then, for $m = 5$ one gets $\binom{3}{2} = 3$ distinct unordered sector-pairs whose demand- and supply-based distances coincide – equivalently, six equal off-diagonal distance matrix entries, since each pair corresponds to $\{i, j\}$ and $\{j, i\}$. In fact, as $m' = m - 2 = 3$, then $\binom{m'}{2} = \frac{3(3-1)}{2} = \frac{3\times 2}{2} = 3$.

Under cost minimization, for a given level of composite intermediate x_i, the conditional demand for intermediate input-s is given by the standard CES demand system. Define the expenditure share of variety-s in the composite intermediate by $e_{is} = \frac{\alpha_{is} p_s^{1-\sigma}}{\sum_{s\in\Phi} \alpha_{is} p_s^{1-\sigma}}$, with $\sum_{s\in\Phi} e_{is} = 1$. A well-known property of the CES demand system is the (log) price elasticities for conditional demands:

$$\frac{\partial \log x_{is}}{\partial \log p_s} = -\sigma\Big[1 - e_{is}\Big] \quad , \quad \frac{\partial \log x_{is}}{\partial \log p_s} = \sigma\, e_{is^{**}} \quad \forall s^{**} \neq s$$

The former, own-price elasticity depends on the substitution parameter and on the expenditure share, while the latter, cross-price elasticity is proportional to the share of the other intermediate input considered.

Focus on a particular upstream sector-s that supplies an intermediate used by several downstream buyers. Consider an idiosyncratic positive demand shock in some sector-j that increases aggregate demand for the good produced by s. Let x_{js} denote the demand shifter (or demand level) associated with the shocked sector-j from sector-s. The upstream price of the good produced by s responds to the shock according to a (log) pass-through parameter $\psi^{fd} = \frac{\partial \log p_s}{\partial \log x_{js}}$ of demand changes into prices. By the chain rule, and noting that only p_s changes in response to the shock (holding other upstream prices fixed in this partial equilibrium exercise), one obtains

$$\frac{\partial \log x_{is}}{\partial \log x_{js}} = \frac{\partial \log x_{is}}{\partial \log p_s} \cdot \frac{\partial \log p_s}{\partial \log x_{js}} = -\sigma\ \left[1 - e_{is}\right] \cdot \psi^{fd}$$

Three points are worth nothing: (i) the elasticity of substitution, σ, magnifies the response – the larger σ, the stronger buyer-i substitutes away from an input whose price rises; (ii) the expenditure share, e_{is}, attenuates the own-price effect – if buyer-i spends only a small fraction on the input from s, then $\big[1 - e_{is}\big] \approx 1$ and the full $-\sigma$ factor operates, while if the share is large, the own-price sensitivity is mechanically smaller in magnitude; and (iii) the upstream pass-through, ψ^{fd}, transmits the original downstream demand shock into a price change at the supplier – without such pass-through ($\psi^{fd} = 0$) the substitution channel vanishes.

Hence, the sign of the log-response is the sign of $-\sigma\big[1 - e_{is}\big]\psi^{fd}$. Because $\sigma > 0$ and $1 - e_{is} > 0$ (for an interior share), the sign is governed by ψ^{fd}: if $\psi^{fd} > 0$ (upstream price rises) and $\sigma > 0$, buyer-i reduces its demand for s's input, thereby delivering opposite comovement; by contrast, if $\psi^{fd} \approx 0$ (perfectly elastic upstream supply), the price channel is absent and does not generate opposite comovement via this substitution mechanism. Its logic is shown in the upper panel of Figure A.1.

□

(Proof of Proposition 2) *Consider an upstream supplier-i that sells a finite set of intermediate inputs to a downstream sector-s. Let the composite intermediate input be a CES aggregator $x_s = \left(\sum_{i\in\Phi} \alpha_{si}^{\frac{1}{\sigma}} x_{si}^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}$, where x_{si} is the quantity of the intermediate supplied by upstream sector-s to downstream buyer-s, $\alpha_{si} \in (0,1)$ is the network intensity, and $\sigma > 0$ the (constant) elasticity of substitution across inter-*

mediate inputs. Associated is the (unit) price index from the usual CES formula, $P_s = \left(\sum_{i\in\Phi} \alpha_{si}\, p_i^{1-\sigma}\right)^{\frac{1}{1-\sigma}}$, with p_i denoting the price of input from upstream supplier-i.

Under cost minimization, for a given level of composite intermediate x_s, the conditional demand for intermediate input-i is given by the standard CES demand system. Define the expenditure share of variety-i in the composite intermediate by $e_{si} = \frac{\alpha_{si}\, p_s^{1-\sigma}}{\sum_{i\in\Phi} \alpha_{si}\, p_i^{1-\sigma}}$, with $\sum_{i\in\Phi} e_{si} = 1$. A well-known property of the CES demand system is the (log) price elasticities for conditional demands:

$$\frac{\partial \log x_{si}}{\partial \log p_i} = -\sigma\left[1 - e_{si}\right] \quad , \quad \frac{\partial \log x_{si}}{\partial \log p_j} = \sigma\, e_{sj} \quad \forall j \neq s$$

The former, own-factor demand elasticity is decreasing in own price, scaled by substitution elasticity and expenditure share, while in the latter, cross-price elasticities, demand for i rises if another input-j becomes more expensive relative to j's share.

Focus on a particular downstream sector-s that buys an intermediate used by several upstream sellers. Consider an idiosyncratic positive demand shock in some sector-j that increases aggregate supply for the good purchased by s. Let x_{sj} denote the supply shifter (or supply level) associated with the shocked sector-j to sector-s. The upstream price of the good produced by j responds to the shock according to a (log) pass-through parameter $\psi^{fs} = \frac{\partial \log p_j}{\partial \log x_{sj}}$ of supply changes into prices.

By the chain rule, and noting that only p_i changes in response to the shock (holding other upstream prices fixed in this partial equilibrium exercise), obtaining

$$\frac{\partial \log x_{si}}{\partial \log x_{sj}} = \frac{\partial \log x_{si}}{\partial \log p_j}\frac{\partial \log p_j}{\partial \log x_{sj}} = \sigma\, e_{sj} \cdot \psi^{fs}$$

Three transparent points: (i) the elasticity of substitution, σ, magnifies the response – a larger σ amplifies the substitution away from i and toward j when p_j falls; (ii) the expenditure share of the common downstream sector, e_{sj}, attenuates the price effect – the cheaper input j represents a bigger share of s's bundle; and (iii) the downstream pass-through, ψ^{fs}, transmits the original upstream supply shock into a price change at the buyer – without such pass-through ($\psi^{fs} = 0$) the substitution channel vanishes. If the downstream sector s uses only a single input from an upstream supplier (i.e. no possibility of substitutability across inputs), the elasticity of substitution is effectively zero ($\sigma = 0$). In this case, there is no scope for reallocation of expenditures across inputs: a price change in j does not alter the demand for i: $\frac{\partial \log x_{si}}{\partial \log p_j} = 0$. Therefore, unlike the demand-driven case, the single-input supply case does not yield a sufficient statistic, as there is no mechanism generating comovement.

Hence, the sign of the log-response is the sign of $\sigma\, e_{sj}\, \psi^{fs}$. Because $\sigma > 0$ and $e_{sj} > 0$ (for an interior share), the sign is governed by ψ^{fs}: if $\psi^{fs} < 0$, the product $\sigma\, e_{sj}\, \psi^{fs}$ is negative, implying opposite comovement between i and j through their common buyer; by contrast, if $\psi^{fs} \approx 0$ (perfectly elastic downstream demand), the price channel is absent and does not generate opposite comovement via this substitution mechanism. Its logic is shown in the lower panel of Figure A.1.

□

(Proof of Theorem 3) *This part provides a purely intuition-driven mathematical proof of Theorem 3, as it relies on a structural distinction between horizontal and vertical operators on production In this sense, the theorem is proved as a "structural impossibility result". I then provide some elements for the proof.*

Economic intuition as a mathematical principle.– *The economic intuition underlying Theorem 3 can be summarized as follows: horizontal interactions capture network similarity across peers (shared upstream suppliers and downstream buyers), whereas vertical interactions capture production inputs flows along idiosyncratic supply chains. Economically, horizontal effects operate within a layer of the production network, while vertical effects operate across layers. Mathematically, this implies that*

"horizontal effects depend on similarity of local neighbourhoods, whereas vertical effects depend on paths of arbitrary length".

Theorem 3 is therefore a structural statement: operators that measure similarity of neighbourhoods cannot be weighted by operators that aggregate along paths.

Minimal graph-theoretic setup.– *Let $G = (m, \Lambda)$ be a finite directed weighted graph, with indicator-m representing the number of nodes in the network, and $\Lambda = \{i, j \in m \times m : \alpha_{ij} \geq 0\}$ the set of directed edges representing non-negative linkages between nodes. Let $\boldsymbol{\mathcal{A}} = [\alpha_{ij}] \in \mathbb{R}^{m \times m}$ denote the adjacency (Input-Output) matrix of G, with α_{ij} representing the inputs that node-i sources from node-j. When the spectral radius $rds(\boldsymbol{\mathcal{A}}) < 1$, the Leontief inverse $\boldsymbol{\mathcal{L}} := (\boldsymbol{I} - \boldsymbol{\mathcal{A}})^{-1}$ exists. Production is sequential at the k-th round of input requirements, $\boldsymbol{\mathcal{L}} \equiv \sum_{k=0}^{\infty} \boldsymbol{\mathcal{A}}^k$.*

Then, the need is to define what horizontal and vertical operators are. Matrix $\boldsymbol{\mathcal{D}} = [d_{ij}] \in \mathbb{R}^{m \times m}$ is called an "horizontal (distance) operator" if $d_{ij} = f^d(\alpha_{\cdot i}, \alpha_{\cdot j})$, where $\alpha_{\cdot i}$ denotes the immediate neighbourhood of node-i (common upstream suppliers or downstream buyers, that is the entire input shares/sales of node-i), and $f^d(\cdot)$ is a function that depends only on the overlap or similarity of these neighbourhoods (i.e., a fixed similarity or distance functional): horizontal effects are thus defined as finite-profile (column/row) dependence.[VII] *Differently, vertical effects are fundamentally different from profile comparison and involve repeated application of $\boldsymbol{V} = \{\boldsymbol{\mathcal{A}}, \boldsymbol{\mathcal{L}}\}$. Indeed, an operator is called "vertical" if it aggregates information along paths of length greater than or equal to one, as it depends on recursive compositions of the Input-Output matrix – i.e., if it can be written as $\sum_{k \geq 1} \boldsymbol{V}^k A$ for some non-negative matrix A. The network matrix indicator $\boldsymbol{V}$ is the canonical vertical operator: it sums contributions from all (directed and indirected) paths in the network, thereby aggregating the effects of shocks along all production chains of arbitrary length: vertical effects are thus defined as recursive composition of production technologies.*

Locality versus propagation.– *Here is the mathematical statement. In particular, through a series of lemmas, I am going to define the nature of both vertical and horizontal anatomies, and then prove the theorem by contradiction.*

[VII] Horizontal distances therefore depend on *finite-dimensional technological profiles*. They compare how two sectors source inputs (or sell outputs), but they do not involve repeated application of the Input-Output matrix.

Lemma 2 *(Locality of horizontal operators)* *If a certain horizontal entry is $d_{ij} = f^d\left(\alpha_{\cdot i}, \alpha_{\cdot j}\right)$, then horizontal matrix $\boldsymbol{\mathcal{D}}$ depends only on a finite-dimensional representation of $\mathbf{V}$ and does not involve recursive application of $\mathbf{V}$.*

Proof: each entry d_{ij} is constructed from the vectors $\alpha_{\cdot i}$ and $\alpha_{\cdot j}$ through a fixed functional $f^d(\cdot)$. No term involves powers $\mathbf{V}^k$ with $k \geq 2$, nor any infinite series.

Lemma 3 *(Non-locality of the Input-Output weights)* *Any vertical operator involving $\mathbf{V}$ depends on all powers $\mathbf{V}^k$, for $k \geq 0$.*

Proof: by definition, $\mathbf{V} \equiv \sum_{k=0}^{\infty} \mathbf{V}^k$. Each power $\mathbf{V}^k$ represents k-fold composition of technologies. Hence, any object involving $\mathbf{V}$ depends on recursively propagated Input-Output relations.

Lemma 4 *(Incompatibility)* *No local operator contains a global path operator.*

Proof: assume by contradiction that a local operator $\boldsymbol{\mathcal{D}}$ could be written as an equilibrium object $\mathbf{B} = \mathbf{V}'\widetilde{\boldsymbol{\mathcal{D}}}$ for some $\widetilde{\boldsymbol{\mathcal{D}}}$, so that horizontal effects enter the equilibrium multiplied by the Input-Output weights.[VIII] *By contrast, given the non-locality of $\mathbf{V}$, the object $\mathbf{B}$ depends on arbitrarily long production paths on all recursive compositions $\mathbf{V}^k$, besides the local neighbourhood similarity of $\boldsymbol{\mathcal{D}}$. Then, $\mathbf{B}$ would inherit the non-local dependence of $\mathbf{V}$, contradicting the locality established above: by the incompatibility lemma, such a representation is impossible.*

A re-written version of the incompatibility lemma delivers Theorem 3: horizontal distance operators cannot be multiplicatively weighted by the Input-Output weights.

Characterizing intuition.– *Turning to a conceptual part of the proof, it is essential to recall that the horizontal dimension of the network, as captured by its network "economic" distances, is not confined to sectors already connected – directly or indirectly – through inter-sectoral trade. Rather, it also encompasses sectors without such direct or indirect linkages, which are nonetheless related through similar demand or supply relationships arising from common upstream suppliers or downstream buyers. This reasoning implies that certain sectors may exhibit a strictly positive distance linkage despite the absence of any Input-Output connection. In other words, considers two sectors, say $\{s, s'\}$, with zero (direct and Leontief inverse) inter-sectoral trade intensity – $\alpha(s, s') = \alpha(s', s) = 0$ and $\ell(s, s') = \ell(s', s) = 0$ –, held together by their demand/supply relationship with a common sector, say s^*, so that $d_{s^* \to}[s', s''] = d_{\to s^*}[s', s''] > 0$. In this case, the Input-Output matrix and the distance matrix would be given by $\boldsymbol{\mathcal{A}} \equiv \boldsymbol{\mathcal{L}} = \begin{bmatrix} \cdot & 0 \\ 0 & \cdot \end{bmatrix}$ and $\boldsymbol{\mathcal{D}}^j = \begin{bmatrix} \cdot & > 0 \\ > 0 & \cdot \end{bmatrix}$, for $j = \{fd, fs\}$. Accordingly, multiplying the distance matrix by the production network matrix would effectively neutralise the role of distances. This rationale underpins Theorem 3.*

$\square$

[VIII] By construction, each entry of $\boldsymbol{\mathcal{D}}$ depends only on finite-dimensional technological profiles $\{\alpha_{\cdot i}, \alpha_{\cdot j}\}$. Therefore, $\mathbf{V}'\widetilde{\boldsymbol{\mathcal{D}}}$ depends on recursive compositions of production technologies, while $\widetilde{\boldsymbol{\mathcal{D}}}$ itself depends only on static comparisons of production technologies. However, horizontal effects arise from substitutability across technologies, not from iterated production chains: there is no economic or algebraic mechanism that generates higher-order compositions of $\mathbf{V}$ inside $f^d(\cdot)$. As a result, $\mathbf{V}'\widetilde{\boldsymbol{\mathcal{D}}}$ cannot arise from the equilibrium derivation.

(Corollary to Theorem 3) *To make the result transparent to a broader theory, I now reformulate the "structural impossibility" argument to prove Theorem 3 using: (i) the language of Spectral Graph theory; and (ii) Green's functions / resolvent operators.*

Spectral Graph theory interpretation. *– In Spectral Graph theory, powers of the adjacency matrix $\boldsymbol{V}^k$ encode walks of length-$k \in [0, \infty)$ on the network, and is therefore a global spectral object sensitive to the full eigenvalue distribution of $\boldsymbol{V}$. Horizontal distance operators, by contrast, are kernel operators defined on the space of columns (or rows) of $\boldsymbol{V}$. They depend only on first-order technological profiles and do not involve eigenvalue amplification or long-run diffusion. Theorem 3 therefore states that a kernel on the column / row space of $\boldsymbol{V}$ cannot be composed multiplicatively with the resolvent of $\boldsymbol{V}$ without changing its mathematical nature.*

Green's functions and resolvents. *– The Input-Output indicator $\boldsymbol{V}$ is the Green's function (or resolvent) associated to a linear operator. It solves propagation problems of the form $\boldsymbol{V}'y = x$, by transmitting shocks through all production chains. An horizontal operator $\boldsymbol{\mathcal{D}}$ is not a Green's function. They do not solve propagation problems and do not correspond to inverses of linear operators. Instead, they define a metric or similarity structure on the space of technologies. From this perspective, Theorem 3 asserts that a metric kernel on technologies (horizontal geometry) cannot appear inside a Green's function propagator (i.e., a vertical production dynamics). The two objects live in distinct mathematical categories: one is a similarity kernel, the other a resolvent of a linear operator.*

$\square$

(Proof of Theorem 4) *Let $G = (m, \Lambda)$ be a finite directed weighted graph, with indicator-m representing the number of nodes in the network, and $\Lambda = \{i, j \in m \times m : \alpha_{ij} \geq 0\}$ the set of directed edges representing non-negative linkages between nodes. Let $\boldsymbol{\mathcal{A}} = [\alpha_{ij}] \in \mathbb{R}^{m \times m}$ denote the adjacency (Input-Output) matrix of G, with α_{ij} representing the inputs that node-i sources from node-j. Decompose as $\boldsymbol{\mathcal{A}}_m = \boldsymbol{\mathcal{V}}_m \cup \boldsymbol{\mathcal{D}}_m$, where $\boldsymbol{\mathcal{V}}_m$ encodes vertical Input-Output relations, that is the vertical supplier-buyer linkages, and $\boldsymbol{\mathcal{D}}_m$ encodes horizontal complementarities, meaning pairwise similarity between sectors based on shared suppliers or buyers, or shared upstream / downstream network structures. The decomposition isolates the vertical and horizontal components, in accordance with Theorem 3.*

To characterize the asymptotic structure of the production network and analyse large networks, I embed the decomposed Input-Output matrix – $\boldsymbol{\mathcal{A}}_m$, $\boldsymbol{\mathcal{V}}_m$, and $\boldsymbol{\mathcal{D}}_m$ – into graphon spaces – $\boldsymbol{\mathcal{W}}_{\boldsymbol{\mathcal{A}}_m}$, $\boldsymbol{\mathcal{W}}_{\boldsymbol{\mathcal{V}}_m}$, and $\boldsymbol{\mathcal{W}}_{\boldsymbol{\mathcal{D}}_m}$ –, representing the global geometry of a large graph, where networks are represented by functions rather than finite matrices. For each $m \times m$ matrix $\boldsymbol{\mathcal{A}}_m$ its associated graphon $\boldsymbol{\mathcal{W}}_{\boldsymbol{\mathcal{A}}_m}$ is a step-function on a measurable space $U : [0, 1]^2$. Graphon convergence is measured in the "cut norm" $\|U\|_{\square}$, identified with subscript-$\square$, a way to capture global structural differences between graphs (i.e., convergence metric: how the network structure behaves in the limit). In fact, graphons allow to capture the "limiting geometry" of very large networks: if $\boldsymbol{\mathcal{W}}_{\boldsymbol{\mathcal{A}}_m} \to \boldsymbol{\mathcal{W}}$, then the global structure stabilizes to a continuous object $\boldsymbol{\mathcal{W}}$. The cut norm controls the *"spectral norm" (i.e., propagation metric: how big the shock propagation can be), the latter identified with subscript-2, up to a universal constant (i.e., the Lovász-Szegedy inequality):* $\|\boldsymbol{\mathcal{A}}_m\|_2 \leq \sigma \|\boldsymbol{\mathcal{W}}_{\boldsymbol{\mathcal{A}}_m}\|_\square$ *for any specified constant* σ*: if a network is sparse in structure, then it cannot amplify shocks much.*

Assume the following. (1) vertical sparsity: supply chains do not scale with network size-m*, and each node has a certain number of suppliers/buyers; the vertical graphon becomes negligible in the limit,* $\|\boldsymbol{\mathcal{W}}_{\boldsymbol{\mathcal{V}}_m}\|_\square \xrightarrow[m\to\infty]{} 0$*, implying* $\|\boldsymbol{\mathcal{V}}_m\|_2 \xrightarrow[m\to\infty]{} 0$*; (2) convergence of horizontal complementarities: network "economic" distances converge to a non-degenerate (non-zero) graphon, as they remain dense and globally relevant in the limit; in particular, there exists a symmetric graphon* $\boldsymbol{\mathcal{W}} : [0,1]^2 \to [0,1]$ *such that* $\|\boldsymbol{\mathcal{W}}_{\boldsymbol{\mathcal{D}}_m} - \boldsymbol{\mathcal{W}}\|_\square \xrightarrow[m\to\infty]{} 0$ *and* $\|\boldsymbol{\mathcal{W}}\|_\square > 0$*, which implies* $\liminf_m \|\boldsymbol{\mathcal{D}}_m\|_2 > 0$*; this is the dominant network structure in the limit, as it creates the non-trivial eigenvalues responsible for persistent (and large) shock propagation; and (3) vector of bounded shocks:* $\|\boldsymbol{\varepsilon}_m\|_\infty \in [0,\infty)$*, which ensures the consistency in scaling the results.*

Finally, define the shock's propagation operators. Given the $m \times m$ *Input-Output matrix* $\boldsymbol{\mathcal{A}}_m$*, define* $\boldsymbol{\mathcal{P}}_m = f^{\mathcal{P}}(\boldsymbol{\mathcal{A}}_m)$ *as the operator that gives the total propagation of an initial idiosyncratic, independent and asymmetric shock to a certain node transmitting to its (direct and indirect) network connections. Similarly, by separating vertical and horizontal channels of propagation, define* $\boldsymbol{\mathcal{P}}_m^{(\mathcal{V})} = f^{\mathcal{P}}(\boldsymbol{\mathcal{V}}_m)$ *and* $\boldsymbol{\mathcal{P}}_m^{(\mathcal{D})} = f^{\mathcal{P}}(\boldsymbol{\mathcal{D}}_m)$*: the former measures propagation due solely to vertical buyer-supplier network ties; the latter transmits shocks through horizontal complementarities.* $f^{\mathcal{P}}(\cdot)$ *is a linear operator, continuous in the norm of its argument: for* $f^{\mathcal{P}}(\cdot) : \mathbb{R}^{m\times m} \to \mathbb{R}^{m\times m}$*, then* $\|a - b\|_2 \to 0$ *implies* $\|f^{\mathcal{P}}(a) - f^{\mathcal{P}}(b)\|_2 \to 0$*. This assumption ensures that whenever* $\boldsymbol{\mathcal{A}}_m, \boldsymbol{\mathcal{V}}_m, \boldsymbol{\mathcal{D}}_m \to 0$ *in spectral norm, the corresponding propagator also vanishes.*

Step 1: Convergence of the full network geometry*.– Given the coexistence of vertical and horizontal geometries, the graphon for the total Input-Output matrix* $\boldsymbol{\mathcal{A}}_m$ *is* $\boldsymbol{\mathcal{W}}_{A_m} = \boldsymbol{\mathcal{W}}_{V_m} \cup \boldsymbol{\mathcal{W}}_{H_m}$*. It follows that*

$$\|\boldsymbol{\mathcal{W}}_{\boldsymbol{\mathcal{A}}_m} - \boldsymbol{\mathcal{W}}\|_\square \leq \|\boldsymbol{\mathcal{W}}_{\boldsymbol{\mathcal{V}}_m}\|_\square \cup \|\boldsymbol{\mathcal{W}}_{\boldsymbol{\mathcal{D}}_m} - \boldsymbol{\mathcal{W}}\|_\square$$

For $m \to \infty$ *vertical part disappears,* $\|\boldsymbol{\mathcal{W}}_{\boldsymbol{\mathcal{V}}_m}\|_\square \xrightarrow[m\to\infty]{} 0$*, and the horizontal part converges,* $\|\boldsymbol{\mathcal{W}}_{\boldsymbol{\mathcal{D}}_m} - \boldsymbol{\mathcal{W}}\|_\square \xrightarrow[m\to\infty]{} 0$*. Thus, the whole network converges,* $\boldsymbol{\mathcal{W}}_{\boldsymbol{\mathcal{A}}_m} \xrightarrow[m\to\infty]{} \boldsymbol{\mathcal{W}}$*: The large-scale geometry of production networks is asymptotically determined entirely by horizontal complementarities.*

Step 2: Vertical propagation vanishes*.– As vertical sparsity implies* $\|\boldsymbol{\mathcal{V}}_m\|_2 \to 0$ *for* $m \to \infty$*, and since the transmission operator is continuous in its norm,* $\boldsymbol{\mathcal{P}}_m^{(\mathcal{V})} = f^{\mathcal{P}}(\boldsymbol{\mathcal{V}}_m)$*, then* $\left\|\boldsymbol{\mathcal{P}}_m^{(\mathcal{V})}\right\|_2 = \|f^{\mathcal{P}}(\boldsymbol{\mathcal{V}}_m)\|_2 \to \|f^{\mathcal{P}}(0)\|_2 = 0$*. Since* $\boldsymbol{\mathcal{P}}_m^{(\mathcal{V})} = f^{\mathcal{P}}(\boldsymbol{\mathcal{V}}_m) \to f^{\mathcal{P}}(0) = 0$*, continuity implies* $\left\|\boldsymbol{\mathcal{P}}_m^{(\mathcal{V})}\boldsymbol{\varepsilon}_m\right\|_2 \xrightarrow[m\to\infty]{} 0$*. Vertical spillovers die out because vertical links are too sparse to amplify anything when* n *is large enough, that is when vertical connections are too sparse to support long-range propagation.*

***Step 3: Horizontal propagation persists**.– As the graphon $\mathcal{W}$ has positive mass: (i) the horizontal network is asymptotically connected; (ii) the expected weighted degree converges to a strictly positive constant; and (iii) spectral norm is bounded away from zero, $\liminf_m \|\mathcal{D}_m\|_2 > 0$. Since $\liminf_m \|\mathcal{D}_m\|_2 > 0$, and due to operator's continuity, then $\liminf_m \left\|\mathcal{P}_m^{(\mathcal{D})}\, \boldsymbol{\varepsilon}_m\right\|_2 > 0$. Thus, the horizontal geometry amplifies the shock as the network enlarges, so that $\left\|\mathcal{P}_m^{(\mathcal{D})}\, \boldsymbol{\varepsilon}_m\right\|_2 > 0$. Horizontal complementarities form a dense backbone through which node-specific shocks propagate as $m \to \infty$.*

***Step 4: Ratio dominance**.– Since $\left\|\mathcal{P}_m^{(\mathcal{V})}\, \boldsymbol{\varepsilon}_m\right\|_2 \to 0$ and $\left\|\mathcal{P}_m^{(\mathcal{D})}\, \boldsymbol{\varepsilon}_m\right\|_2 > 0$ for $m \to \infty$, then $\frac{\left\|\mathcal{P}_m^{(\mathcal{V})}\, \boldsymbol{\varepsilon}_m\right\|_2}{\left\|\mathcal{P}_m^{(\mathcal{D})}\, \boldsymbol{\varepsilon}_m\right\|_2} \xrightarrow[m\to\infty]{} 0$. Vertical spillovers disappear relative to the horizontal.*

***Characterizing intuition**.– In large economies with static production networks, the number of direct upstream suppliers or downstream buyers per nodes does not scale with the network size-m. Thus, vertical propagation channel become negligible in the limit and vertical shock propagation cannot scale: vertical effects remain local and thin, thereby explaining the vanishing vertical transmission mechanism. In contrast, network participants tend to share connections with many others, especially when the number of nodes grows (i.e., that is for larger and larger networks). Shared buyers/suppliers become more likely, horizontal similarity becomes dense, and horizontal complementarities increase with network size-m. Thus, horizontal effects remain global and strong, thereby resulting in a global structure that persists in the limit. As a result, in very large production networks, horizontal effects tend to dominate the transmission of shocks, whereas vertical chain effects become negligible. While vertical propagation fades away, horizontal effects persist: large Input-Output economies are dominated by network similarity effects, not supply chain effects.*

□

(Proof of Corollary 2 for Theorem 4) *Let $\mathcal{W}_{\mathcal{V}_m}$ and $\mathcal{W}_{\mathcal{D}_m}$ denote the graphons associated with the vertical and horizontal components of the network, respectively. Shock propagation along these two is given by the linear operators $\mathcal{P}_m^{(\mathcal{V})}$ and $\mathcal{P}_m^{(\mathcal{D})}$.*

By assumption, $\|\mathcal{W}_{\mathcal{V}_m}\|_\square$ is non-decreasing in network size-m, and horizontal complementarities must satisfy $\|\mathcal{W}_{\mathcal{D}_m}\|_\square \geq a\ \|\mathcal{W}_{\mathcal{V}_m}\|_\square$ far a given $a > 0$. Even if vertical linkages increase with network size-m, the horizontal complementarities induced by shared suppliers or buyers grow faster, because each new vertical relation creates new horizontal connections among nodes that share Input-Output partners. Henceforth:

$$\|\mathcal{W}_{\mathcal{V}_m}\|_\square \Big/ \|\mathcal{W}_{\mathcal{D}_m}\|_\square \leq \frac{1}{a} \tag{T3.1}$$

The Lovasz-Szegedy inequality provides universal constants $0 < b \leq c < \infty$, that is $\left\|\mathcal{P}_m^{(\mathcal{V})}\right\|_2 \leq c\,\|\mathcal{W}_{\mathcal{V}_m}\|_\square$ and $\left\|\mathcal{P}_m^{(\mathcal{D})}\right\|_2 \geq b\,\|\mathcal{W}_{\mathcal{D}_m}\|_\square$. When dividing these inequalities, $\frac{\left\|\mathcal{P}_m^{(\mathcal{V})}\right\|_2}{\left\|\mathcal{P}_m^{(\mathcal{D})}\right\|_2} \leq \frac{c}{b}\frac{\|\mathcal{W}_{\mathcal{V}_m}\|_\square}{\|\mathcal{W}_{\mathcal{D}_m}\|_\square}$. Plugging in eq. (T3.1) yields the uniform bound

$$\left\|\mathcal{P}_m^{(\mathcal{V})}\right\|_2 \Big/ \left\|\mathcal{P}_m^{(\mathcal{D})}\right\|_2 \leq \frac{c}{b}\frac{1}{a} \tag{T3.2}$$

Let the shock vector $\boldsymbol{\varepsilon}_m \in \mathbb{R}^m$ *be uniformly bounded, so that* $\|\boldsymbol{\varepsilon}_m\|_2 \leq a_\varepsilon$ *for some constant* $a_\varepsilon > 0$ *independent of network size-*m*. By submultiplicativity of the operator norm,* $\left\|\mathcal{P}_m^{(\mathcal{V})}\boldsymbol{\varepsilon}_m\right\|_2 \leq \left\|\mathcal{P}_m^{(\mathcal{V})}\right\|_2 \times \|\boldsymbol{\varepsilon}_m\|_2 \leq a_\varepsilon \left\|\mathcal{P}_m^{(\mathcal{V})}\right\|_2$*. Similarly,* $\left\|\mathcal{P}_m^{(\mathcal{D})}\boldsymbol{\varepsilon}_m\right\|_2 \leq a_\varepsilon \left\|\mathcal{P}_m^{(\mathcal{D})}\right\|_2$*. Hence, for all network size-*m*,* $\frac{\left\|\mathcal{P}_m^{(\mathcal{V})}\boldsymbol{\varepsilon}_m\right\|_2}{\left\|\mathcal{P}_m^{(\mathcal{D})}\boldsymbol{\varepsilon}_m\right\|_2} \leq \frac{\left\|\mathcal{P}_m^{(\mathcal{V})}\right\|_2}{\left\|\mathcal{P}_m^{(\mathcal{D})}\right\|_2}$*. By construction, as a network with size-*m *grows, the horizontal component becomes strictly denser than the vertical one, so the denominator grows faster than the numerator. If combined with eq. (T3.2) I obtain a uniform,* m*-independent, strictly finite upper bound. Hence* $\left\|\mathcal{P}_m^{(\mathcal{V})}\boldsymbol{\varepsilon}_m\right\|_2 \Big/ \left\|\mathcal{P}_m^{(\mathcal{D})}\boldsymbol{\varepsilon}_m\right\|_2 \xrightarrow[m\to\infty]{} 0$ *still holds. This establishes that, even under endogenous expansion of the vertical component, the horizontal dimension remains asymptotically dominant in determining the total propagation of shocks. So, in large networks, a growth in vertical edges produces a related growth in horizontal complementarities: the dominance of horizontal transmission survives when vertical linkages grow endogenously. In other words, even if network participants form more vertical linkages as the economy grows, the combinatorial explosion of shared upstream suppliers and downstream buyers ensures that horizontal complementarities still become the overwhelmingly dominant channel of shock propagation.*

□

(Proof of Theorem 5) *I want to formalize the role of horizontal propagation in generating aggregate output fluctuations in the absence of central or dominant sectors.*

Let $G = (m, \Lambda)$ *be a finite directed weighted graph, with indicator-*m *representing the number of nodes in the network, and* $\Lambda = \{i, j \in m \times m : \alpha_{ij} \geq 0\}$ *the set of directed edges representing non-negative linkages between nodes. Let* $\boldsymbol{\mathcal{A}} = [\alpha_{ij}] \in \mathbb{R}^{m\times m}$ *denote the adjacency (Input-Output) matrix of* G*, with* α_{ij} *representing the inputs that node-*i *sources from node-*j*. When the spectral radius* $rds(\boldsymbol{\mathcal{A}}) < 1$*, the Leontief inverse* $\boldsymbol{\mathcal{L}} := (\boldsymbol{I} - \boldsymbol{\mathcal{A}})^{-1}$ *exists. Production is sequential at the* k*-th round of input requirements,* $\boldsymbol{\mathcal{L}} \equiv \sum_{k=0}^{\infty} \boldsymbol{\mathcal{A}}^k$*.*

Let each sector potentially experiences a an asymmetric, idiosyncratic and independent productivity shock. Let $\mathbb{P}$ *be a probability space. Define the productivity-related random variable* $\{\varepsilon_1, \ldots, \varepsilon_i, \varepsilon_j \ldots, \varepsilon_m\} : \mathbb{P} \to \mathbb{R}$ *such that*

$$\mathbb{E}[\varepsilon_i] = 0 \quad , \quad \mathbb{E}\left[\varepsilon_i^2\right] = \sigma^2 < \infty \quad , \quad \mathbb{E}[\varepsilon_i\,\varepsilon_j] = 0 \quad \text{for} \quad i \neq j$$

The vector of micro-level shocks be $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_m)'$*. The covariance matrix of shocks is therefore* $\sigma^2\boldsymbol{I}$*. In this sense, all correlations in equilibrium outcomes must come from the network, not from correlated shocks.*

Let $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_m)' \in \mathbb{R}^m$ *be the sectoral Domar weights, satisfying* $\lambda_i > 0$ *with* $\sum_{i=1}^m \lambda_i = 1$*. Define aggregate log-output as* $Y := \boldsymbol{\lambda}'\,\boldsymbol{\mathcal{L}}\,\boldsymbol{\varepsilon}$ *or, equivalently,*

$$Y = \sum_{i=1}^m v_i\,\varepsilon_i \qquad \boldsymbol{v} := \boldsymbol{\mathcal{L}}'\,\boldsymbol{\lambda}$$

where v_i measures the general equilibrium importance of sector-i. This is the standard Acemoglu et al. (2012) representation: aggregate output is a weighted sum of sectoral production levels after vertical network propagation system.

Given this configuration, it is then possible to define the variance of aggregate output from micro-level determinants under classical supply chain amplification: independent and idiosyncratic shocks generate correlated sectoral outputs through Leontief inverse-based vertical chains.

LEMMA 5 *(Vertical variance of aggregate output)* *In an Input-Output economy, aggregate volatility depends on how concentrated the network-adjusted weights are*

$$var\,(Y) = \sigma^2\, \boldsymbol{\lambda}'\, \boldsymbol{\mathcal{L}}\, \boldsymbol{\mathcal{L}}'\, \boldsymbol{\lambda}$$

Proof: central idea is to compute the baseline aggregate output variance, ignoring horizontal linkages (i.e., network "economic" distances) between vertexes, to show the contribution of vertical chains alone. By definition, $Y = \sum_{i=1}^{m} v_i \varepsilon_i$, with variance $var\,(Y) = \mathbb{E}\,\left[\left(\sum_{i=1}^{m} v_i\, \varepsilon_i\right)^2\right]$. Expanding the square,

$$\begin{aligned} var\,(Y) &= \mathbb{E}\left[\sum_{i=1}^{m} v_i^2 \varepsilon_i^2 + \sum_{i \neq j} v_i v_j \varepsilon_i \varepsilon_j\right] && \text{where, using independence, then} \\ &= \sum_{i=1}^{m} v_i^2 \mathbb{E}\left[\varepsilon_i^2\right] + \sum_{i \neq j} v_i v_j \mathbb{E}\left[\varepsilon_i \varepsilon_j\right] && \text{Since the cross-terms vanish, then} \\ &= \sigma^2 \sum_{i=1}^{m} v_i^2 \end{aligned}$$

Since $\boldsymbol{v} = \boldsymbol{\mathcal{L}}'\, \boldsymbol{\lambda}$, then $\sum_{i=1}^{m} v_i^2 = \|\boldsymbol{v}\|^2 = \boldsymbol{\lambda}'\, \boldsymbol{\mathcal{L}}\, \boldsymbol{\mathcal{L}}'\, \boldsymbol{\lambda}$. Aggregate volatility is the quadratic form induced by the propagation of micro-level shocks along vertical supply chains.

It is now required to formalize the horizontal propagation channel. Let, for each sector-i, the upstream set (i.e., the suppliers of sector-i) and the "upstream (demand-based) similarity" be defined by

$$\Phi_i^{up} = \{m \in \Phi : \alpha_{im} > 0\} \quad , \quad d_{ij}^{fd} = \frac{\left|\Phi_i^{up} \cap \Phi_j^{up}\right|}{\left|\Phi_i^{up} \cup \Phi_j^{up}\right|}$$

If $\{i, j\}$ buy from the same upstream sectors, then $d_{ij}^{fd} > 0$. This measures shared supplier exposure. Analogously let, for each sector-i, the downstream set (i.e., the buyers of sector-i) and the "downstream (supply-based) similarity" be defined by

$$\Phi_i^{dw} = \{m \in \Phi : \alpha_{mi} > 0\} \quad , \quad d_{ij}^{fs} = \frac{\left|\Phi_i^{dw} \cap \Phi_j^{dw}\right|}{\left|\Phi_i^{dw} \cup \Phi_j^{dw}\right|}$$

If $\{i, j\}$ sell to the same downstream sectors, then $d_{ij}^{fs} > 0$. This measures shared buyer exposure. Combined horizontal similarity is then $\boldsymbol{\mathcal{D}} = \boldsymbol{\mathcal{D}}^{fd} \cup \boldsymbol{\mathcal{D}}^{fs}$. This matrix captures the full horizontal propagation logic. Recall that, by Theorem 2, matrix $\boldsymbol{\mathcal{D}}^{j}$, for $j = \{fd, fs\}$, is symmetric by construction, so is $\boldsymbol{\mathcal{D}}$. Horizontal linkages generate correlation between sectors sharing suppliers or buyers, independently of vertical propagation. Entry $d_{ij} > 0$ means that shocks hitting sector-i partially affect sector-j through common upstream suppliers or downstream buyers, independently of vertical *propagation mechanism. The stronger the overlap, the stronger the induced correlation. This implies that even small sectors can transmit shocks horizontally if they share key upstream or downstream partners.*

Let the total propagation operator to be defined by

$$\boldsymbol{\mathcal{P}} := \boldsymbol{\mathcal{L}}\boldsymbol{\mathcal{L}}' \cup \boldsymbol{\mathcal{D}}$$

which is the total covariance matrix reflecting the union of vertical and horizontal effects. Given the vertical propagation, $\boldsymbol{\mathcal{L}}\boldsymbol{\mathcal{L}}'$, and the horizontal propagation, $\boldsymbol{\mathcal{D}}$, the union operator $\cup$ implies that each channel contributes independently to aggregate variance, with no weighting or multiplicative interaction.[IX] *Separated structure makes the proof sharper, because horizontal propagation does not need vertical propagation to generate aggregate fluctuations (no dependence of horizontal propagation on vertical chains). In terms of economic intuition, aggregate fluctuations are driven by the superposition of two independent channels: shocks propagate both vertically along chains and horizontally via shared Input-Output connections.*

LEMMA 6 *(Variance of aggregate output)* *In an Input-Output economy, aggregate volatility depends on the combination of vertical and horizontal propagation*

$$var\,(Y) = \sigma^2\,\boldsymbol{\lambda}'\,\boldsymbol{\mathcal{P}}\,\boldsymbol{\lambda}$$

Proof: substitute the total amplification operator into Lemma 5.

Let me, finally, impose some assumptions.

ASSUMPTION 2 *(Bounded vertical amplification)* *There exists $\chi > 0$ such that for all network size-m and all $\boldsymbol{x} \in \mathbb{R}^m$, $\boldsymbol{x}'\boldsymbol{\mathcal{L}}\,\boldsymbol{\mathcal{L}}'\boldsymbol{x} \geq \chi\|\boldsymbol{x}\|^2$.*

It guarantees that vertical propagation is finite and non-degenerate. Element $\boldsymbol{x}$ is an arbitrary real vector of dimension equal to the number of sectors (think of it as a hypothetical pattern of sectoral exposure or weights, that is a subset of sectors, a weighted combination of sector outputs, a direction along which shocks are aggregated, …).[X] *No matter how to aggregate sectors, vertical propagation does not collapse that aggregation to zero. In other words, the Input-Output network is not degenerate: there are no directions in which shocks completely die out, and vertical propagation has full rank and strength. This assumption is not required for the horizontal-only result and is included for completeness.*

ASSUMPTION 3 *(Diffuse Domar weights)* *There exist constants $q, r > 0$ such that, for all network size-m and every sector-i, $\frac{q}{m} \leq \lambda_i \leq \frac{r}{m}$.*

[IX] By Theorem 3, horizontal propagation is not weighted by vertical Input-Output weights. Henceforth, an operator of the form $\boldsymbol{\mathcal{P}} := \boldsymbol{\mathcal{D}} \circ (\boldsymbol{\mathcal{L}}\boldsymbol{\mathcal{L}}')$, denoting the Hadamard (entry-wise) product, is not consistent. Moreover, this would imply that strong (weak) vertical propagation generates stronger (weaker) horizontal covariance, thereby pushing towards the same direction.

[X] Later in the proof I will impose that $x_i = \lambda_i$ is the GDP-weight of sector-i, representing the aggregate exposure of its horizontal linkages.

It ensures that no sector dominates fluctuations in aggregate GDP, thereby sharpening the analysis of aggregate fluctuations in large economies with no superstar sectors. In particular, it rules out granularity since $\lambda_i \sim \frac{1}{m}$. *If aggregate volatility survives, it is not because one sector/node is big or dominates; rather, it is purely a network phenomenon.*

ASSUMPTION 4 ***(Macroscopic horizontal cluster)*** *There exists a finite set* $\Psi \subseteq \{1, \ldots, m\}$ *with cardinality* $|\Psi|$ *such that, for the fraction-*η *of sectors belonging to the horizontal cluster: (i)* $\frac{|\Psi|}{m} \to \eta > 0$*; and (ii)* $d_{ij} \geq \underline{d} > 0$, $\forall\, i, j \in \Psi$.

It ensures that a positive fraction of sectors are strongly connected horizontally, sufficient to generate non-vanishing variance. The first condition imposes that the cluster of network participants connected horizontally does not vanish as the economy grows (by Theorem 4). If $\eta = 0$, *then the cluster becomes negligible, its contribution to aggregate output vanishes, and the horizontal propagation cannot generate aggregate fluctuations. Henceforth,* $\eta > 0$ *ensures macroeconomic relevance, and determines the order of magnitude of the horizontal cluster's aggregate weight (more later on). Moreover,* η *is conceptually different from "large or important sectors" (e.g., Acemoglu et al. 2012). The latter own a large* λ_i*, generating a concentration of GDP. Differently, the horizontal clusters may imply small* λ_i *but a large number of sectors. Aggregate effect comes from extensive margin, not intensive margin. The second condition imposes positive horizontal connections (by Theorem 2).*

Step 0: Aggregate variance via vertical and horizontal propagation*.– Recall that, in Lemma 5, I compute baseline variance ignoring horizontal links, thereby determining the contribution of vertical supply chains alone to aggregate output,* $var\,(Y) = \sigma^2\, \boldsymbol{\lambda}'\, \boldsymbol{\mathcal{L}}\, \boldsymbol{\mathcal{L}}'\, \boldsymbol{\lambda}$*: classical supply chain amplification, where independent shocks generate correlated sectoral outputs through Leontief chains. In Lemma 6, I introduce additional variance from sectors sharing upstream suppliers or downstream buyers, thereby capturing the key mechanism of horizontal propagation independent of vertical chains,* $var\,(Y) = \sigma^2\, \boldsymbol{\lambda}'\, \boldsymbol{\mathcal{P}}\, \boldsymbol{\lambda}$. *These two independent channels coexist, and horizontal linkages generate aggregate variance regardless of vertical amplification.*

Step 1: Focus on horizontal cluster*.– Isolate network participants with strong horizontal linkages (that is that substantially connected horizontally with the others):*

$$x_i = \begin{cases} \lambda_i, & i \in \Psi \\ 0, & i \notin \Psi \end{cases} \quad \textit{thus implying} \quad \boldsymbol{\lambda}'\, \boldsymbol{\mathcal{P}}\, \boldsymbol{\lambda} \geq \boldsymbol{x}'\, \boldsymbol{\mathcal{P}}\, \boldsymbol{x}$$

Only nodes in the cluster contribute substantially to aggregate variance; other sectors' contributions are smaller. This allows a lower bound on variance.

Step 2: Bound contribution of the cluster*.– In what follows I am going to focus only on the horizontal propagation, as dropping the component* $\boldsymbol{\mathcal{L}}\boldsymbol{\mathcal{L}}'$ *in* $\boldsymbol{\mathcal{P}}$ *gives a lower bound to fluctuations in aggregate output (i.e., considering only the sources* *of aggregate cycles from the horizontal geometry of a network; even if vertical chains were shut down, shared suppliers/buyers still induce correlation). From this, and expanding the previous step, then*

$$\boldsymbol{x}' \boldsymbol{\mathcal{D}}\, \boldsymbol{x} = \sum_{i,j \in \Psi} \lambda_i \, \lambda_j \, d_{ij}$$

which, by the cluster assumption ($d_{ij} \geq \underline{d}$),

$$\sum_{i,j \in \Psi} \lambda_i \, \lambda_j \, d_{ij} \geq \underline{d} \sum_{i,j \in \Psi} \lambda_i \, \lambda_j \quad \textit{with the latter being} \quad \underline{d} \left(\sum_{i \in \Psi} \lambda_i \right)^2$$

Strong horizontal linkages induce correlated shocks within the cluster, thereby ensuring non-vanishing variance. Each pair of sectors co-moves, and aggregate correlation scales with the square of total weight.

***Step 3: Bound cluster weight**.– I now bound λ_i, $\forall i \in \Psi$, using the diffuse weights assumption, $\lambda_i \geq \frac{q}{m}$. So, $\sum_{i \in \Psi} \lambda_i \geq |\Psi| \times \frac{q}{m}$. From the cluster assumption,*

$$\left(\sum_{i \in \Psi} \lambda_i \right)^2 \geq \left(|\Psi| \, \frac{q}{m} \right)^2 \to \left(\eta \, q \right)^2 > 0$$

A positive fraction of the economy is in the horizontal cluster: aggregate contribution from horizontal linkages persists as $m \to \infty$. In fact, even if each sector is small, together they matter since the fact that many of them are horizontally connected induces substantial horizontal effect.

***Step 4: Combine bounds**.– When combining everything, it holds that*

$$var\,(Y) \geq \sigma^2 \, \underline{d} \left(\eta \, q \right)^2 \qquad \Longrightarrow \qquad \liminf \; var\,(Y) > 0$$

Horizontal propagation alone generates persistent aggregate fluctuations even if no sector is dominant. Vertical propagation is not required for this effect. The aggregate variance from horizontal effects is strictly positive, independent of the network size-m, does not rely on large sectors, and does not rely on vertical propagation. Aggregate volatility is positive when horizontal clusters scale proportionally with the size of the economy ($\Psi \propto m$), that is $\liminf_{m \to \infty} var\,(Y_m) > 0$, as also resulting from Theorem 4.

What the theorem states is that horizontal linkages (i.e., network "economic" distances) creates systematic correlation across many vertexes. Aggregate volatility potentially emerges without shocks to most central sectors, and the horizontal network geometry alone is sufficient. This is a new origin of aggregate fluctuations, distinct from: granular shocks to certain nodes, vertical chains, and superstar sectors/firms.

□

(Remark to Theorem 5) *Clarification on the definition of the covariance operator and the role of horizontal propagation. When proving Theorem 5, Appendix A.2, I refer to a covariance operator $\boldsymbol{\mathcal{P}}$ as the "union" of vertical and horizontal propagation mechanisms, denoted $\boldsymbol{\mathcal{P}} := \boldsymbol{\mathcal{L}}\boldsymbol{\mathcal{L}}' \cup \boldsymbol{\mathcal{D}}$. Since the operator $\cup$ is not standard in linear* *algebra, I clarify its precise meaning and its role in the proof.*

The purpose of introducing $\boldsymbol{\mathcal{P}}$ is not to impose a specific functional form for how vertical and horizontal propagation interact, but rather to formalize the coexistence of two distinct channels (by Theorem 3) through which sectoral shocks may generate aggregate comovement. In particular, my results rely exclusively on lower-bound arguments for aggregate variance. Accordingly, it is sufficient to define $\boldsymbol{\mathcal{P}}$ as any symmetric positive semidefinite (PSD) matrix satisfying the partial order conditions $\boldsymbol{\mathcal{P}} \succeq \boldsymbol{\mathcal{D}}$ and $\boldsymbol{\mathcal{P}} \succeq \boldsymbol{\mathcal{L}}\boldsymbol{\mathcal{L}}'$, where $\succeq$ denotes the Loewner order on symmetric matrices.

Equivalently, for any vector of Domar weights $\boldsymbol{\lambda}$, this definition ensures $\boldsymbol{\lambda}'\boldsymbol{\mathcal{P}}\boldsymbol{\lambda} \geq \boldsymbol{\lambda}'\boldsymbol{\mathcal{D}}\boldsymbol{\lambda}$ and $\boldsymbol{\lambda}'\boldsymbol{\mathcal{P}}\boldsymbol{\lambda} \geq \boldsymbol{\lambda}'\boldsymbol{\mathcal{L}}\boldsymbol{\mathcal{L}}'\boldsymbol{\lambda}$. This definition formalizes the notion that vertical and horizontal propagation channels coexist without one mechanically weighting or scaling the other, and it is sufficient for establishing lower bounds on aggregate volatility. Importantly, no assumption is made that the two channels combine additively; the argument only requires that the contribution of horizontal propagation is not attenuated by the vertical structure of production.

For expositional simplicity, one may alternatively define $\boldsymbol{\mathcal{P}} := \boldsymbol{\mathcal{L}}\boldsymbol{\mathcal{L}}' + \boldsymbol{\mathcal{D}}$, which corresponds to the special case in which the two channels contribute additively to output covariance. This formulation is mathematically standard and yields identical lower-bound results. However, the more general definition above is preferred in order to remain consistent with the earlier result that vertical and horizontal propagation coexist but do not weight each other.

Finally, the proof makes use of the inequality $\boldsymbol{x}'\boldsymbol{\mathcal{D}}\boldsymbol{x} \geq 0$ for a vector $\boldsymbol{x}$ supported on a horizontally connected cluster of sectors. While similarity matrices such as those based on shared suppliers or buyers (e.g., Jaccard-type measures) are not positive semidefinite in general, positive semi-definiteness is not required globally. The argument only relies on the restriction of $\boldsymbol{\mathcal{D}}$ to a subset Ψ of sectors forming a horizontal cluster, on which $\boldsymbol{\mathcal{D}}$ is symmetric and element-wise positive. Since the quadratic form is evaluated on vectors supported on Ψ, non-negativity follows directly. This restricted use of $\boldsymbol{\mathcal{D}}$ is sufficient to establish the lower bound on aggregate volatility driven by horizontal propagation.

$\square$

(Proof of Corollary 3 for Theorem 5) *Closely following the logic of Theorem 5, allowing for heterogeneous Domar weights does not alter its conclusion.*

Step 0: Horizontal cluster and variance representation. *– From the proof of Theorem 5, the variance of aggregate output can be bounded from below using the quadratic form restricted to the horizontally connected cluster Ψ:*

$$var\,(Y) \geq \sigma^2\, \boldsymbol{x}'\boldsymbol{\mathcal{D}}\boldsymbol{x}, \qquad with \quad x_i = \begin{cases} \lambda_i, & i \in \Psi \\ 0, & i \notin \Psi \end{cases}$$

Here, Ψ satisfies $|\Psi|/m \to \eta > 0$ and $d_{ij} \geq \underline{d} > 0$ for all $i, j \in \Psi$. This step isolates the horizontal propagation, ignoring vertical chains.

***Step 1: Lower bound on horizontal contribution**.– By construction of $\boldsymbol{x}$ and the properties of $\boldsymbol{\mathcal{D}}$ on Ψ,*

$$\boldsymbol{x}'\boldsymbol{\mathcal{D}}\boldsymbol{x} = \sum_{i,j\in\Psi} \lambda_i\lambda_j d_{ij} \;\geq\; \underline{d} \sum_{i,j\in\Psi} \lambda_i\lambda_j = \underline{d}\left(\sum_{i\in\Psi}\lambda_i\right)^2$$

In Theorem 5, Domar weights are not allowed to be heterogeneous and may include large sectors, $\lambda_i \approx 1/m$, so $\sum_{i\in\Psi}\lambda_i \sim |\Psi|/m$. When allowing for granular Domar weights, some λ_i would be larger than others. If a shock hits one of these sectors, then $\sum_{i\in\Psi}\lambda_i \geq \bar{\lambda} \gg |\Psi|/m$. Horizontal propagation is amplified proportionally to the total weight in the cluster: large sectors make the quadratic form bigger, thereby increasing aggregate variance.

***Step 2: Combine bounds**.– Combining the previous inequalities yields*

$$var\,(Y) \geq \sigma^2\boldsymbol{x}'\boldsymbol{\mathcal{D}}\boldsymbol{x} \geq \sigma^2\,\underline{d}\,\bar{\lambda}^2 > 0$$

Compared to the diffuse Domar weight case (i.e., the original theorem), the presence of large sectors increases $\bar{\lambda}$, hence amplifying the aggregate effect of the horizontal geometry. Horizontal propagation generates non-vanishing aggregate fluctuations in all cases, with granularity magnifying this effect. In other words, horizontal propagation generates non-vanishing aggregate output fluctuations even when Domar weights are heterogeneous and some sectors are large.

□

(Proof of Corollary 4 for Theorem 5) *Consider the economy described in proving Theorem 5, made of $\{1,\ldots,i,j,\ldots,m\} \in \Phi$ sectors, characterized by micro-level shocks, $\boldsymbol{\varepsilon} = (\varepsilon_1,\ldots,\varepsilon_m)'$, and sectoral Domar weights, $\boldsymbol{\lambda} = (\lambda_1,\ldots,\lambda_m)' \in \mathbb{R}^m$.*

***Step 1: Variance decomposition**.– The standard variance decomposition of aggregate output can be written as*

$$var\,(Y) = \sum_{i=1}^{m}\lambda_i^2\; var\,(y_i) + \sum_{i\neq j}\lambda_i\lambda_j\; cov\,(y_i, y_j)$$

This separates two sources of aggregate volatility. The first term captures the contribution of individual sectoral volatility, as sectors fluctuate independently one another: this is the standard diversification channel. The second term is where the network structure matters, capturing comovement as sectors move in response. Hence, any vertical or horizontal effects from the network affects the covariance term.

***Step 2: Contribution of the individual variance terms**.– Since $\lambda_i \asymp \frac{1}{m}$, according to the standard diversification aggregate variance $var\,(y_i)$ is bounded:*

$$\sum_{i=1}^{m}\lambda_i^2\; var\,(y_i) \asymp \frac{1}{m}$$

This corresponds to the standard Lucas (1977) diversification argument: when av-

eraging an asymmetric, idiosyncratic and independent sectoral shock across many independent sectors, $m \to \infty$, aggregate volatility goes to zero. So, without any network-based effects, the GDP fluctuations are smoother the larger is the number of sectors. The next steps provide the baseline effects such that this benchmark is break.

***Step 3: Horizontal propagation and covariance structure**.– Let $|\Psi|$ denote the size of the set Ψ as horizontal cluster, defined as the number of sectors whose productions respond jointly to a given shock due to shared Input-Output structures. Within each cluster, sectoral responses are positively correlated, $cov\left(y_i, y_j\right) > 0$, for sectors $\{i, j\}$ belonging to the same cluster. Across clusters, sectoral responses are approximately independent. The total number of clusters in the economy is therefore approximately $\frac{m}{|\Psi|}$, and each cluster contains roughly $|\Psi|^2$ ordered pairs of correlated sectors, so the total number of correlated pairs in the economy is of order $m\ |\Psi|$.*

***Step 4: Contribution of covariance terms**.– Each covariance term is weighted by $\lambda_i\, \lambda_j$ which, from the diffuse Domar weight's assumption, satisfies $\lambda_i\, \lambda_j \asymp \frac{1}{m^2}$. This is the weight of each pair in the economy. Multiplying it by the number of correlated pairs, $m\ |\Psi|$, one obtains that the covariance term into the variance decomposition is*

$$\sum_{i \neq j} \lambda_i \lambda_j\ cov\left(y_i, y_j\right) \asymp \frac{|\Psi|}{m}$$

This step shows how sectoral comovement from horizontal propagation accumulates in aggregate. Even if individual covariance is small, a shock in $\boldsymbol{\varepsilon}$ propagating through the horizontal geometry does not average out; rather, it produces systematic comovement since horizontal complementarities induce many correlated pairs.

***Step 5: Rate of decay of aggregate volatility**.– Plug the variance and covariance (steps 2 and 4, respectively) contributions into the variance decomposition:*

$$var\left(Y\right) = \mathcal{O}\left(\frac{1}{m}\right) + \mathcal{O}\left(\frac{|\Psi|}{m}\right) \quad \longleftrightarrow \quad var\left(Y\right) = \mathcal{O}\left(\frac{1 + |\Psi|}{m}\right)$$

where $\mathcal{O}\left(\cdot\right)$ describes the asymptotic rate at which aggregate GDP volatility decays as the number of network participants increases. Because horizontal propagation induces comovement, the covariance terms within cluster-$|\Psi|$ are positive and cannot be cancelled. This implies that

$$var\left(Y\right) \geq c\ \frac{|\Psi|}{m} \quad \longleftrightarrow \quad var\left(Y\right) = \Omega\left(\frac{1}{m/\,|\Psi|}\right)$$

In other words, there exists a constant $c > 0$ such that aggregate volatility is positive under horizontal propagation, and variance cannot decay faster than $\frac{|\Psi|}{m}$. For $m \to \infty$, if $|\Psi| \to \infty$ while $|\Psi| \ll m$, aggregate volatility is positive (matching with Theorem 5), but it decays at a slower rate than the standard diversification argument, since $\Omega\left(\frac{|\Psi|}{m}\right) > \mathcal{O}\left(\frac{1}{m}\right)$. If instead the size of the horizontal cluster grows with m (so that $\Delta\,|\Psi| \propto \Delta m$), aggregate volatility is persistent and does not decay, as a growing *portion of the economy is horizontally-related, since $\frac{|\Psi|}{m}$ grows at a constant rate. By Theorem 2 all sectors belong to the same horizontal cluster ($|\Psi| = \Phi$): the economy behaves as a single macro-unit and aggregate volatility does not vanish asymptotically.*

□

A.3. Model derivation

(Households problem) *The household-i's utility problem is to maximize utility function subject to its budget constraint:*

$$\max_{c_i(s),\, n_i(s)} \quad \mathcal{U}_i \left(\{ c_i(s) \}_{\forall s \in \Phi} ; n_i(s) \right) := \prod_{s \in \Phi} \left[c_i(s)^{\omega(s)} - \frac{n_i(s)^{1+\phi}}{1+\phi} \right]$$

$$s.t. \qquad \sum_{s \in \Phi} p(s)\, c_i(s) = w(s)\, n_i(s) + \sum_{s \in \Phi} \pi_i(s)$$

Sectoral consumption and labour supplied are $c_i(s)$ and $n_i(s)$, respectively, and each is determined by a price, $p(s)$ and $w(s)$. Parameter $\omega(s) \in (0,1)$ identifies the weight of each sectoral good in household-i's consumption basket and $\sum_{s \in \Phi(s)} \omega(s) = 1$, ϕ is the inverse Frisch labour supply elasticity, measuring the elasticity of hours worked to the wage rate under a constant marginal utility of income. Finally, $\pi_i(s)$ is the constant share of sector-specific profits flowing from sector-s to household-i.

Utility maximization implies the Lagrangian function to be

$$\underset{c_i(s),\, n_i(s)}{L} := \prod_{s \in \Phi} \left[c_i(s)^{\omega(s)} - \frac{n_i(s)^{1+\phi}}{1+\phi} \right] + \\ - \psi^L \left[\sum_{s \in \Phi} p(s)\, c_i(s) - \left(w(s)\, n_i(s) + \sum_{s \in \Phi} \pi_i(s) \right) \right]$$

with ψ^L being the penalty multiplier. Optimality conditions are in order

$$\frac{p(s)\, c_i(s)}{\omega(s)} = \frac{p(s')\, c_i(s')}{\omega(s')}$$

$$w(s) = n_i(s)^{\phi}\, p(s)\, c_i(s) \left[\omega(s) \;\; c_i \right]^{-1}$$

which, once aggregated across households, $\frac{p(s) \int_i c_i(s) \mathrm{d}\, i}{\omega(s)} = \frac{p(s') \int_i c_i(s') \mathrm{d}\, i}{\omega(s')}$ and $w(s) = \int_i n_i(s)^{\phi} \mathrm{d}\, i\, p(s) \int_i c_i(s) \mathrm{d}\, i \left[\omega(s) \;\; \int_i c_i \mathrm{d}\, i \right]^{-1}$, would report the sector-level optimality conditions for consumption and labour supply. The first condition for consumption defines how the total expenditure from households for sectoral goods gives rise to the relative importance of each good in total consumption, C. This is an aggregator over sectoral consumption levels:

$$C = f^c \left(c(s), c(s'), c(s''), \ldots, c(m) \;\; ; \;\; \{ \omega(s) \}_{s=1}^{m} \right)$$

Assumption 5 ***(Final consumption requirements)*** *As main characteristics of a final consumption aggregator: (i) it is strictly quasi-concave, non-decreasing and homogeneous of degree one in each of the sectoral consumption levels; (ii) consumption* *goods are normal, so that their demand increases with households' income; and (iii) the weight of sectoral goods in total consumption is driven by $\omega(s)$, which is the relative weight that households give in the consumption of the good produced by sector-s.*

Moreover, denote by $\boldsymbol{c} = \left[c(s)\right]$ the $m \times 1$ vector of consumption levels over sectoral goods, with generic element defined as $c(\cdot) > 0$. Under this specification and the regularities in Assumption 5, it holds that $C = \prod_{s\in\Phi} c(s)^{\omega(s)} \propto \boldsymbol{\omega}'\boldsymbol{c}$.

□

(Sectoral optimization) *The production side of the economy is made of a finite set of sectors, $\{s, s', \ldots, m\} \in \Phi$, populated by a representative firm whose output is the result of a Cobb-Douglas technology satisfying Assumption 1:*

$$y(s) = z(s)\left(n(s)\right)^{\alpha(s)} \prod_{s'\in\Phi}\left(x(s,s')\right)^{\alpha(s,s')} \varkappa_j(s) \tag{A.3}$$

where $\varkappa_j(s)$, for $j = \{fd, fs\}$, is a normalization constant ruling out double counting of intermediate inputs when considering both factor input demand and supply network distances between sectors; absent any of these two cases, then $\varkappa_d(s) = 1$.

A perfectly competitive representative firm in sector-s maximizes total revenues from production net of costs of inputs of production:

$$\max_{n(s),\, x(s,s')} \quad y(s) = z(s)\left(n(s)\right)^{\alpha(s)} \prod_{s'\in\Phi}\left(x(s,s')\right)^{\alpha(s,s')} \varkappa_j(s)$$

$$\text{s.t.} \qquad \mathcal{C}(s) := w(s)\, n(s) + \sum_{s'\in\Phi} p(s')\, x(s,s')$$

where $\alpha(s) + \sum_{s'} \alpha(s,s') = 1$, and $\varkappa_d(s) = 1$ for $j = \{fd, fs\}$. Profit maximization then implies that optimal competitive factor demands of sector-s relative to sector-s' are given by $n(s) = \alpha(s)\,\frac{p(s)\,y(s)}{w(s)}$ and $x(s,s') = \alpha(s,s')\,\frac{p(s)\,y(s)}{p(s')}$.

□

(Equilibrium characterization) *In equilibrium, the model should specify the clearing conditions of labour, circulating intermediate inputs, and goods markets. Starting from the labour market, equating labour demand and labour supply would deliver the labour market equilibrium condition, $n(s) = \left[\alpha(s)\ \frac{C}{c(s)}\ \omega(s)\ y(s)\right]^{\frac{1}{1+\phi}}$. Moreover, total labour demand is found by aggregating labour optimality condition of sectors, $N^{dem} = \sum_{s\in\Phi} n^{dem}(s)$, and total labour supply is found by aggregating labour optimality condition of households, $N^{sup} = \sum_{s\in\Phi} n^{sup}(s)$. Labour market clearing then requires that $N^{dem} = N^{sup}$. For the equilibrium in the circulating intermediate inputs market, total demand of intermediates from sectors is $x^{dem}(s) = \sum_{s'\in\Phi} x(s,s')$. Analogously, its total supply of intermediate inputs is $x^{sup}(s) = \sum_{s'\in\Phi} x(s',s)$ so that, in equilibrium it must be true that $x^{dem}(s) = x^{sup}(s)$ which, in aggregate, simply states that $\sum_{s\in\Phi} x^{dem}(s) = \sum_{s\in\Phi} x^{sup}(s) \equiv X^{sup} = X^{dem}$. The equilibrium in the good market is given by summing over all sectors the sectoral equilibrium in which total production must equal its total final consumption (by households) and its circulating intermediate input bought by other sectors: $\sum_{s\in\Phi} y(s) = \sum_{s\in\Phi}\left(\int_i c_i(s)\,\mathrm{d}i\right) +$*

$\sum_{s\in\Phi}\left(\sum_{s'\in\Phi} x\left(s,s'\right)\right)$. *Finally, by aggregating the households' budget constraints over households and sectors, and imposing the clearing conditions so far, the aggregate resource constraint for this economy reads as* $P^c C = WN + \pi$ *which equals total output, defined by* $Y \propto \boldsymbol{\omega}'\boldsymbol{y}$. *According to the structure of the model, equilibrium conditions read as follows. A competitive equilibrium for this efficient economy is defined by a set of sectoral prices for labour and intermediate inputs,* $\left\{w\left(s\right), p\left(s\right)\right\}_{s\in\Phi}$, *a set of sectoral production input quantities,* $\left\{n\left(s\right), x\left(s,s'\right)\right\}_{s,s'\in\Phi}$, *exogenous sectoral productivities,* $\left\{z\left(s\right)\right\}_{s\in\Phi}$, *and a set of aggregate quantities,* $\Omega = \left(Y, C, N, X, \pi\right)$, *such that (i) each household satisfies its optimality conditions, (ii) the representative firm of each sector maximizes profits, and (iii) all markets clear, shaping* Ω.

$\square$

(Derive aggregate fluctuations result of eq. (4)) *Following exactly the same operating steps in vom Lehn and Winberry (2022) to characterize the dynamical changes in aggregate Gross Domestic Product (GDP), it is first necessary to determine the existing relation between sectoral intermediate output,* $y\left(s\right)$, *and the Divisia Index.*[XI] *Notice that constant returns to scale and homogeneity of degree one in Assumption 1 and in the production function in eq. (A.3) implies the following zero-profit condition*

$$p\left(s\right) y\left(s\right) = w\left(s\right) n\left(s\right) + \sum_{s'\in\Phi} p\left(s'\right) x\left(s,s'\right)$$

where $p\left(s\right) y\left(s\right)$ *is the (gross) nominal output of sector-*s*, and from where the national accounting definition of nominal value added is just* $p\left(s\right) y\left(s\right) - p^s x^s = w\left(s\right) n\left(s\right)$, *where I define* $p^s x^s = \sum_{s'\in\Phi} p\left(s'\right) x\left(s,s'\right)$ *the total expenditure of sector-*s *for intermediate inputs from other connected sectors. Define the left-hand side of the previous equation as* $p^Y\left(s\right) y^Y\left(s\right) \equiv p\left(s\right) y\left(s\right) - p^s x^s$, *i.e., the nominal value-added.*

In order to construct a single, aggregate measure of real output (or value added) across multiple sectors, I use a Divisia index. This index combines the growth rates of individual sectoral outputs into an overall growth rate, but does so in a way that accounts for each sector's economic importance – measured by its share in total nominal value added. Specifically, the Divisia index computes a weighted average of the growth rates of sectoral real outputs, where the weights are each sector's share in total nominal value added. This allows us to track how the economy's output is changing over time, adjusting dynamically as the composition of output across sectors evolves.

Henceforth, the Divisia Index requires to differentiate the nominal value added under constant price levels associated to variables:

[XI] This indicator provides a theoretically consistent measure of growth. It is a composite index, particularly suited to analyse variables made of multiple and differentiated elements, measuring changes in certain aggregate quantity of a given variable, and weighting all its defining components according to their relevance on that variable, that is it expresses the overall rate of output change as a weighted average of the growth rates of individual components, with time-varying weights corresponding to each component's share in total output. Conceptually, it captures the notion that aggregate GDP growth reflects both the differential growth of its components and their evolving relative importance within the economy. This formulation offers an improvement over fixed-weight indices by accommodating structural changes in the composition of economic activity over time. Refer to Oulton (2022) for further details.

$$
\begin{aligned}
& p^Y(s)\ d\,log\,y^Y(s) = p(s)\ d\,log\,y(s) - p^s\,d\,log\,x^s \\
\rightarrow\quad & p^Y(s)\,y^Y(s)\ d\log y^Y(s) = p(s)\,y(s)\ d\log y(s) - p^s x^s\,d\log x^s \\
\rightarrow\quad & \alpha(s)\ d\log y^Y(s) = d\log y(s) - \alpha^s\,d\log x^s \\
\rightarrow\quad & \alpha(s)\ d\log y^Y(s) = \Big(d\log z(s) + \alpha(s)\ d\log n(s) + \alpha^s\,d\log x^s\Big) - \alpha^s\,d\log x^s \\
\longrightarrow\quad & d\log y^Y(s) = \frac{1}{\alpha(s)}\,d\log z(s) + d\log n(s)
\end{aligned}
$$

where $\alpha^s = \sum_{s\in\Phi}\alpha(s,s')$. Plug the above sector-level nominal value added growth, $d\log y^Y(s)$, into the Divisia Index, $d\log Y = \sum_{s\in\Phi}\left(\frac{p^Y(s)y^Y(s)}{P^Y Y}\right)d\log y^Y(s)$, so that it is possible to obtain a non-complete equation for real GDP growth:

$$
d\log Y = \sum_{s\in\Phi}\left(\frac{p^Y(s)\,y^Y(s)}{P^Y Y}\right)\left[\frac{1}{\alpha(s)}\,d\log z(s) + d\log n(s)\right] \tag{A.4}
$$

Summing over all sectors the sectoral optimality conditions for intermediate inputs, $\sum_{s'\in\Phi}x(s,s') = \sum_{s'\in\Phi}\alpha(s,s')\,[p(s)\ y(s)]\,(p(s'))^{-1}$, noticing that $\alpha(s) = 1-\alpha^s$, and inserting the zero-profit condition would yield $\alpha(s) = \frac{p^Y(s)y^Y(s)}{p(s)y(s)}$, which is the ratio of sectoral value-added to sectoral intermediate (gross) output.

By plugging into eq. (A.4) the condition for $\alpha(s)$, and making all the adjustments to simplify the elements multiplying the sectoral productivity, it is then possible to get eq. (4), which determines the drivers of real aggregate GDP fluctuations.

□

(Labour market equilibrium with network "economic" distances) *Beyond the equilibrium definition, rearranging optimality conditions for labour market variables from both households and sectors, the following equilibrium condition for sectoral labour force is detected:*

$$
n(s) = \left[\alpha(s)\ \frac{C}{c(s)}\ \omega(s)\ \ y(s)\right]^{\frac{1}{1+\phi}} \tag{A.5}
$$

It defines that employment is a positive function of both output and consumption (recall that total consumption C increases with sector-specific consumption).

To account for the role of changes to other sectors' employment on sectoral employment level, notice that when sector-s buys intermediate inputs from sector-s', it is introjecting also the quantities of production inputs in such sector. Henceforth, each element in the intermediate inputs bundle of sector-s can be taught as $x(s,s') = \vartheta(s,s')\ \ y(s')$, $\forall s' \in \Phi(s)$, which states that each intermediate input is just a given share $\vartheta(s,s')$ of other sector's output. Labour market equilibrium in eq. (A.5) is thus

$$
n(s) = \left[\alpha(s)\ \frac{C}{c(s)}\ \omega(s)\ \ z(s)\Big(n(s)\Big)^{\alpha(s)}\prod_{s'\in\Phi_s}\Big(\vartheta(s,s')\ \ y(s')\Big)^{\alpha(s,s')}\right]^{\frac{1}{1+\phi}} \tag{A.6}
$$

so that, once plugging in the output of sector-s', sectoral employment levels are transparently related through the production network structure. As a result, the above labour market condition allows to study the general equilibrium comovement of employment levels across sectors.

Once inserting the production function of eq. (A.3) in the labour market equilibrium condition of eq. (A.6), then optimal quantities of a sector are related to that of the other ones. To incorporate factor input demand distance, in Appendix A.3 I show that manipulation in general equilibrium would result in

$$n\,(s) = \left[\alpha\,(s)\ \frac{C}{c(s)}\ \omega\,(s)\ \ z(s)\ n(s)^{\alpha(s)}\ \ \aleph^{fd}\ \boldsymbol{\Lambda}^{fd}\ \boldsymbol{\Xi}^{fd}\right]^{\frac{1}{1+\phi}} \tag{A.7}$$

where $\aleph^{fd} = \prod_{s'\in\Phi}\left(\prod_{s\in\Phi} x\,(s',s)^{\alpha(s',s)}\right)^{\frac{\alpha(s,s')}{1+\phi}}$ *identifies the set of intermediate inputs bought by sector-s and all the other sectors connected to it, the element* $\boldsymbol{\Lambda}^{fd} = \prod_{s'\in\Phi}\left(\vartheta\,(s,s')\,z(s')\,n(s')^{\alpha(s)}\right)^{\alpha(s,s')}$ *captures the relevance of other sectors' productivities and employment levels on sector-s working through the production network, and the component* $\boldsymbol{\Xi}^{fd} = \prod_{s'\in\Phi}\left[\prod_{s\in\Phi}\left(\frac{x(s',s)}{x(s,s)}\right)^{\frac{\alpha(s',s)}{\alpha(s,s)}}\right]$ *determines the ratio among the intermediate inputs between sector-s and each of the other sectors, when these are buying from the same upstream sector.*

Manipulation in general equilibrium to include factor input supply distance for pairs of sectors would result in

$$n\,(s) = \left[\alpha\,(s)\ \frac{C}{c(s)}\ \omega\,(s)\ \ z(s)\ n(s)^{\alpha(s)}\ \ \aleph^{fs}\ \boldsymbol{\Lambda}^{fs}\ \boldsymbol{\Xi}^{fs}\right]^{\frac{1}{1+\phi}} \tag{A.8}$$

where $\aleph^{fs} = \prod_{s'\in\Phi}\left(\prod_{s'\in\Phi} x\,(s',s')^{\alpha(s',s')}\right)^{\frac{\alpha(s,s')}{1+\phi}}$ *is the compounded set of intermediate inputs,* $\boldsymbol{\Lambda}^{fs} = \prod_{s'\in\Phi}\left(\vartheta\,(s,s')\,z(s')\,n(s')^{\alpha(s)}\right)^{\alpha(s,s')}$ *considers sectoral productivities and employment levels, and* $\boldsymbol{\Xi}^{fs} = \prod_{s'\in\Phi}\left[\prod_{s\in\Phi}\left(\frac{x(s',s)}{x(s',s')}\right)^{\frac{\alpha(s',s)}{\alpha(s',s')}}\right]$ *determines the ratio among intermediate inputs when sectors are selling to the same downstream sector.*

□

A.4. Network-based propagation to sectoral employment

PROPOSITION 4 (Propagation of variations to intermediate inputs) *Consider a market economy characterized by Input-Output linkages as in eq. (A.3), whose labour market equilibrium is eq. (A.5). Then, the response of sectoral employment to changes in sectoral intermediate inputs is a first-order (log-linear) approximation*

$$d\log\boldsymbol{n} = \Theta\left\{d\log C + \underbrace{d\log\boldsymbol{z} - d\log\boldsymbol{c}}_{d\log\boldsymbol{S}} + \boldsymbol{\mathcal{A}}\,d\log\mathbf{X}\right\} \tag{A.9}$$

where:

(a) $\boldsymbol{n}$ *identifies sectoral employment levels;*

(b) $\Theta = \frac{1}{1+\phi+\boldsymbol{\alpha}}$ *is a compound of structural parameters, and* $\boldsymbol{\alpha} = [\alpha(s), \ldots, \alpha(m)]$*;*

(c) C *is aggregate consumption;*

(d) $\mathcal{S}$ *identifies sector-specific changes in productivity and consumption;*

(e) $\mathcal{A}$ *is the Input-Output matrix, comprising sectoral weight on other sectors, influencing the elements of* $\mathbf{X}$*, a matrix with intermediate inputs quantities.*

Proof in Appendix A.5.

PROPOSITION 5 (Direct propagation of variations to sectoral employment) *Consider an Input-Output economy defined by a labour market equilibrium as stated in eq. (A.6). Then, the response of sectoral employment to other sectors' employment is a first-order (log-linear) approximation*

$$d\log \boldsymbol{n} = \Theta \left\{ d\log C + d\log \mathcal{S} + \underbrace{\mathcal{A}\left(\Phi_{\{s,s'\}}\right)\left[d\log \mathbf{z} + \boldsymbol{\alpha}\, d\log \boldsymbol{n}\right] + \mathcal{E}\, d\log \mathbf{X}}_{d\log \mathcal{V}} \right\} \tag{A.10}$$

where:

(a) $\boldsymbol{n}$ *identifies sectoral employment levels;*

...

(e) $\mathcal{V}$ *identifies the production network effect of other sectors' changes in productivities, employment levels, and intermediate inputs usage, where:*

- $\mathcal{A}(\Phi_{\{s,s'\}})$ *is the Input-Output matrix, comprising the weight of each sector on other sectors, whose entries are set to 0 whenever* $s = s'$*;*
- $\mathcal{E} = \mathcal{A}(\Phi_{\{s,s'\}})'\mathcal{A}(\Phi_{\{s',s\}})$ *is a compounded network effect, made of the inner product of the Input-Output matrix, influencing the elements of* $\mathbf{X}$*.*

Proof in Appendix A.5.

PROPOSITION 6 (Direct propagation under factor input demand distance) *Consider an Input-Output economy defined by a labour market equilibrium as stated in eq. (A.7). Then, the response of sectoral employment to other sectors' employment is a first-order (log-linear) approximation*

$$d\log \boldsymbol{n} = \Theta \left\{ d\log C + d\log \mathcal{S} + d\log \mathcal{V} + \underbrace{\mathcal{D}^{fd}\left[d\log \boldsymbol{n} - \left(\mathcal{I}'_{\Phi}\, d\log \boldsymbol{n}\right)\mathbf{1}\right]}_{d\log \mathcal{D}(fd)} \right\} \tag{A.11}$$

where:

(a) $\boldsymbol{n}$ *identifies sectoral employment levels;*

...

(f) $\mathcal{D}^{fd}$ *identifies all the demand-based distances across any pair of sectors;*

(g.i) $\mathcal{D}(fd)$ *identifies the production network distance effect of other sectors' variations in employment levels.*

Proof in Appendix A.5.

PROPOSITION 7 (Direct propagation under factor input supply distance) *Consider an Input-Output economy defined by a labour market equilibrium as stated in eq. (A.8). Then, the response of sectoral employment to other sectors' employment is a first-order (log-linear) approximation*

$$d\log \boldsymbol{n} = \Theta \left\{ d\log C + d\log \boldsymbol{\mathcal{S}} + d\log \boldsymbol{\mathcal{V}} + \underbrace{\boldsymbol{\mathcal{D}}^{fs}\Big[d\log \boldsymbol{p} - \left(\boldsymbol{\mathcal{I}}_{\Phi}'\, d\log \boldsymbol{p}\right)\mathbf{1}\Big]}_{d\log \boldsymbol{\mathcal{D}}(fs)} \right\} \tag{A.12}$$

where:

(a) $\boldsymbol{n}$ *identifies sectoral employment levels;*

...

(f) $\boldsymbol{\mathcal{D}}^{fs}$ *identifies all the supply-based distances across any pair of sectors;*

(g.ii) $\boldsymbol{\mathcal{D}}\left(fs\right)$ *identifies the production network distance effect of other sectors' variations in employment levels.*

Proof in Appendix A.5.

A.5. *Proofs for model derivation*

(Derive labour market equilibrium of eqs. (A.7) and (A.8)) *From the sector-level equilibrium definition of the labour market,* $n\left(s\right) = \left[\alpha\left(s\right)\ \frac{C}{c(s)}\ \omega\left(s\right)\ y\left(s\right)\right]^{\frac{1}{1+\phi}}$*, including the condition* $x\left(s,s'\right) = \vartheta\left(s,s'\right)\ y\left(s'\right)$ *would imply eq. (A.6), that is*

$$n\left(s\right) = \left[\alpha\left(s\right)\ \frac{C}{c(s)}\ \omega\left(s\right)\ z\left(s\right)\left(n\left(s\right)\right)^{\alpha(s)} \prod_{s'\neq s}\left(\vartheta\left(s,s'\right)\ y\left(s'\right)\right)^{\alpha(s,s')}\ \varkappa_j\left(s\right)\right]^{\frac{1}{1+\phi}} \tag{A.13}$$

where it should be inserted the production function of eq. (A.3) for sector-s'*, and* $\varkappa_j\left(s\right) \neq 1$*, with* $j = \{fd, fs\}$*, for the purpose of this derivation. The resulting equation can be simplified by imposing an infinite inverse Frisch elasticity of labour,* $\phi \to \infty$*. In fact, in the model of Section 3, only the extensive margin of employment (i.e., total number of workers) is important, while it is not the intensive margin (i.e., total hours worked), which is ruled out under* $\phi \to \infty$*; refer to Rogerson (1988). However, for the sake of generality, I will proceed with* $\phi \in \left[0, \infty\right)$*.*

Factor input demand.*– Assume the non-common (constant) element in the production function to be given by*[XII]

[XII] In eq. (A.14), the component $\tau_{fd}\left(s\right) = \prod_{s'\in\Phi}\left[\prod_{s\in\Phi}\left(\frac{x(s',s)}{x(s,s)}\right)^{\frac{\alpha(s',s)}{\alpha(s,s)}}\right]^{\frac{1}{\alpha(s,s')}}$ allows to ensure that the distance effects are not influenced by the Leontief inverse weight. Specifically, incorporating the condition $x\left(s,s'\right) = \vartheta\left(s,s'\right)\ y\left(s'\right)$ in the labour market equilibrium of eq. (A.13) would result in a double counting of the values from the Input-Output matrix: one of them is used to capture sectoral distances, while the other would have been multiplied by distance matrix $\boldsymbol{\mathcal{D}}^{fd}$ when analysing the effect of sectoral propagation in Proposition 6. Within this context, it is essential to highlight that, starting from the baseline labour market equilibrium in eq. (A.5), and deriving the results presented in this section and in Proposition 6 without including the exogenous condition $x\left(s,s'\right) = \vartheta\left(s,s'\right)\ y\left(s'\right)$ would produce an outcome reflecting the effect of network distances *without* the influence of the Input-Output Leontief inverse weights. In other words, as discussed in the main text

$$\varkappa_{fd}(s) = \tau_{fd}(s) \prod_{s'} \left[\frac{\prod_s x(s',s)^{\frac{1}{\alpha(s,s)}+\alpha(s',s)}}{\prod_s x(s,s)^{\frac{\alpha(s',s)}{\alpha(s,s)}}} \right]^{\frac{\alpha(s,s')}{1+\phi}} \tag{A.14}$$

Then, multiply and divide both sides of the above labour market equilibrium, including the production function, by $\prod_{s'} \left\{ \left[\prod_s x(s',s)^{\frac{1}{\alpha(s,s)}} \right] \Big/ \left[\prod_s x(s,s)^{\frac{\alpha(s',s)}{\alpha(s,s)}} \right] \right\}^{\frac{\alpha(s,s')}{1+\phi}}$. *Finally, making all the required adjustments once including* $\varkappa_{fd}(s)$*, the resulting condition will deliver eq. (A.7). All the multiplications are over* Φ.

Factor input supply.– *Assume now the non-common (constant) element in the production function to be given by*

$$\varkappa_{fs}(s) = \tau_{fs}(s) \prod_{s'} \left[\frac{\prod_s x(s',s)^{\frac{1}{\alpha(s,s)}}}{\prod_{s'} x(s',s')^{\frac{\alpha(s',s)}{\alpha(s',s')}-\alpha(s',s')}} \right]^{\alpha(s,s')} \tag{A.15}$$

Multiply and divide both sides of the above labour market equilibrium, including the production function, by $\prod_{s'} \left\{ \left[\prod_s x(s',s)^{\frac{1}{\alpha(s',s')}} \right] \Big/ \left[\prod_{s'} x(s',s')^{\frac{\alpha(s',s)}{\alpha(s',s')}} \right] \right\}^{\frac{\alpha(s,s')}{1+\phi}}$. *Making all the required adjustments once including* $\varkappa_{fs}(s)$*, the resulting condition will deliver eq. (A.8). All the multiplications are over* Φ.

□

(Proof of Proposition 3) *The proof directly follows Propositions 6 and 7.* □

(Proof of Proposition 4) *The interest is in characterizing the way in which sectoral employment changes in response to variations in sector-specific intermediate inputs usage. In order to inspect these changes, it is necessary to express the equilibrium conditions in terms of log-linear conditions. Starting from the sector-*s *labour market equilibrium of eq. (A.5), i.e.,* $n(s) = \left[\alpha(s) \frac{C}{c(s)} \omega(s)\ y(s) \right]^{\frac{1}{1+\phi}}$*, the associated log-linear form is* $\widetilde{n}(s) = \frac{1}{1+\phi} \left\{ \widetilde{y}(s) + \widetilde{C} - \widetilde{c}(s) \right\}$*, with tilded variables,* $\tilde{\cdot}$*, expressing the deviation from their respective steady state. Such log-linearized equation contains the log-linearized output, obtained by log-differentiating the production function in eq. (A.3):* $\widetilde{y}(s) = \widetilde{z}(s) + \alpha(s)\widetilde{n}(s) + \sum_{s'\in\Phi} \alpha(s,s')\widetilde{x}(s,s')$*. Substituting out these two equations, one easily gets that log-linearized sectoral employment is a function of*

$$\widetilde{n}(s) = \frac{1}{1+\phi} \left\{ \widetilde{z}(s) + \alpha(s)\widetilde{n}(s) + \sum_{s'\in\Phi} \alpha(s,s')\widetilde{x}(s,s') + \widetilde{C} - \widetilde{c}(s) \right\}$$

(Theorem 3), the effect of the network "economic" distance across pairs of sectors is independent of intersectoral trade intensities characterizing the production network; imposing the $\tau_{fd}(s)$ term due to the condition $x(s,s') = \vartheta(s,s')\ y(s')$ allows to avoid that such concept is violated.

expressing the elements whose variations induce changes in employment level of sector-s. In vectorial notation, one can rewrite the inner summation as $\boldsymbol{a}(s)\widetilde{\boldsymbol{x}}(s)$ which, when stacked over all sectors, can be written in matrix form:

$$\widetilde{\boldsymbol{n}} = \frac{1}{1+\phi}\left\{\widetilde{C} - \widetilde{\boldsymbol{c}} + \widetilde{\boldsymbol{z}} + \boldsymbol{\alpha}\widetilde{\boldsymbol{n}} + \boldsymbol{\mathcal{A}}'\widetilde{\mathbf{X}}\right\}$$

where $\widetilde{\boldsymbol{n}}$ is an $m \times 1$ vector of changes in sector-specific employment levels, identifies changes in aggregate consumption, $\widetilde{\boldsymbol{c}}$ is an $m \times 1$ vector of changes in sector-specific final consumption by households, $\widetilde{\boldsymbol{z}}$ is an $m \times 1$ vector of changes in sectoral productivities, $\boldsymbol{\mathcal{A}}$ is an $S \times S$ squared Input-Output matrix identifying the structure of intersectoral trade, and $\widetilde{\mathbf{X}}$ is an $m \times 1$ vector of changes in sector-specific set of intermediate inputs bought within the production network. Finally, ϕ is a scalar for aggregate inverse Frisch elasticity of labour supply to wage level, and $\boldsymbol{\alpha} = [\alpha(s), \alpha(s'), \ldots, \alpha(m)]$ comprises sector-specific labour force as a share of its intermediate output.

Bringing the vector of sectoral employments, $\widetilde{\boldsymbol{n}}$, on the left-hand side, rewriting $\widetilde{\cdot} = d\log(\cdot)$, and rearranging terms, then one obtains eq. (A.9) in Proposition 4.

□

(Proof of Proposition 5) *The interest is in characterizing the way in which sectoral employment changes in response to variations in other sectors' employment levels. In order to inspect these changes, it is necessary to express the equilibrium conditions in terms of log-linear conditions. Starting from the sector-s labour market equilibrium of eq. (A.5), i.e., $n(s) = \left[\alpha(s)\ \frac{C}{c(s)}\ \omega(s)\ y(s)\right]^{\frac{1}{1+\phi}}$, the associated log-linear form is $\widetilde{n}(s) = \frac{1}{1+\phi}\left\{\widetilde{y}(s) + \widetilde{C} - \widetilde{c}(s)\right\}$, with tilded variables, $\widetilde{\cdot}$, expressing the deviation from their respective steady state. Then, use the condition $x(s,s') = \vartheta(s,s')\ y(s')$, which states that each intermediate input is just a given share $\vartheta(s,s')$ of other sector's output. Inserting its log-linearized form $\widetilde{x}(s,s') = \widetilde{y}(s')$, along the log-differentiation of the production function in eq. (A.3), $\widetilde{y}(s) = \widetilde{z}(s) + \alpha(s)\widetilde{n}(s) + \sum_{s'\in\Phi}\alpha(s,s')\widetilde{x}(s,s')$, into the log-linear labour market equilibrium, one gets*

$$\widetilde{n}(s) = \frac{1}{1+\phi}\left\{\begin{array}{l} \widetilde{C} - \widetilde{c}(s) + \widetilde{z}(s) + \alpha(s)\widetilde{n}(s) \quad + \\ + \displaystyle\sum_{s'\in\Phi_s}\alpha(s,s')\left[\widetilde{z}(s') + \alpha(s')\widetilde{n}(s') + \sum_{s\in\Phi_{s'}}\alpha(s',s)\widetilde{x}(s',s)\right] \end{array}\right\}$$

expressing the elements whose variations induce changes in employment level of sector-s. Note that using the above condition $x(s,s') = \vartheta(s,s')\ y(s')$ implies that summations are not over the whole set of sectors, but rather it should be excluded the sector whose one is summing for. To this aim, denote Φ the set of all sectors; then, both Φ_s and $\Phi_{s'}$ denotes improper subsets of Φ, since they are excluding sector-s and sector-s', respectively. In other words, all the elements in $\{\Phi_s, \Phi_{s'}\}$ are contained in Φ but $\{\Phi_s, \Phi_{s'}\}$ and Φ are not equal, $\Phi_s \subseteq \Phi$ with $\Phi_s \nsubseteq \Phi$, and $\Phi_{s'} \subseteq \Phi$ with $\Phi_{s'} \nsubseteq \Phi$. When stacking the above condition over all sectors (i.e., matrix notation), the Input-Output matrix from $\alpha(s,s')$ and $\alpha(s',s)$ are not full but rather are set to zero whenever $s = s'$, i.e., using the $\Phi_{\{s,s'\}}$ as a matrix indicator.

Using such notation, and expressing the above log-linearized labour market equa-

tion in vectorial form and then in matrix form (as in the Proof of Proposition 4), and solving for $d\log n(s)$*, one obtains eq. (A.10) of Proposition 5.*

$\square$

(Proof of Proposition 6) *The interest is in characterizing the way in which sectoral employment changes in response to variations in other sectors' employment levels according to their factor input demand network distance relationships. Taking the logarithm of eq. (A.7) would result in*

$$\log n(s) = \frac{1}{1+\phi}\left\{\begin{array}{l} \log\alpha(s) + \log C + \log z(s) + \alpha(s)\,\log n(s) - \log c(s) \quad + \\ + \sum_{s'\in\Phi}\alpha(s,s')\left[\log\vartheta(s,s') + \log z(s') + \alpha(s')\,\log n(s')\right] \quad + \\ + \sum_{s'\in\Phi}\alpha(s,s')\left\{\sum_{s\in\Phi}\alpha(s',s)\log x(s',s)\right\} + \log\omega(s) \quad + \\ + \sum_{s'\in\Phi}\left\{\frac{\sum_{s\in\Phi}\alpha(s',s)}{\sum_{s\in\Phi}\alpha(s,s)}\log\left[\frac{\sum_{s\in\Phi}x(s',s)}{\sum_{s\in\Phi}x(s,s)}\right]\right\} \end{array}\right\}$$

Inserting the condition in eq. (A.1) used to derive demand-distance matrix in the $log[\cdot]$ *component in the last row, compute the associate total differentiation, following closely the steps to derive Proposition 5, and using the notation associated to network distance matrix* $\mathcal{D}^{fd}$*, then one obtains eq. (A.11) characterizing Proposition 6.*

$\square$

(Proof of Proposition 7) *The interest is in characterizing the way in which sectoral employment changes in response to variations in other sectors' employment levels according to their factor input supply network distance relationships. Taking the logarithm of eq. (A.8) would result in*

$$\log n(s) = \frac{1}{1+\phi}\left\{\begin{array}{l} \log\alpha(s) + \log C + \log z(s) + \alpha(s)\,\log n(s) - \log c(s) \quad + \\ + \sum_{s'\in\Phi}\alpha(s,s')\left[\log\vartheta(s,s') + \log z(s') + \alpha(s')\,\log n(s')\right] \quad + \\ + \sum_{s'\in\Phi}\alpha(s,s')\left\{\sum_{s'\in\Phi}\alpha(s',s')\log x(s',s')\right\} + \log\omega(s) \quad + \\ + \sum_{s\in\Phi}\left\{\frac{\sum_{s'\in\Phi}\alpha(s',s)}{\sum_{s'\in\Phi}\alpha(s',s')}\log\left[\frac{\sum_{s'\in\Phi}x(s',s)}{\sum_{s'\in\Phi}x(s',s')}\right]\right\} \end{array}\right\}$$

Inserting the condition in eq. (A.2) used to derive supply-distance matrix in the $log[\cdot]$ *component in the last row, compute the associate total differentiation, following closely the steps to derive Proposition 5, and using the notation associated to network distance matrix* $\mathcal{D}^{fs}$*, then one obtains eq. (A.12) characterizing Proposition 7.*

$\square$

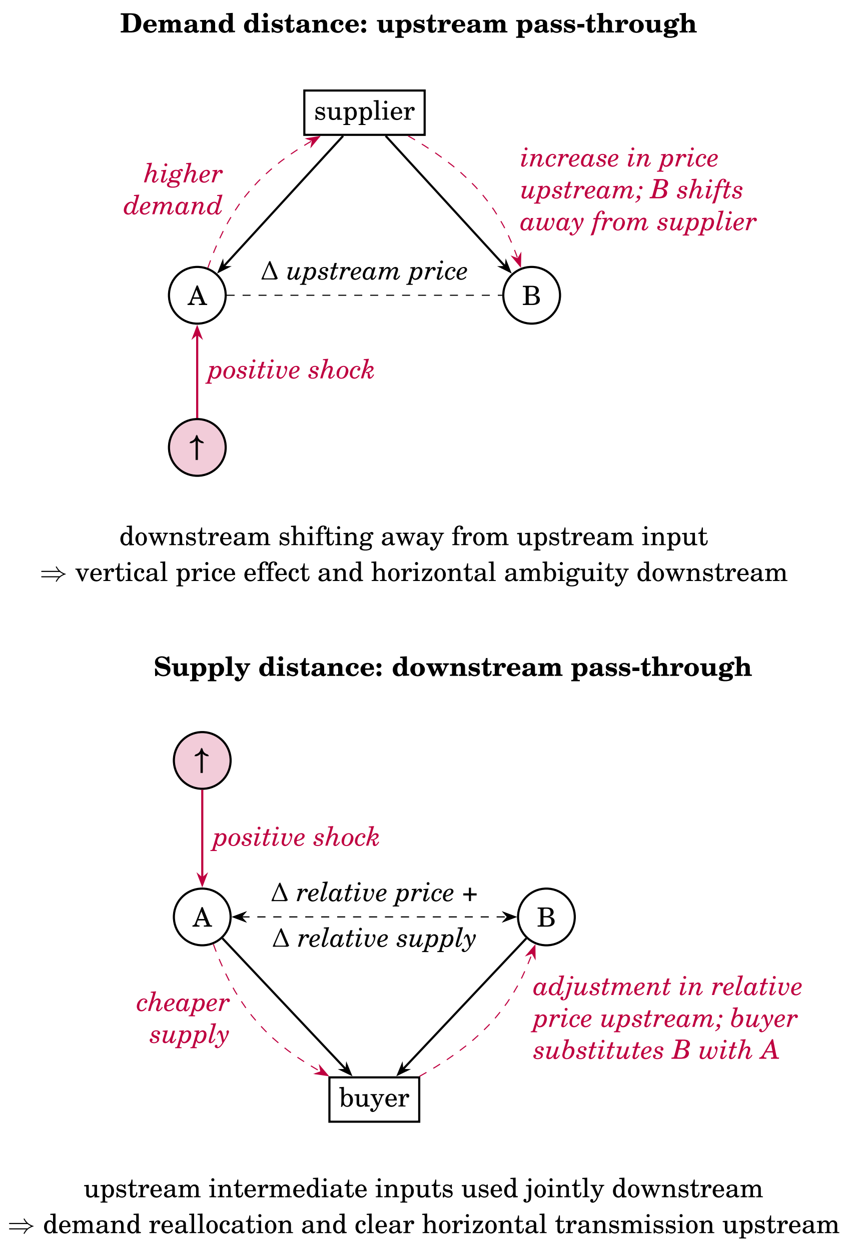


FIGURE A.1: DISTANCE-LEAD COMOVEMENT

Note: graphical visualization of Propositions 1-2 for comovement patterns within a production network based on network "economic" distances from resembling Input-Output structure across participants.

B. Data, figures and tables

(Data details) *In order to build the main dataset with a panel of 65 private 3-digit U.S. 2017 North American Industry Classification System (NAICS) sectors used in Sections 4-5, I rely on different data sources from the Bureau of Economic Analysis (BEA), otherwise stated.*[I] *For years from 1998 to 2022, these information are:*

(a) *employment (Persons Engaged in Production by Industry, in thousands. Last revised on: September 29, 2023);*

(b) *value-added (Value-Added by Industry, in billions of U.S. dollars. Last revised on: December 19, 2024);*

(c) *aggregate Total Factor Productivity (from Fernald 2014) and Government Consumption Expenditures and Gross Investment (e.g., Ramey and Zubairy 2018);*[II]

(d) *inter-sectoral trade and Input-Output linkages, in particular:*

- *sectoral share of own production, and thus the sectoral share of intermediate inputs from other sectors (Industry-by-Commodity Market Share Matrix, After Redefinitions – Summary, in producers' prices);*
- *total inputs required (directly and indirectly) in order to deliver one dollar of output to final users (Commodity-by-Commodity Total Requirements, After Redefinitions – Summary, in producers' prices);*
- *sector-specific imports and exports of goods and services, government (total, defence, non-defence) expenditure shares, as well as changes in private inventories (The Use of Commodities by Industries, Before Redefinitions – Summary, in producers' prices and in millions of U.S. dollars).*

Note that "Summary" defines the level of sectoral disaggregation considered in the analysis (3-digit). The BEA also provides information on Input-Output matrices to a more detailed level (6-digit), but these are on a five-yearly basis (thus it would impossible to run the analysis in Subsection 5.1), and data on other variables are not collected. Moreover, the sample begins in 1998 so to have all sectors classified given the NAICS system, and ends in 2022, the last year available when I started this project.[III] *Last time I accessed these public network data was in March 2025.*

Turn to the manipulation of Input-Output matrices. To recover the production share of intermediate inputs of a given sector coming from the production of other sectors, each cell of the yearly squared matrix contains the intermediate output share

[I] Note that I refer to 3-digit but, according to the BEA classification, four sectors are at 2-digit ("construction", "management of companies and enterprises", "educational services", and "other services, except government"), while five sectors related to finance and insurance are at 4-digit level.

[II] Here is the `website` for the TFP series. The U.S. government expenditure series is downloaded from the Federal Reserve Economic Data (FRED) database, with reference name *GCEA*; such series, net of federal defence expenditure (named *FDEFX*), represents the non-defence expenditure. Last revised on: June 15, 2026.

[III] The 3-digit classification is the most granular if one wants to keep track of the evolution in the set of intermediate inputs, and 1998 is the second year in which the NAICS system has been adopted. Before 1997, classification follows the U.S. Standard Industrial Classification (SIC) system.

from a given sector to another and, on the main diagonal, the share of intermediate output of a given sector directly produced by that sector. Henceforth, labelling by $\mathbf{X} = \left[x\left(s,s'\right)\right]$, with $x\left(s,s'\right) \geq 0$, this Input-Output matrix I compute, for each sector, $x\left(s\right) = 1 - x\left(s,s\right)$. In other words, to extract the total share of intermediate inputs that sector-s buys from other sectors, I compute $\boldsymbol{x} = \mathbf{1} - diag\left(\mathbf{X}\right)$, where $\mathbf{1}$ is an $m \times 1$ vector of ones; the resulting $m \times 1$ vector $\boldsymbol{x}$ reports, for every considered year, the total production of a sectoral good which is due to intermediate inputs bought through Input-Output relationships. For what concerns the production network structure, I rely on the main Input-Output matrix (Commodity-by-Commodity Total Requirements, After Redefinitions) which, consistently with the main text, I label as $\boldsymbol{\mathcal{A}} = \left[\alpha\left(s,s'\right) \geq 0\right]$.[IV] *The Leontief inverse matrix (labelled with $\boldsymbol{\mathcal{L}}$ in the main text) investigates its deep characteristics and, following the literature (e.g., Acemoglu et al. 2012; Carvalho 2014) I disregard small transactions across sectors: this allows me to account for a narrower network amplification due to the removal of bias from meaningless inter-sectoral trade intensity. Disregarding small transactions implies that $\alpha\left(s,s'\right) = \frac{\alpha\left(s,s'\right)}{\alpha\left(s\right)}$ is set to 0 if $\alpha\left(s,s'\right) \leq 0.01$, where $\alpha\left(s\right) = \sum_{s' \in row} \alpha\left(s,s'\right)$ is sector-s's total input purchases (i.e., row sum of matrix $\boldsymbol{\mathcal{A}}$), or $\alpha\left(s\right) = \sum_{s' \in column} \alpha\left(s,s'\right)$ is sector-s's total input sales (i.e., column sum of matrix $\boldsymbol{\mathcal{A}}$).*[V] *The resulting matrix would therefore display each cell to be greater than 0.01 – i.e., larger than 1%. Leontief inverse matrix, upstream and downstream sectors, and the weight associated to each of them (as in Subsections 5.1-5.2) are identified through it.*

Differently, to compute network distances between sectors I rely on the promptly-downloadable BEA Input-Output table (i.e., the Commodity-by-Commodity Total Requirements – After Redefinitions), which reflects the aggregate output of each commodity required to satisfy one unit of final demand for any given commodity. These values incorporate both direct and indirect flows within the production network, but they do not result from a formal inversion of a normalized technical coefficients matrix. In other words, while indirect dependencies are present, the network represented by this raw table does not inherently amplify interconnections in the manner of a true Leontief inverse, which recursively propagates direct inputs throughout the entire network. By contrast, when inverting it, then the classic Leontief inverse is generated, where even small direct flows are amplified through all indirect pathways, leading to inflated measures of sectoral interdependence. Therefore, when I derive Euclidean distances (eqs. B.1-B.2) between sectors using the raw table from BEA requirements, I am capturing "immediate" network relationships, that is observed direct and indirect linkages, without introducing the additional amplification inherent to the fixed technical coefficients of the Leontief inverse. From this I compute intensive margin values for both network "economic" distance metrics.

[IV] Row sectors identify (upstream) suppliers, while column sectors the (downstream) buyers: row sector is the origin and column sector is the destination of the circulating intermediate input.

[V] The "sales-based" Leontief inverse (in the main text) is a normalization of the baseline Input-Output matrix by sectoral total sales, so that each entry represents the share of a sector's inputs relative to its output.

(Shortest path algorithm, technical but suitable for coding) *Given element $\mathcal{G}$ being an $m \times m$ adjacency matrix with non-negative discrete values, the idea is to compute a distance matrix $\mathcal{D}$ where each cell $d[s, s']$ is the length of the shortest path between node-s and node-s'. The numerical algorithm is as follows:*

(a) from the Input-Output matrix $\mathcal{A}$, build an associated identity matrix, $\mathbf{I}$;

(b) set the initial length of the path, $k = 1$, and build a length matrix $\mathcal{G}_k = \mathcal{G}$;

(c) build a Boolean matrix, $\mathbf{B}$, whose elements are $b(s, s') = 1$ if $\mathcal{G}_k \neq 0$;

(d) as long as matrix $\mathbf{B}$ contains a value different from zero (i.e., there are connections among nodes not considered):

- *update identity matrix $\mathbf{I}$ by adding k whenever the condition on $\mathbf{B}$ is true, that is, $I(s, s') = I(s, s') + k$ if $b(s, s') = 1$;*
- *update lengths, $k = k + 1$;*
- *algebraic matrix calculation: $\mathcal{G}_k = \mathcal{G}_k' \mathcal{G}$;*
- *update $\mathbf{B}$ with new paths: set $b(s, s') = 1$ if $\mathcal{G}_k(s, s') \neq 0$ and $d[s, s'] = 0$;*

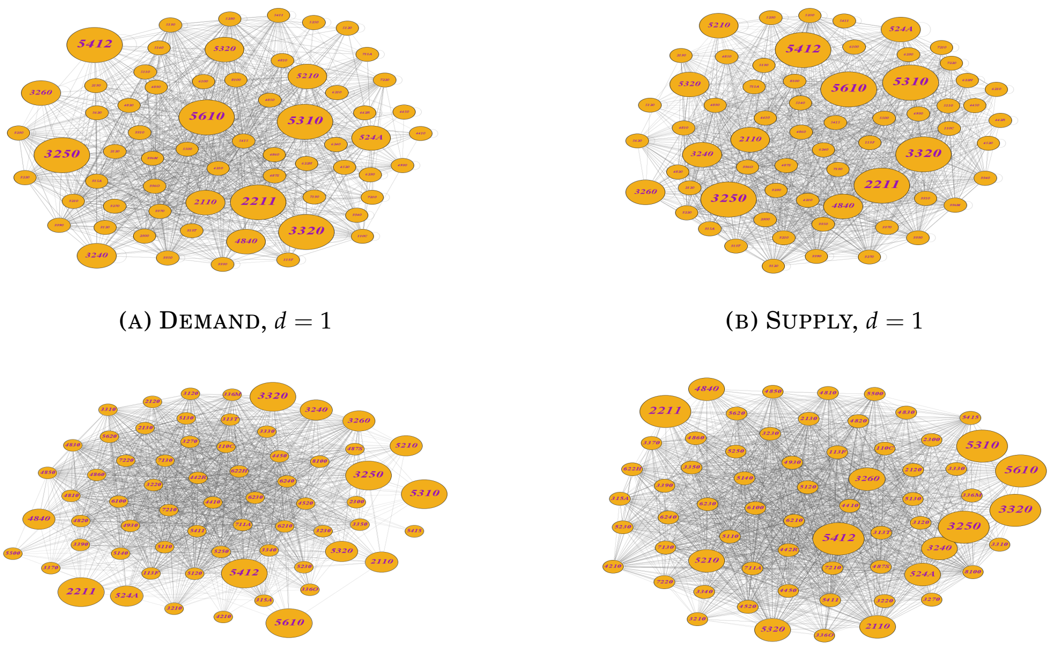
(e) once this loop concludes, set to infinite all the disconnected nodes, that is $d(s, s') = \infty$ if $d[s, s'] = 0$, to check consistency;

(g) set to the minimum distance the elements on the main diagonal, $\mathcal{D} = \mathcal{D} - \mathbf{I}$.

Outside this routine set to zero all the disconnected nodes (those at ∞). Matrix $\mathcal{D}_{ext}$, whose values are at the extensive margin and identify network distances among any pair $\{s, s'\}$, is built. The generic element identifying the existence of a link between node-s and node-s' in the distance matrix $\mathcal{D}$ is then $d_{ext}[s, s'] = \{1, 2, \ldots, d_{max}\}$.

(Characteristics of the distance networks) . *To illustrate the constructed distance matrices Figure B.1 displays, for each extensive margin value, the horizontal linkages among sectors from both common demand and supply network ties. Importantly, each connection does not necessarily represent inter-sectoral trade between two sectors, but rather it reflects an horizontal linkage based on demand- or supply-driven network distance relationships.*

A few concatenated points are then worth noting. First, the distribution of horizontal complementarities is uneven across distance levels. At very short range ($d = 1$), the demand-based network appears slightly denser than the supply-based one, suggesting that sectors are more frequently linked through common upstream suppliers than through shared downstream buyers. However, this asymmetry vanishes as the distance increases ($d = 2$), where both types of horizontal ties spread more diffusely across the network. Second, sectors with the highest number of Input-Output linkages tend to concentrate at the very core of the production system when considering only close connections ($d = 1$). As shown in Panels B.1a and B.1b, both the largest hubs (top 10% by connectivity) and the mid-sized hubs (top 20%) are similarly central, occupying short distances from the network's nucleus. Third, these same highly connected



(A) DEMAND, $d = 1$ (B) SUPPLY, $d = 1$

(C) DEMAND, $d = 2$ (D) SUPPLY, $d = 2$

FIGURE B.1: VISUALIZATION OF PRODUCTION NETWORK DISTANCES

Note: each panel of the figure represents the distance-based inter-sectoral production network corresponding to the (*commodity-by-commodity total direct requirements*) U.S. Input-Output matrix in the year 2007. *Factor input demand* and *factor input supply* network distances are computed implementing the *shortest path algorithm* on distance matrices whose values are at the intensive margin, as well derived from the *directed* configuration of the production network. Panels B.1a and B.1c correspond to the demand-based network distances, with $d = 1$ for low distance (closest sectors) and $d = 2$ for major distance (further sectors), among any pair of sectors; analogously it is for the supply-based network distances in Panels B.1b and B.1d. There are 65 3-digit U.S. 2017 NAICS private sectors, and each orange node corresponds to one of them. Two sectors connected under $d = 1$ cannot be linked under $d = 2$, and otherwise; constructed distance matrices are full (Theorem 2). Biggest nodes correspond to top 10% of mostly connected sectors (*i.e.*, major number of inter-sectoral connections), while intermediate ones the additional top 20%. Figure drawn with the software package *Gephi*, version 0.10, exploiting the *ForceAtlas2* layout algorithm. *Source*: BEA and own calculations.

sectors tend to become relatively more peripheral once longer distances ($d = 2$) are considered. Panels B.1c and B.1d show that, at this range, their positions are more evenly distributed across the network, reflecting the fact that their trade structures – both upstream and downstream – are highly similar to those of other sectors. In other words, most connected sectors are less unique at longer distances, as commonalities in demand and supply linkages become more widespread.

Altogether, these observations highlight that the structure of horizontal complementarities depend crucially on the notion of distance. When short, horizontal linkages cluster around the key sectors whereas, at longer distances, such complementarities are more evenly distributed. This pattern confirms that sectors typically share both upstream suppliers and downstream buyers at smaller but not at a greater distance, and the same vertical production system generates distinct horizontal geometries (Theorem 2).

TABLE B.1: LIST OF CONSIDERED SECTORS AND THEIR CODES

denomination	codes: NAICS 2017	codes: BEA
Farms	111, 112	111CA
Forestry, Fishing and Related Activities	113, 114, 115	113FF
Oil and Gas Extraction	211	211
Mining, except Oil and Gas	212	212
Support Activities for Mining	213	213
Utilities	22	22
Construction	23	23
Wood Products	321	321
Nonmetallic Mineral Products	327	327
Primary Metals	331	331
Fabricated Metal Products	332	332
Machinery	333	333
Computer and Electronic Products	334	334
Electrical Equipment, Appliances, and Components	335	335
Motor Vehicle, Bodies and Trailers, and Parts	3361-3	3361MV
Other Transportation Equipment	3364-9	3364OT
Furniture and Related Products	337	337
Miscellaneous Manufacturing	339	339
Food and Beverage and Tobacco Products	311, 312	311FT
Textile Mills and Textile Product Mills	313, 314	313TT
Apparel and Leather and Allied Products	315, 316	315AL
Paper Products	322	322
Printing and Related Support Activities	323	323
Petroleum and Coal Products	324	324
Chemical Products	325	325
Plastics and Rubber Products	326	326
Wholesale Trade	42	42
Motor Vehicle and Parts Dealers	441	441
Food and Beverage Stores	445	445
General Merchandise Stores	452	452
Other Retail	442-4, 446-8, 45 ex. 452	4AO
Air Transportation	481	481
Rail Transportation	482	482
Water Transportation	483	483
Truck Transportation	484	484
Transit and Ground Passenger Transportation	485	485

...

Sectors are defined according to the nomenclature used by the U.S. Bureau of Economic Analysis (BEA).

TABLE B.1: LIST OF CONSIDERED SECTORS AND THEIR CODES (*continued*)

denomination	***codes***	
	NAICS 2017	*BEA*
...		
Pipeline Transportation	486	486
Other Transportation and Support Activities	487, 488, 492	487OS
Warehousing and Storage	493	493
Publishing Industries, except Internet (includes Software)	511	511
Motion Picture and Sound Recording industries	512	512
Broadcasting and Telecommunications	515, 517	513
Data Processing, Internet Publishing, and Other Information Service	518, 519	514
Federal Reserve Banks, Credit Intermediation, and Related Activities	521, 5221, 5222, 5223	521CI
Securities, Commodity Contracts, and Investments	523	523
Insurance Carriers and Related Activities	5241, 5242	524
Funds, Trusts, and Other Financial Vehicles	525	525
Housing (not considered in the analysis)	·	HS
Other Real Estate	531	ORE
Rental and Leasing Services and Lessors of Intangible Assets	532, 533	532RL
Legal Services	5411	5411
Computer Systems Design and Related Services	5415	5415
Miscellaneous Professional, Scientific, and Technical Services	541 ex. 5411, 5415	5412OP
Management of Companies and Enterprises	55	55
Administrative and Support Services	561	561
Waste Management and Remediation Services	562	562
Educational Services	61	61
Ambulatory Health Care Services	621	621
Hospitals	622	622
Nursing and Residential Care Facilities	623	623
Social Assistance	624	624
Performing Arts, Spectator Sports, Museums, and Related Activities	711, 712	711AS
Amusement, Gambling, and Recreation industries	713	713
Accommodation	721	721
Food Services and Drinking Places	722	722
Other Services, except Government	81	81

Sectors are defined according to the nomenclature used by the U.S. Bureau of Economic Analysis (BEA).

(Network distances at the intensive margin) *Section 4 of the main text characterizes distance matrices at their "extensive margin", $\boldsymbol{\mathcal{D}}^{fd}_{ext}$ and $\boldsymbol{\mathcal{D}}^{fs}_{ext}$, whose values are $d^{j}_{ext}[s,s'] = \{1,2,\ldots,d_{max}\}$ for $j = \{fd, fs\}$, exploiting the shortest path algorithm on the "intensive margin" distances from the directed production network, $\boldsymbol{\mathcal{A}}$, that accounts for immediate (i.e., direct and indirect linkages not amplified by Leontief inverse transformation) sectoral connections only.*

Indeed, to characterize the "intensive margin" matrices $\boldsymbol{\mathcal{D}}^{fd}$ and $\boldsymbol{\mathcal{D}}^{fs}$ I exploit the system of Cartesian coordinates typical of any production network (e.g., Conley and Dupor 2003). Given $\alpha^{fd}(s,s') = \frac{\alpha(s,s')}{\sum_{s' \in row} \alpha(s,s')}$, the generic element of factor input demand distance matrix is computed as follows:

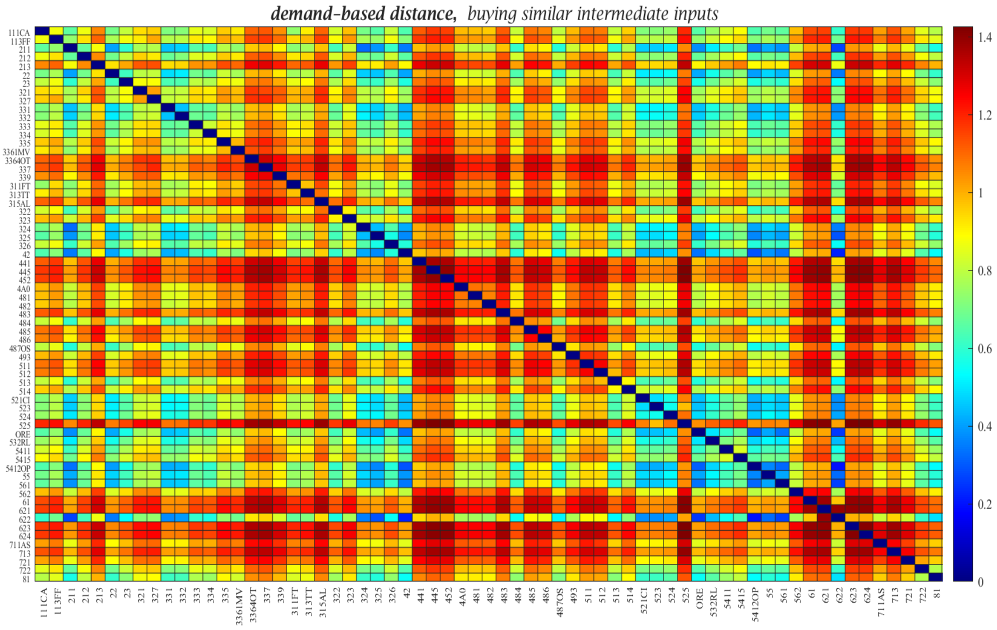
$$d^{fd}[s,s'] = \left\{ \sum_{s^* \in \Phi} \left[\alpha^{fd}(s,s^*) - \alpha^{fd}(s',s^*) \right]^2 \right\}^{\frac{1}{2}} \tag{B.1}$$

which defines the length of the line segment (i.e., the Euclidean distance) connecting sectors s and s' when both are buying from the same other sector-s^*, $\forall s^* \in \Phi$, so that $\boldsymbol{\mathcal{D}}^{fd} = \left[d^{fd}[s,s'] \right]$ with generic element $d^{fd}[s,s'] > 0$. Demand distance is constructed by dividing each cell by its row sum. Why? A row sum represents the total sales of the sector in question, so dividing each entry by its associated row sum yields the share of that (row) sector's output purchased by each of the column sectors. This normalization allows for a meaningful comparison across dyads of sectors: by examining these shares, one can assess the relative distance between two (column) sectors in terms of their purchases from common upstream suppliers, thereby pointing to the strength of the upstream pass-through (share of total sales of the common sector).

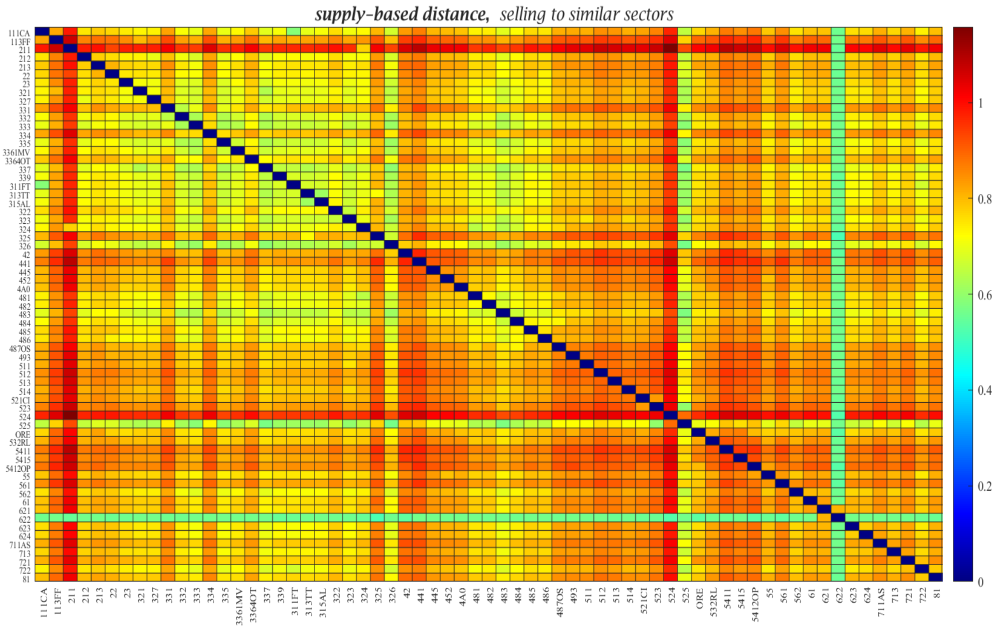
In an analogous way, given $\alpha^{fs}(s,s') = \frac{\alpha(s,s')}{\sum_{s' \in column} \alpha(s,s')}$, the generic element of factor input supply distance matrix is shaped by

$$d^{fs}[s,s'] = \left\{ \sum_{s^* \in \Phi} \left[\alpha^{fs}(s^*,s) - \alpha^{fs}(s^*,s') \right]^2 \right\}^{\frac{1}{2}} \tag{B.2}$$

tracing a Euclidean distance between sectors $\{s,s'\}$ when both are selling to the same sector-s^*, $\forall s^* \in \Phi$, so that $\boldsymbol{\mathcal{D}}^{fs} = \left[d^{fs}[s,s'] \right]$ with $d^{fs}[s,s'] > 0$. Supply distance is constructed by dividing each cell by its column sum. Why? A column sum represents the total purchases of the sector in question, so dividing each entry by its associated column sum yields the share of that (column) sector's input purchased from each of the row sectors. This normalization allows for a meaningful comparison across dyads of sectors: by examining these shares, one can assess the relative distance between two (row) sectors in terms of their traded quantities to common downstream buyers, thereby pointing to the strength of the downstream pass-through (share of total purchases of the common sector).



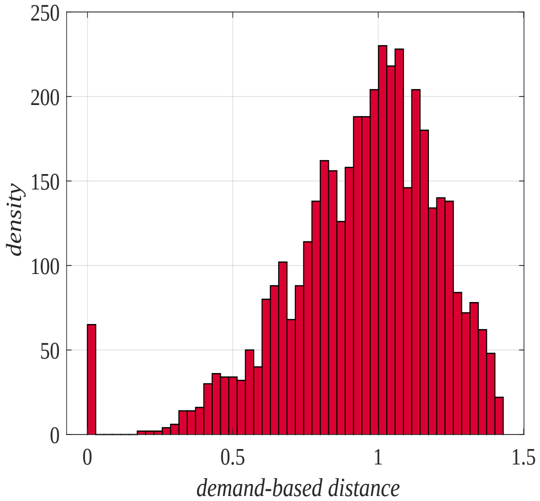
(A) FACTOR INPUT DEMAND



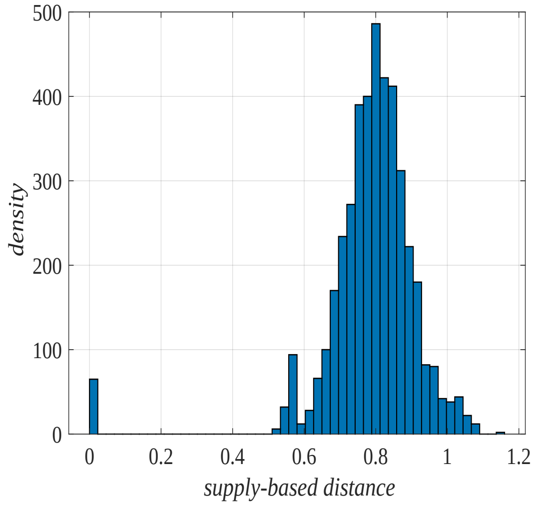
(B) FACTOR INPUT SUPPLY

FIGURE B.2: INTENSIVE MARGIN MATRICES FOR NETWORK ECONOMIC DISTANCE

Note: heatmaps of sectoral distances characterizing the U.S. production network in year 2007 for 3-digit U.S. 2017 NAICS sectors. Entry (s, s') represents the intensive marginal value, $d^j\,[s, s'] \geq 0$ for $j = \{fd, fs\}$, of the horizontal relationship between sector-s and sector-s' under *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers, Panel B.2a) and *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers, Panel B.2b). Diagonal sparsity (*i.e.*, lots of 0s), is a direct consequence of the fact that the difference of inter-sectoral trade intensities between a sector and itself is zero. Distance matrices are at the *intensive margin*, as in eqs. (B.1)-(B.2), computed from the *directed* sectoral production network. *Source*: BEA and own calculations.



(A) FACTOR INPUT DEMAND



(B) FACTOR INPUT SUPPLY

FIGURE B.3: DISTRIBUTION OF NETWORK ECONOMIC DISTANCES

Note: these figures depict the distributions for *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers, Panel B.3a) and *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers, Panel B.3b). Distance values on the x-axis are at the *intensive margin*, as in eqs. (B.1)-(B.2), computed from the *directed* sectoral production network *Source*: BEA and own calculations.

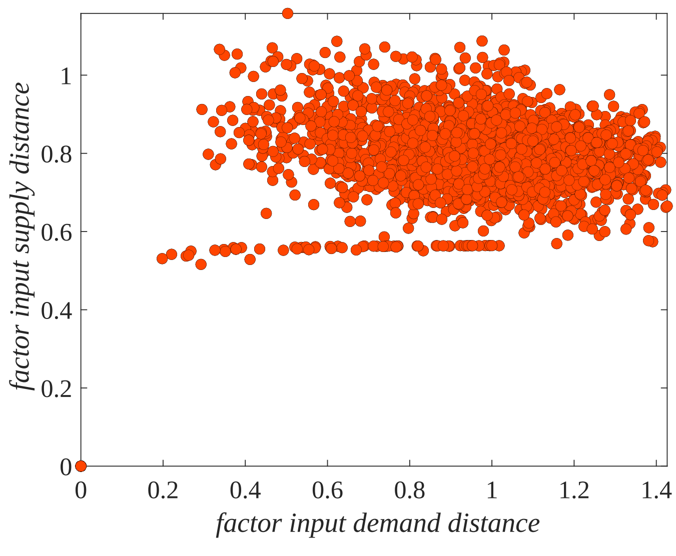
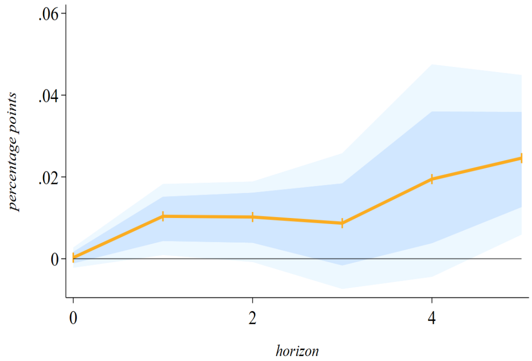


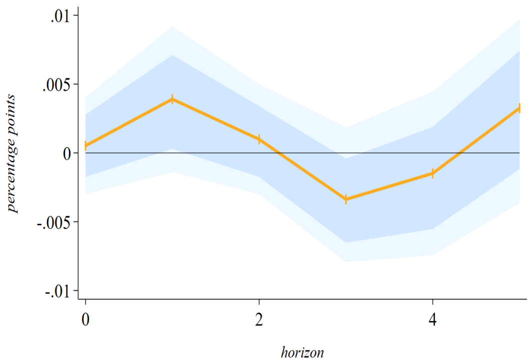
FIGURE B.4: CORRELATION OF NETWORK ECONOMIC DISTANCES

Note: the figure shows the correlation between the measures for *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers, x-axis) and *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers, y-axis). Distances are at the *intensive margin*, as in eqs. (B.1)-(B.2), computed from the *directed* sectoral production network *Source*: BEA and own calculations.

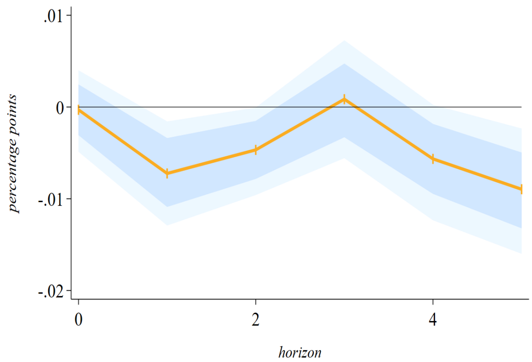
C. Complementary results



(A) Sector-specific



(B) $d = 1$

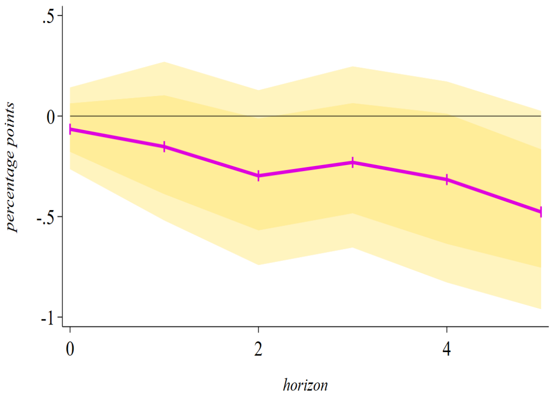


(C) $d = 2$

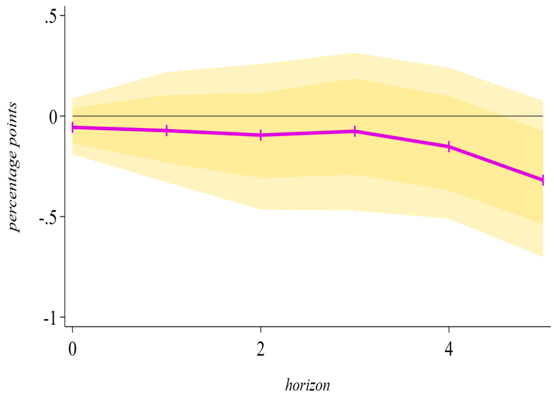
Figure C.1: Sectoral employment response to intermediate inputs

Note: given the *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment to a 1% increase in the sector-specific set of intermediate inputs as a share of its value-added for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. In particular, Panel C.1a represents the sector-level employment response to changes in its intermediate inputs, while Panel C.1b-C.1c to changes in the set of intermediates in closer (distance equal to 1) and further (distance equal to 2) sectors. The solid-orange line corresponds to the average response of employment across sectors, while shadow-blue and shadow-light blue areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eqs. (6)-(7). *Source*: BEA and own calculations.

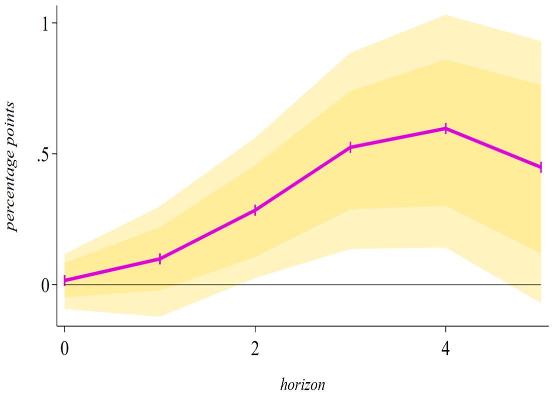
(Plotting separated results for highly interlinked sectors) *Unpacking all the combined plots, below I report the estimated Local Projection (LP) dynamics for top 5%, top 10%, top 20%, and top 30% of more connected sectors in the production network (i.e., major number of Input-Output linkages) separately.*[I] *The purpose is to show the significance of the responses appearing in Panels 11a-11b of Figure 11 in Subsection 5.2 of the main text, and in Panels 9a-9b of Figure 9 in Appendix C.*



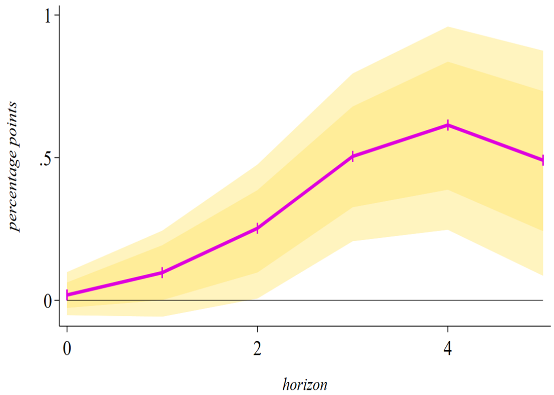
(A) TOP 5%



(B) TOP 10%



(C) TOP 20%

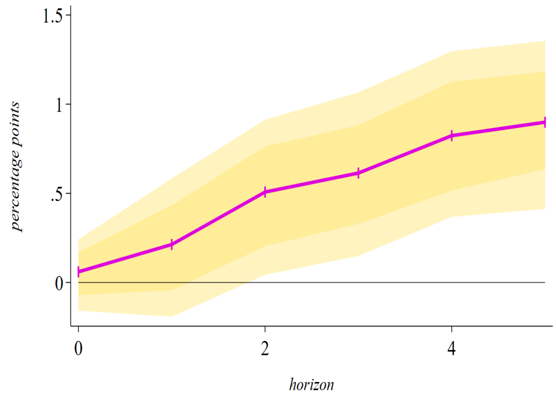


(D) TOP 30%

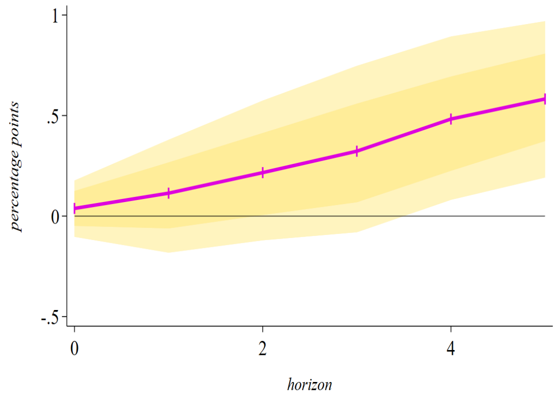
FIGURE C.2: DEMAND ($d = 1$) EFFECTS FOR INTERLINKED SECTORS

Note: given the *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Unpacking Panel 9a of Figure 9, each panel of this figure shows the employment response of top 5% (Panel C.2a), 10% (Panel C.2b), 20% (Panel C.2c), and 30% (Panel C.2d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their closer (distance equal to 1) sectors. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (9). *Source*: BEA and own calculations.

[I] Besides my network-based effects outlined in the main text, further theories beyond employment comovement are in Rogerson (1987) and Boldrin et al. (2001).



(A) TOP 5%



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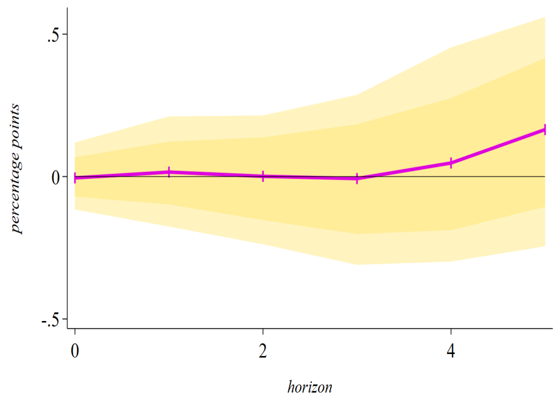
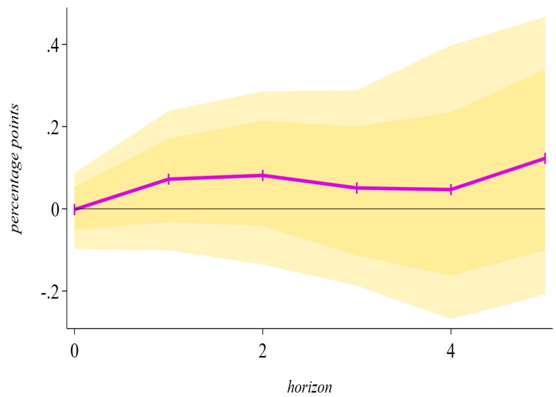


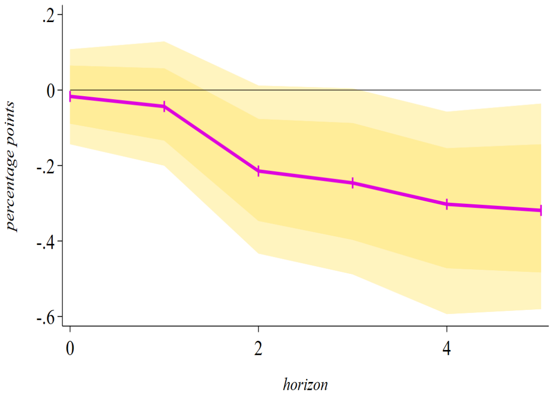
(C) TOP 20%



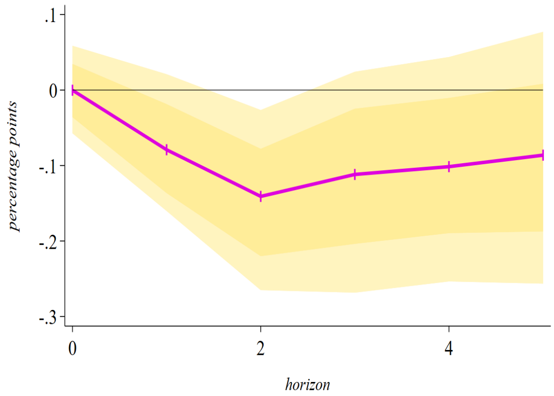
(D) TOP 30%

FIGURE C.3: DEMAND ($d = 2$) EFFECTS FOR INTERLINKED SECTORS

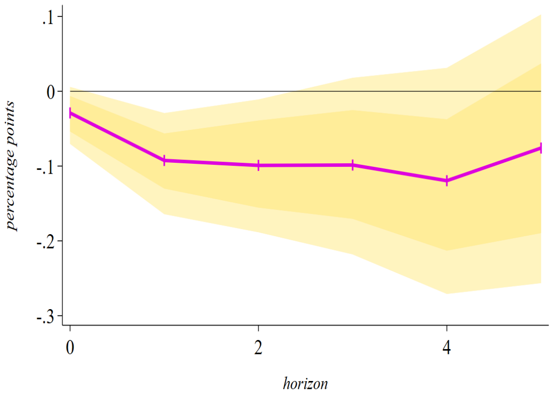
Note: given the *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Unpacking Panel 9b of Figure 9, each panel of this figure shows the employment response of top 20% (Panel C.3c) and 30% (Panel C.3d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) to employment changes in their further (distance equal to 2) sectors. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (9). *Source*: BEA and own calculations.



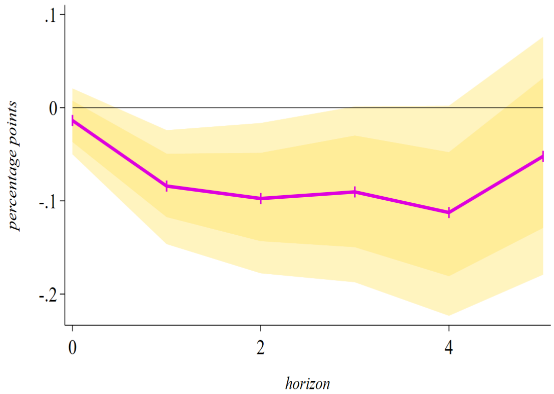
(A) TOP 5%



(B) TOP 10%



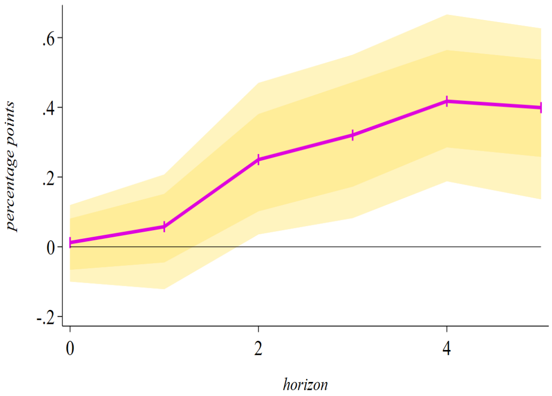
(C) TOP 20%



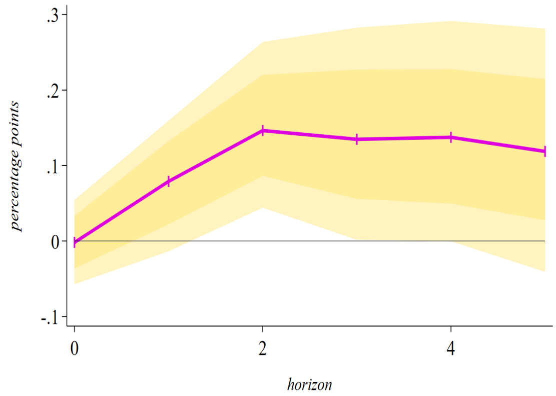
(D) TOP 30%

FIGURE C.4: SUPPLY ($d = 1$) EFFECTS FOR INTERLINKED SECTORS

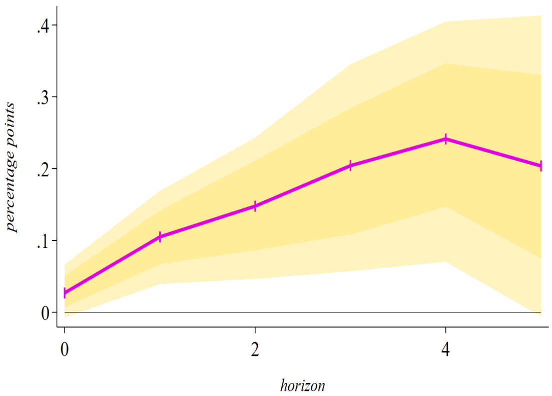
Note: given the *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Unpacking Panel 11a of Figure 11, each panel of this figure shows the employment response of top 5% (Panel C.4a), 10% (Panel C.4b), 20% (Panel C.4c), and 30% (Panel C.4d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their closer (distance equal to 1) sectors. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (9). *Source*: BEA and own calculations.



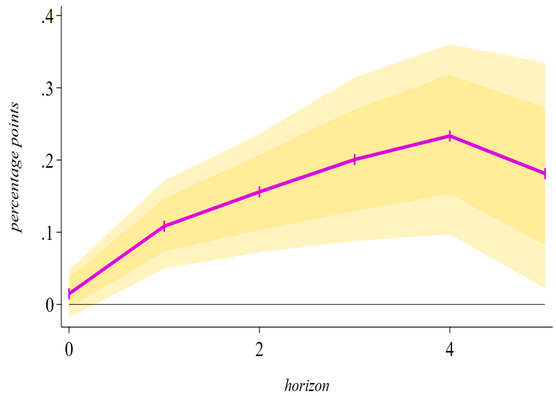
(A) TOP 5%



(B) TOP 10%



(C) TOP 20%



(D) TOP 30%

FIGURE C.5: SUPPLY ($d = 2$) EFFECTS FOR INTERLINKED SECTORS

Note: given the *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Unpacking Panel 11b of Figure 11, each panel of this figure shows the employment response of top 5% (Panel C.5a), 10% (Panel C.5b), 20% (Panel C.5c), and 30% (Panel C.5d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their further (distance equal to 2) sectors. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (9). *Source*: BEA and own calculations.

(Robustness with centrality measures) *To identify the centrality degree of each sector, I follow the approach outlined in Carvalho (2014) to measure the sector-level Bonacich-Katz centrality.*[II] *Using the Leontief-inverse matrix $\boldsymbol{\mathcal{L}}$ for the U.S. production network as of 2007, the Bonacich-Katz centrality $m \times 1$ vector, $\boldsymbol{bk}$, is defined as:*

$$\boldsymbol{bk} = \sum_{k=0}^{\infty} \zeta^k \boldsymbol{\mathcal{L}}^k \mathbf{1} = \left(\boldsymbol{I} - \zeta \boldsymbol{\mathcal{L}}\right)^{-1} \mathbf{1}, \tag{C.1}$$

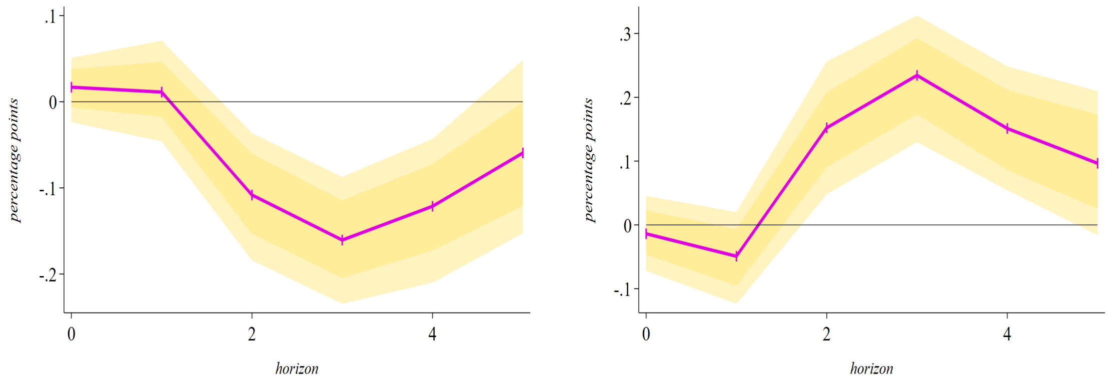
where index-k refers to the length of the paths in the production network – that is, the number of steps (or network "hops") connecting one sector to another – and, given a finite set of sectors $\{s, s', s'', \ldots, m\} \in \Phi$:

- $\mathbf{1}$ *is an $m \times 1$ vector of ones, reflecting the base influence;*
- $\zeta \in \left(0, \frac{1}{v_{max}}\right)$ *is a dampening scalar parameter, with v_{max} being the largest eigenvalue of matrix $\boldsymbol{\mathcal{L}}$;*
- $\boldsymbol{I}$ *is the $m \times m$ identity matrix.*

The value of ζ determines the weight assigned to indirect linkages: smaller values place more emphasis on direct connections, while larger values account for deeper propagation within the network. Following Carvalho (2014), a value of $\zeta = 0.5$ is typically used, which lies safely below the inverse of the largest eigenvalue of most empirically observed Input-Output matrices, ensuring convergence of the series. This measure reflects the idea that a sector is central not only if it is directly connected to many others, but also if it is connected to sectors that are themselves highly central.

This formulation ensures that the centrality score of each sector captures recursive amplification along both its direct connections and its indirect influence through other sectors due to the use of matrix $\boldsymbol{\mathcal{L}} = \left(\boldsymbol{I} - \boldsymbol{\mathcal{A}}\right)^{-1} = \left[\ell\,(s, s') \geq 0\right]$, where $\boldsymbol{\mathcal{A}} = \left[\alpha\,(s, s') \geq 0\right]$ is the directed production network matrix (i.e., that in the BEA Input-Output tables considering direct and indirect connections but not amplification), defining the intensity of good produced by sector-s' in the total intermediate inputs used by sector-s, with $\alpha\,(s, s') = 0$ indicating that sector-s does not make use of the good produced by sector-s' in producing its own intermediate good.

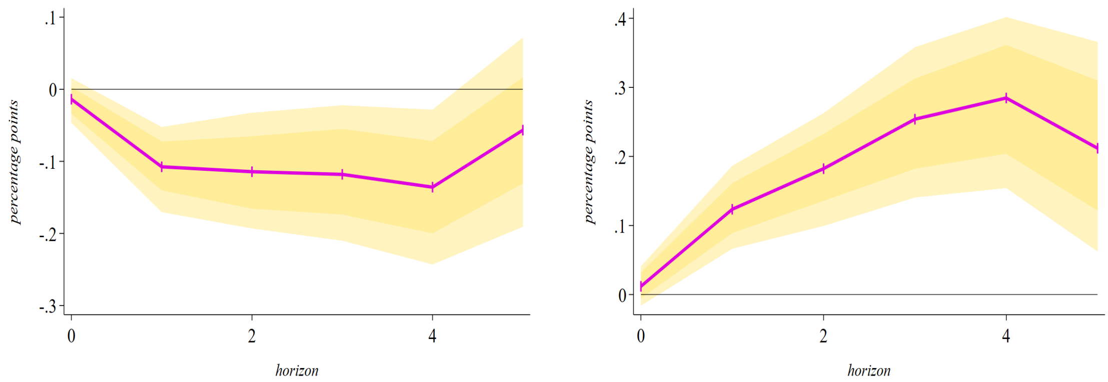
[II] Outlined by Bonacich (1987), it is a measure of a sector's overall importance within an Input-Output system, capturing not only its direct connections to other sectors but also the centrality of those it is connected to. In essence, a sector is considered central if it is linked to other central sectors, creating a recursive structure of influence. This measure goes beyond simple counts of linkages by incorporating indirect connections – weighted by a dampening factor – to reflect the diminishing influence of more distant sectors in the network. In the context of production network, a sector with high Bonacich-Katz centrality is one that plays a key role in the propagation of shocks, as its position allows it to influence (or be influenced by) large portions of the economy through both direct and indirect channels. Compared to other centrality measures, Bonacich-Katz is particularly well-suited for economic applications, as it accounts for the intensity of inter-sectoral relationships and the structure of the network in a realistic and nuanced way.



(A) $d = 1$ (B) $d = 2$

FIGURE C.6: DEMAND LINKAGES AND COMOVEMENT IN PERIPHERAL SECTORS

Note: given the *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of most central sectors (with *centrality* defined according to sector's relevance in final consumption) to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Panel C.6a represents the sector-level employment response to employment changes in closer (distance equal to 1) sectors, while Panel C.6b plots the response to employment changes in further (distance equal to 2) ones. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (9). *Source*: BEA and own calculations.

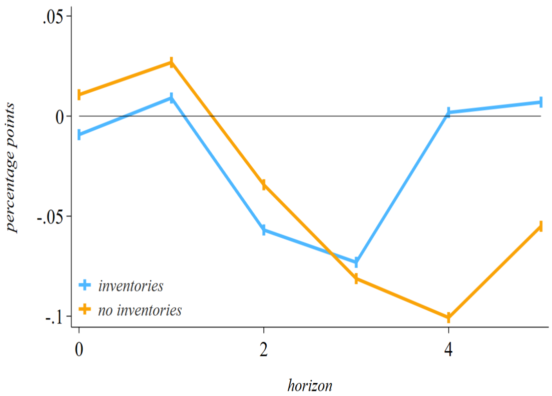


(A) $d = 1$ (B) $d = 2$

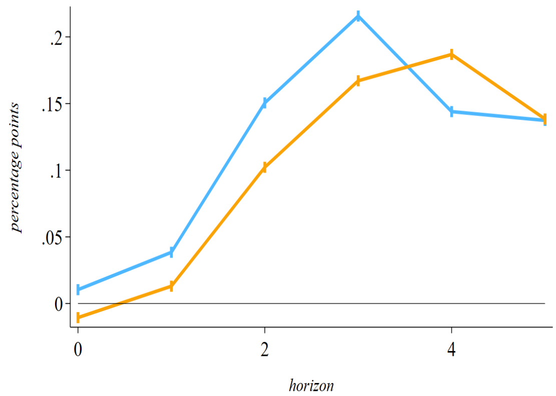
FIGURE C.7: SUPPLY LINKAGES AND COMOVEMENT IN PERIPHERAL SECTORS

Note: given the *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of most central sectors (with *centrality* defined according to sector's relevance in final consumption) to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Panel C.7a represents the sector-level employment response to employment changes in closer (distance equal to 1) sectors, while Panel C.7b plots the response to employment changes in further (distance equal to 2) ones. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (9). *Source*: BEA and own calculations.

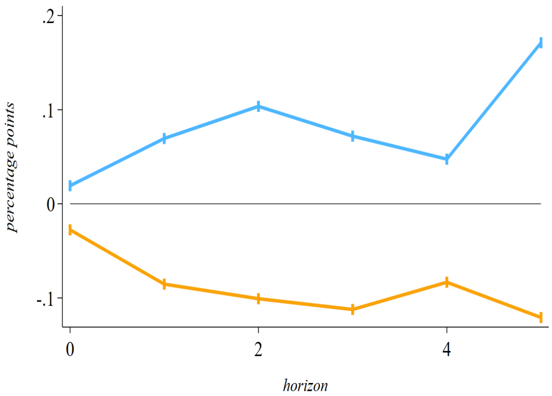
(Robustness with private inventories) *Responses for comovement in sectoral employment levels when controlling for changes in sectoral inventories, distinguishing further between sectors with inventories and those without. The results are shown in Figure C.8. Note that the plots refer dynamic changes in inventories and the Input-Output matrix held at the baseline year 2007.*



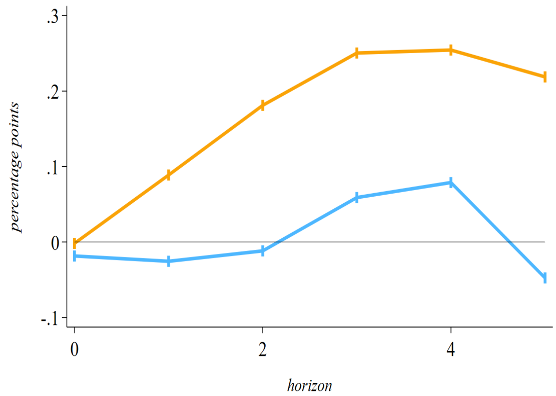
(A) DEMAND LINKAGES, $d = 1$



(B) DEMAND LINKAGES, $d = 2$



(C) SUPPLY LINKAGES, $d = 1$

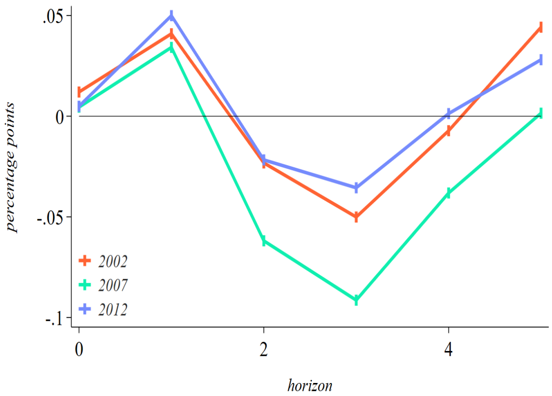


(D) SUPPLY LINKAGES, $d = 2$

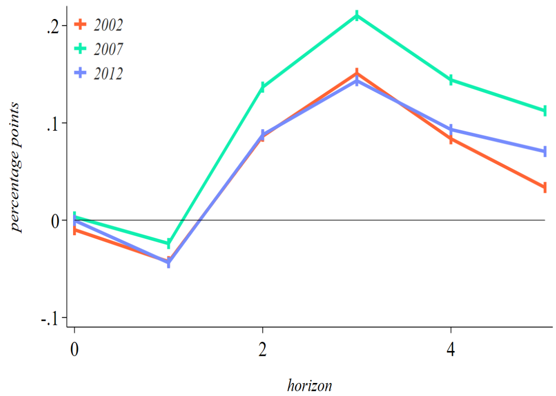
FIGURE C.8: COMOVEMENT UNDER CHANGES IN SECTORAL INVENTORIES

Note: given both the *factor input demand* (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *factor input supply* (*i.e.*, supply linkages across sectors given their common downstream buyers) network distances, and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors to a 1% increase in the employment level of other sectors when controlling for the changes in private inventories, for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Solid lines corresponds to the average response of employment across sectors, given 68% and 90% significance levels of bootstrapped Confidence Interval (CI), computed from eq. (9). Responses are referred to employment changes in closer (distance equal to 1) and further (distance equal to 2) sectors. Since not all sectors have inventories, the panels are disentangled by subgroups that make use or not of private inventories. *Source*: BEA and own calculations.

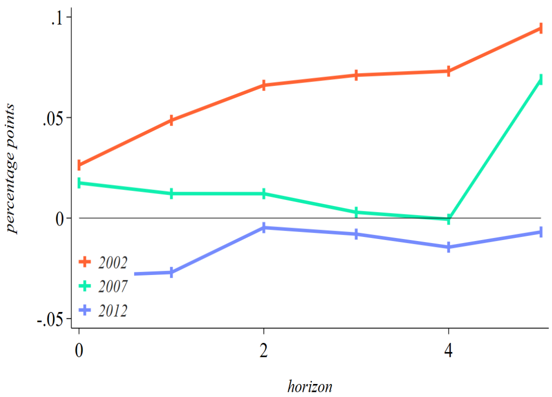
(Robustness with production network at different base years) *Responses for comovement in sectoral employment levels in the main analysis of Section 5 are centred on the implementation of a sectoral production network represented by an Input-Output matrix in the baseline year 2007. Differently, Figures C.9, C.10 and C.11 perform again all the main analysis by referring, instead, to a production network structure five years before (2002) and five years after (2012) the baseline year.*



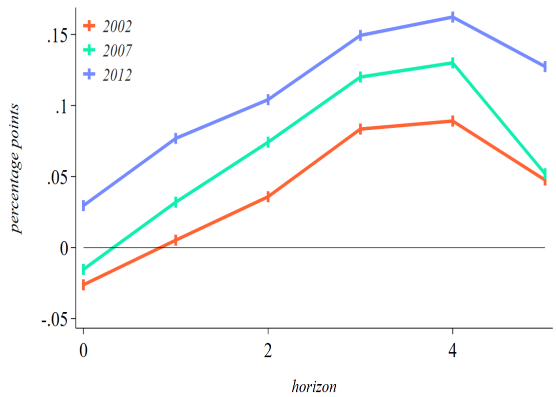
(A) DEMAND LINKAGES, $d = 1$



(B) DEMAND LINKAGES, $d = 2$



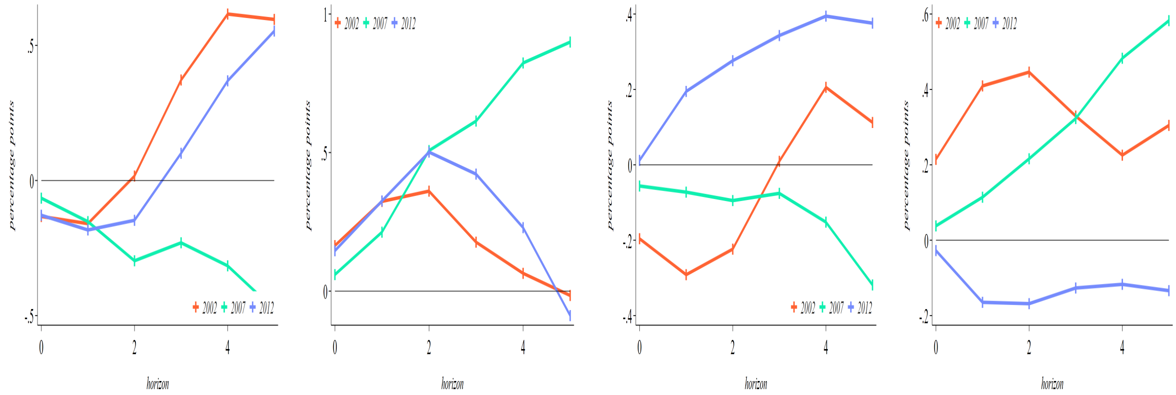
(C) SUPPLY LINKAGES, $d = 1$



(D) SUPPLY LINKAGES, $d = 2$

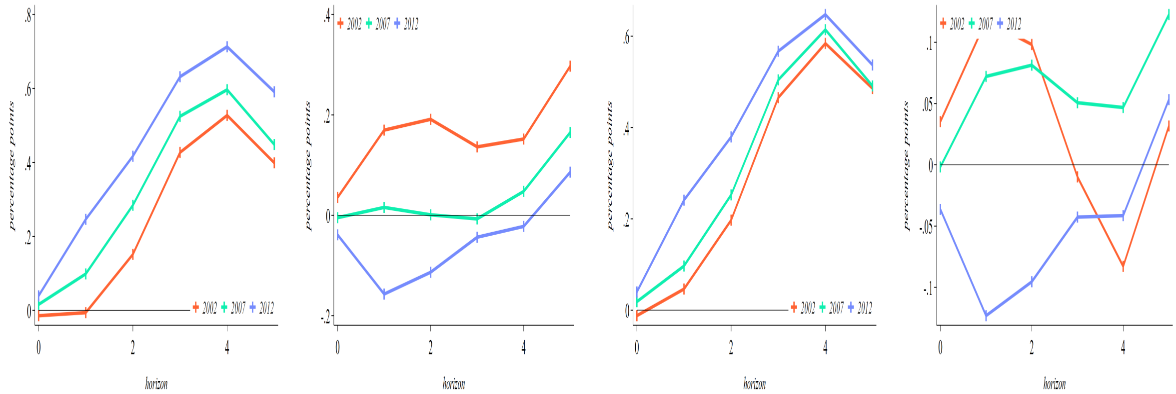
FIGURE C.9: COMOVEMENT OVER DIFFERENT BASELINE YEARS

Note: given both the *factor input demand* (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *factor input supply* (*i.e.*, supply linkages across sectors given their common downstream buyers) network distances, and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network over different years (2002, 2007, and 2012), the figure shows the response of sectoral employment of sectors to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Expanding to different production network structures the results in the main text: Panels C.9a-C.9b expand their relatives of Figure 8, while Panels C.9c-C.9d do so for their relatives in Figure 10. Solid lines corresponds to the average response of employment across sectors, while 68% and 90% significance levels of bootstrapped Confidence Interval (CI), computed from eq. (9), roughly correspond to the associated figures in the main text. Responses are referred to employment changes in closer (distance equal to 1) and further (distance equal to 2) sectors. *Source*: BEA and own calculations.



$d = 1$ $d = 2$ $d = 1$ $d = 2$

(A) TOP 5% (B) TOP 10%

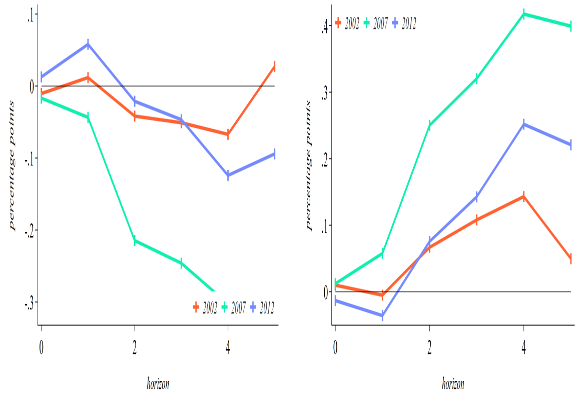


$d = 1$ $d = 2$ $d = 1$ $d = 2$

(C) TOP 20% (D) TOP 30%

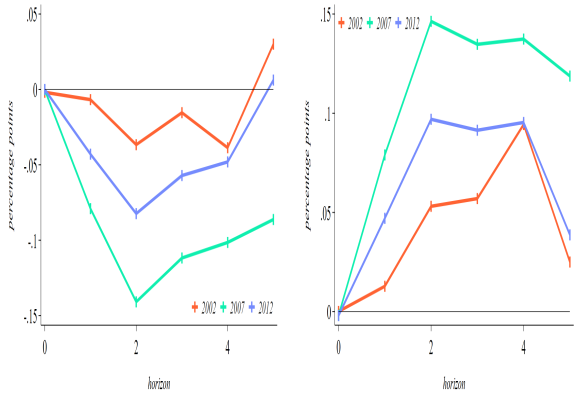
FIGURE C.10: DEMAND LINKAGES AND INTERLINKED SECTORS OVER THE YEARS

Note: given both the *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network over different years (2002, 2007, and 2012), the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Expanding to different production network structures the responses of Panels 9a-9b of Figure 9, each panel of this figure shows the employment response of top 5% (Panel C.10a), 10% (Panel C.10b), 20% (Panel C.10c), and 30% (Panel C.10d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their closer (distance equal to 1) or further (distance equal to 2) sectors. Solid lines corresponds to the average response of employment across sectors, while 68% and 90% significance levels of bootstrapped Confidence Interval (CI), computed from eq. (9), roughly correspond to the associated figure. *Source*: BEA and own calculations.



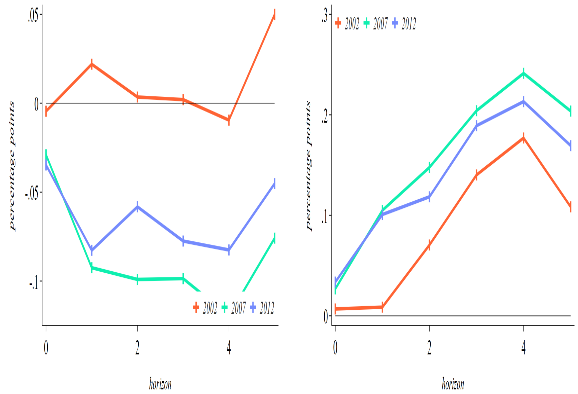
$d = 1$ $d = 2$

(A) TOP 5%



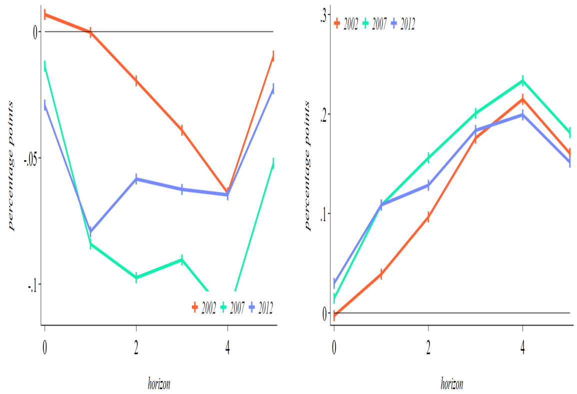
$d = 1$ $d = 2$

(B) TOP 10%



$d = 1$ $d = 2$

(C) TOP 20%



$d = 1$ $d = 2$

(D) TOP 30%

FIGURE C.11: SUPPLY LINKAGES AND INTERLINKED SECTORS OVER THE YEARS

Note: given both the *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network over different years (2002, 2007, and 2012), the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Expanding to different production network structures the responses of Panels 11a-11b of Figure 11, each panel of this figure shows the employment response of top 5% (Panel C.11a), 10% (Panel C.11b), 20% (Panel C.11c), and 30% (Panel C.11d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their closer (distance equal to 1) or further (distance equal to 2) sectors. Solid lines corresponds to the average response of employment across sectors, while 68% and 90% significance levels of bootstrapped Confidence Interval (CI), computed from eq. (9), roughly correspond to the associated figure. *Source*: BEA and own calculations.

(Illustrative episodes of horizontal transmission) *In conclusion, a set of recent real-world episodes that exemplify the formalized mechanisms of horizontal shock transmission are presented. Firms / sectors not directly linked through bilateral Input-Output transactions nonetheless exhibit simultaneous outcomes because they share a common downstream buyer (supply-based distance) or a common upstream supplier (demand-based distance). These episodes illustrate how network-based horizontal economic distance, rather than observed vertical exposure, can generate comovement.*

***Technological shock in Samsung and supply-based spillovers to competing memory suppliers (2026)**.– On February 2, 2026, Samsung Electronics announced significant technological progress in the development and production of next-generation High Bandwidth Memory 4 (HBM4) chips. HBM is a vertically stacked dynamic memory technology optimized for high-speed data processing and is a critical input for Artificial Intelligence (AI) accelerators produced by firms such as NVIDIA. The announcement constituted a positive, firm-specific productivity shock for Samsung in the high-performance memory segment. Because Samsung and other memory producers – most notably Micron and SK Hynix – compete to supply advanced memory chips to the same downstream buyers (including NVIDIA), market expectations regarding future procurement patterns were immediately revised. Financial press coverage reported that investors anticipated a potential reallocation of orders toward Samsung, reflecting its improved technological position. Importantly, Samsung and its rival suppliers do not maintain vertical trade relationships with one another. Nevertheless, the positive shock in Samsung generated adverse valuation effects for competing firms, reflecting expected substitution by the common downstream buyer.*[III]

***Supply-based Apple's reallocation of AirPods assembly orders (2022)**.– In 2022, Apple reduced and partially suspended assembly orders for AirPods 2 at Goertek, a major Chinese contract manufacturer. The shock was idiosyncratic to Goertek, stemming from operational issues that impaired its ability to meet Apple's standards and delivery schedules. Rather than reducing overall AirPods production proportionally, Apple responded by shifting demand from the negatively shocked supplier to its competitor, reallocating a substantial share of the affected orders to Luxshare Precision Industry, another contract manufacturer producing similar components for Apple. As a result, Luxshare experienced an increase in expected output and revenues, while Goertek faced declining orders and downward revisions in market expectations. Goertek and Luxshare are not vertically linked to each other; however, they are horizontally proximate because they depend on the same downstream buyer (Apple). The negative shock in one supplier therefore translated into a positive demand shift toward its competitor: Luxshare experienced an increase in expected orders and revenues despite having no direct vertical linkage with Goertek.*[IV]

[III] Reported by financial outlets such as `Barron's`, `investing.com`, and `MarketWatch` in February 2026.

[IV] Covered by `Yicai Global`, `Euronew` and `MacRumors` in November 2022.

***Supply-based Apple's reallocation for OLED panels (2026)**.– In January 2026, Apple shifted a significant portion of its iPhone OLED (Organic Light-Emitting Diode) panel orders from BOE Technology Group in China to the american company Samsung Display after persistent production and yield issues affected BOE's ability to meet quality and volume requirements. OLED panels are critical display components used in iPhones, and Apple relies on multiple suppliers to ensure supply stability and bargaining power. The operational difficulties at BOE constituted a negative firm-specific shock. In response, Apple reallocated millions of display panel orders to Samsung Display, which had available capacity and superior production reliability. The reallocation of downstream demand increased expected revenues for Samsung Display while reducing BOE's projected sales and market share. As in the AirPods episode, the two suppliers are not connected through vertical production relationships; rather, they are horizontally linked because they sell to the same downstream buyer. The transmission of the shock occurred through Apple's procurement decision, not through direct trade between BOE and Samsung Display.*[V]

***AI-driven surge in memory chips demand (2025)**.– In 2025, the rapid expansion of Artificial Intelligence (AI) applications led to an unprecedented increase in global demand for advanced memory chips, particularly DRAM (Dynamic Random Access Memory) and HBM (High Bandwidth Memory). DRAM constitutes the primary volatile memory used in computing systems, while HBM is a high-performance, vertically stacked memory technology optimized for AI accelerators and data-intensive workloads. The surge in AI-related computing, especially for large language models and generative AI systems, generated a sharp increase in procurement by major cloud service providers and AI hardware producers. Because global production capacity for advanced memory is concentrated among a small number of firms (e.g., Samsung, SK Hynix, Micron), the downstream demand shock tightened upstream supply conditions and increased prices. Downstream sectors relying on these same memory inputs – including consumer electronics manufacturers, enterprise hardware producers, and specialized computing firms – faced rising input costs and constrained availability, even though they were not directly linked to each other through trade.*[VI]

***Cloud-based services expansion (2023-2024)**.– Between 2023 and 2024, large-scale investment in cloud computing infrastructure – driven by AI adoption and data center expansion – significantly increased downstream demand for advanced semiconductors, including graphics processing units (GPUs) and specialized AI accelerators. This expansion placed substantial pressure on upstream semiconductor manufacturing capacity. Because leading-edge chip fabrication is highly concentrated – particularly in firms such as Taiwan Semiconductor Manufacturing Com-*

[V] Reported by `MacRumors`, `TechNode`, and `DisplayDaily` in January 2026.

[VI] Reported in 2025 by `investing.com`, `Business Standard`, and `Reuters`.

pany (TSMC) – the increase in cloud-sector demand reduced effective upstream supply available to other industries relying on the same fabrication processes. Automotive manufacturers, consumer electronics firms, and industrial equipment producers faced delays and supply constraints despite having no direct trade relationship with cloud service providers. The resulting synchronized production adjustments across otherwise unrelated sectors reflected shared exposure to upstream fabrication capacity rather than vertical trade interdependence.[VII]

Reallocation of TSMC supply toward AI customers (2026). *– In 2023, the rapid expansion of Artificial Intelligence (AI) applications triggered a sharp increase in downstream demand for high-performance computing chips, particularly graphics processing units (GPUs). These chips rely on advanced fabrication nodes supplied by TSMC, the world's leading contract chip manufacturer, operating as a pure-play foundry, producing chips designed by numerous fabless semiconductor firms across industries, and whose cutting-edge production capacity is both technologically complex and capacity-constrained. Given the long planning horizon of semiconductor manufacturing – with wafer capacity typically allocated several quarters, and often years, in advance – the AI-driven surge in orders during 2023 induced TSMC to reprioritize production toward high-margin AI customers: in 2024, TMSC reallocated portions of its advanced fabrication capacity toward AI chip designers such as NVIDIA, responding to exceptionally strong demand for AI accelerators. Industry reports indicate that such reallocations affected fabrication schedules extending into 2025–2026, as a growing share of advanced-node capacity was reserved for AI-related chips. Given finite leading-edge fabrication capacity, TMSC's prioritization of AI-related chip production effectively reduced available output for other clients, including smartphone and consumer electronics chip designers. As a consequence, other downstream clients competing for the same foundry capacity faced relatively tighter access or lower priority in allocation. These experienced longer lead times and constrained access to advanced nodes, despite not interacting directly with the AI hardware sector. The shock therefore originated in heightened demand from one downstream segment (AI hardware) but propagated through TSMC as a shared upstream production node, generating horizontal effects on multiple unrelated industries.*[VIII]

Across all episodes, the defining feature is that idiosyncratic variations are mediated by the horizontal geometry of the production network rather than by bilateral trade, where shared downstream buyers (supply-based distance) or shared upstream suppliers (demand-based distance) determine the strength and the direction of micro-level shock transmission. These cases provide qualitative empirical grounding for the paper's central theoretical insight: comovement can arise from horizontal network proximity even in the absence of vertical Input-Output linkages.

[VII] Documented by the `CNBC`, `Reuters`, and `Financial Times` during 2023-2024.

[VIII] Reported in January-February 2026 by `MacRumors`, `Business Insider`, and `AppleInsider`.

References in appendix

Acemoglu, D., V. M. Carvalho, A. Ozdaglar, and A. Tahbaz-Salehi (2012), "The network origins of aggregate fluctuations", *Econometrica*, 80(5):1977–2016

Boldrin, M., L. J. Christiano, and J. D. M. Fisher (2001), "Habit persistence, asset returns, and the business cycle", *American Economic Review*, 91(1):149–166

Bonacich, P. (1987), "Power and centrality: a family of measures", *American Journal of Sociology*, 92(5):1170–1182

Carvalho, V. M. (2014), "From micro to macro via production networks", *Journal of Economic Perspectives*, 28(4):23–48

Conley, T. G. and B. Dupor (2003), "A spatial analysis of sectoral complementarity", *Journal of Political Economy*, 11(2):311–352

Fernald, J. G. (2014), "A quarterly, utilization-adjusted series on Total Factor Productivity", *Federal Reserve Bank of San Francisco Working Paper 2012-19*

Lucas, R. E. (1977), "Understanding business cycles", *Carnegie-Rochester Conference Series on Public Policy*, 5:7–29

Oulton, N. (2022), "The Divisia approach to measuring output and productivity: with an application to the BEA-BLS integrated industry-level production account, 1987-2020", *Discussion Paper 2022-17, Centre For Macroeconomics*

Ramey, V. A. and S. Zubairy (2018), "Government spending multipliers in good times and in bad: evidence from US historical data", *Journal of Political Economy*, 126(2):850–901

Rogerson, R. (1987), "An equilibrium model of sectoral reallocations", *Journal of Politcal Economy*, 95(4):824–834

— (1988), "Indivisible labor, lotteries and equilibrium", *Journal of Monetary Economics*, 21(1):3–16

vom Lehn, C. and T. Winberry (2022), "The investment network, sectoral comovement, and the changing U.S. business cycle", *Quarterly Journal of Economics*, 137(1):387–433